agenda/afternoon – breakout rooms

Session 2: analysis of your own sequencing data using allen institute resources.

Room 1165 – 1185

1:10-1:35 | Using a reference cell type taxonomy to understand Alzheimer's disease

1:35-2:00 | cross-species alignment and comparative analyses

2:00-2:20 | other interactive tools for -omics data exploration (CellxGene, Cytosplore, etc.)

2:20-2:30 | break

2:30-2:40 | ABC Atlas demo & external use case examples

2:40-3:40 | user exploration of allen institute tools

3:40-4:00 | feedback survey

4:00 – 5:00 | additional time for user exploration

github.com/AllenInstitute/SPOCTT/tree/main/2026_Additional_Presentations

Using a reference cell type taxonomy to understand Alzheimer's disease/

Kyle Travaglini, Assistant Investigator



allen institute/

allen institute

Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD)

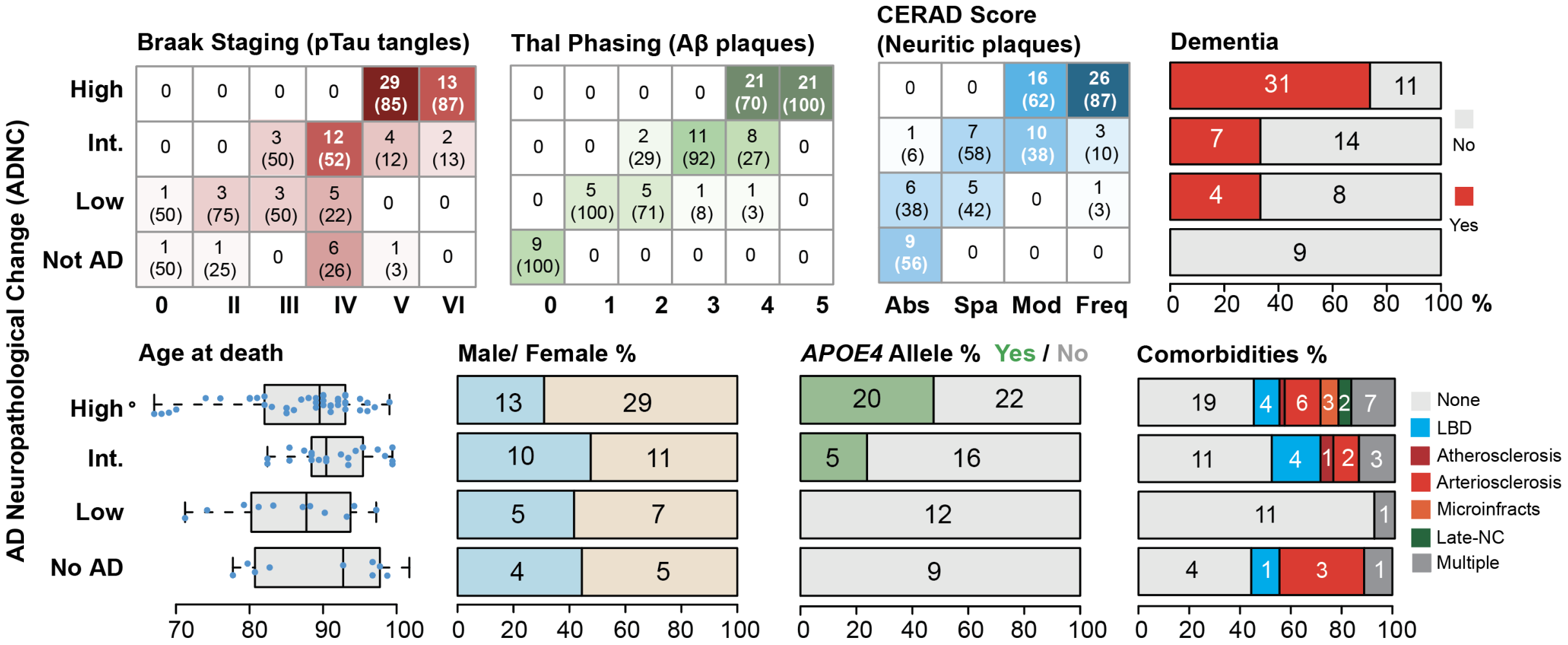


The Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD) consortium is supported by the National Institutes on Aging (NIA) grant 2U19AG060909.

SEA-AD: Primary Goals

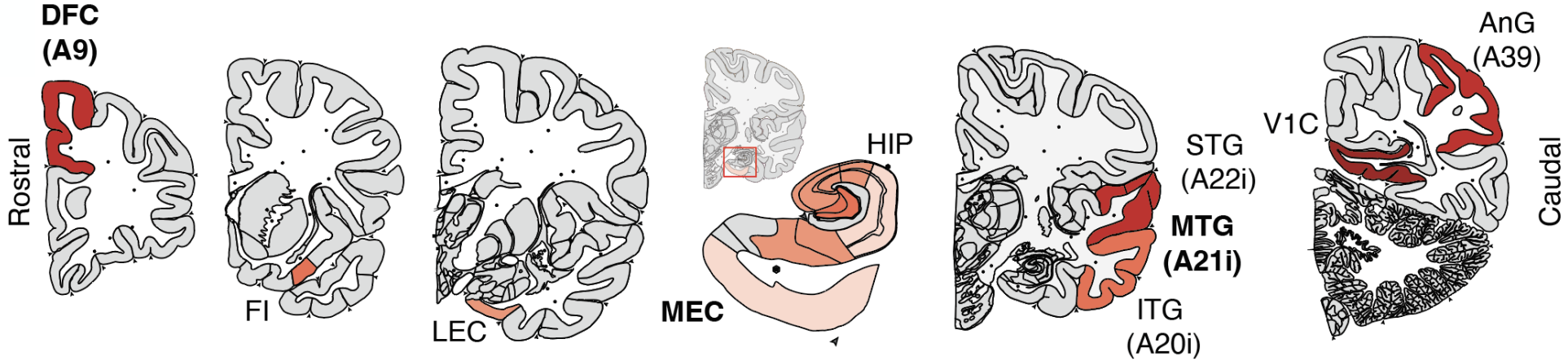
- Move beyond a case-control paradigm using data-driven representation of AD
- Determine and stage the molecular and cellular changes that occur in AD
- Prioritize cellular and molecular changes related to AD initiation and progression
- Nominate intervention strategies for notable cellular and molecular targets

Moving beyond case-control to capture the spectrum of AD pathological and cognitive phenotypes



Gabitto, Travaglini et al (2024)

Multimodal characterization of each brain region

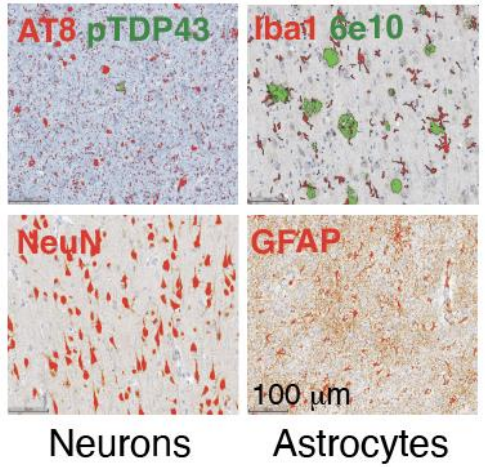


Braak I/II: MEC, CA1 (HIP) Braak III: Sub (HIP), CA2-3 (HIP), LEC Braak IV: CA4 (HIP), ITG, MTG, FI

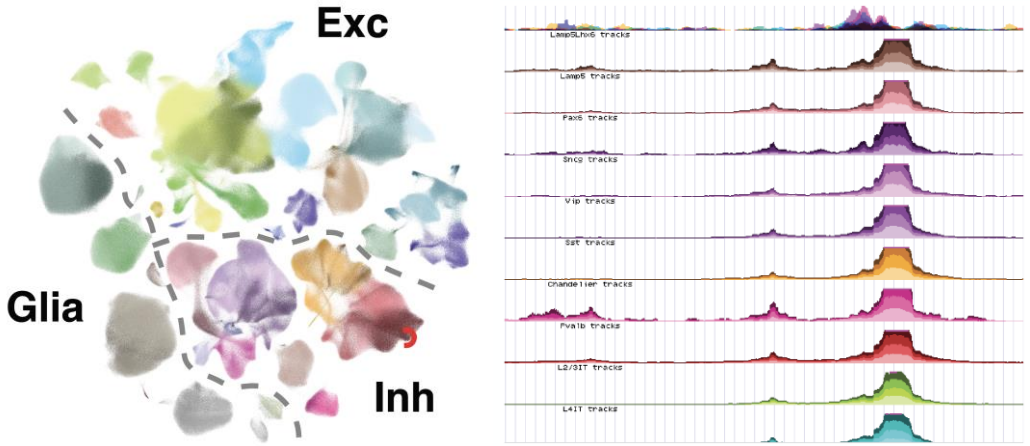
Braak V: STG, DFC, AnG Braak VI: V1C AD spectrum only (n=43) / **AD spectrum and co-morbidities (n=84)**

Quantitative Neuropathology

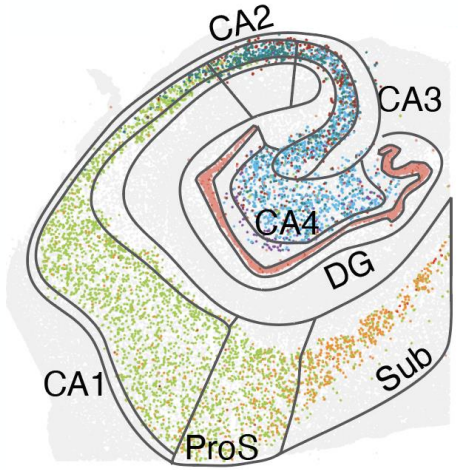
pTau & LATE Aβ & Microglia



Single nucleus multiomics (RNAseq and ATACseq)



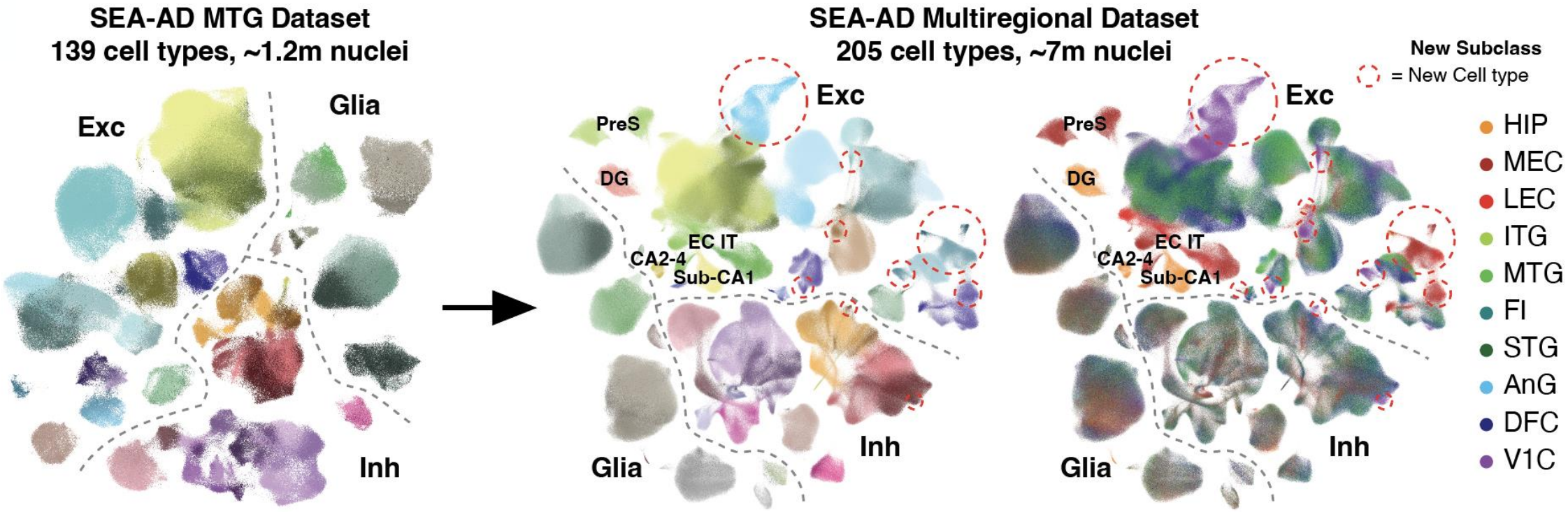
Targeted spatial genomics



Whole genome sequencing

Travaglini, Gabitto, Ding et al (2026)

All -omics data mapped to an expanded BICAN taxonomy to account for regional cell types and AD-specific states



The challenge: need to reliably map roughly 1k libraries from 84 donors across 10 brain regions to the reference taxonomy

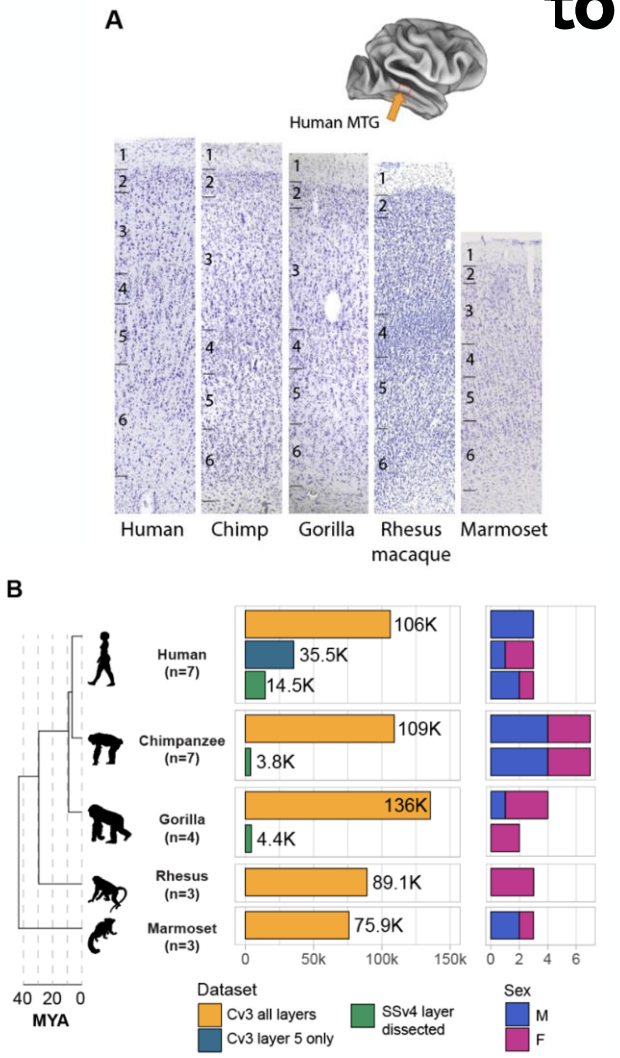
1. Benchmark the reference taxonomy using K-fold validation to determine which types are mappable using deep learning models

2. Hierarchically map SEA-AD data to each cell type taxonomic level
   - 3 classes -> 28 subclasses -> 205 supertypes

3. Identify low quality nuclei in the SEA-AD data using defined metrics

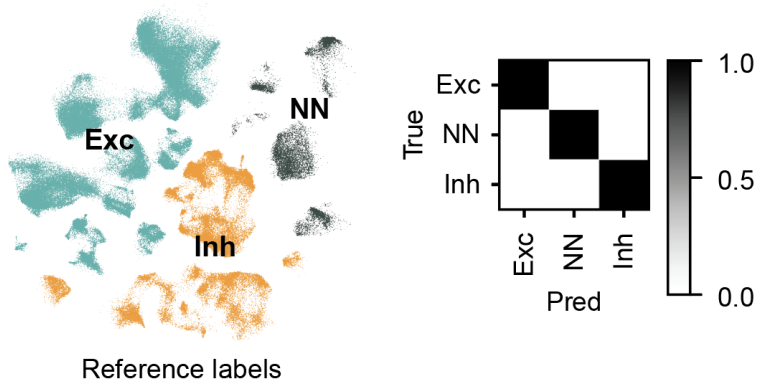
4. Identify new cellular states/types not present in the reference taxonomy
   1. Alzheimer's associated states (Microglia)
   2. Populations not present in the reference due to deeper sampling

Benchmark the reference taxonomy using K-fold validation to determine which types are mappable

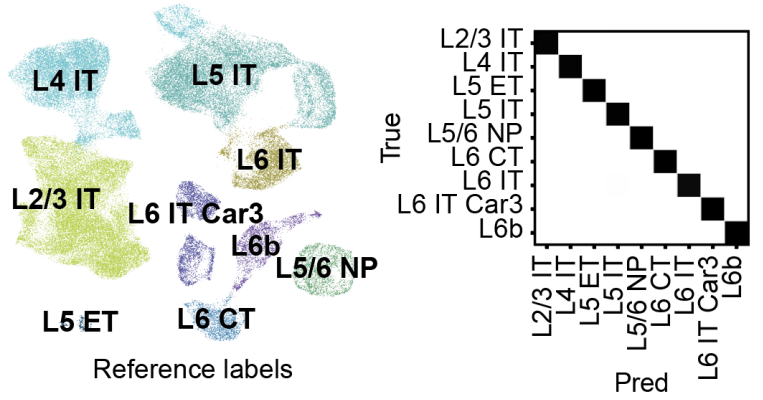


Jorstad et al (2023)

(1) Hold out labels for 1 donor at a time
train on remaining donors, and predict cell class



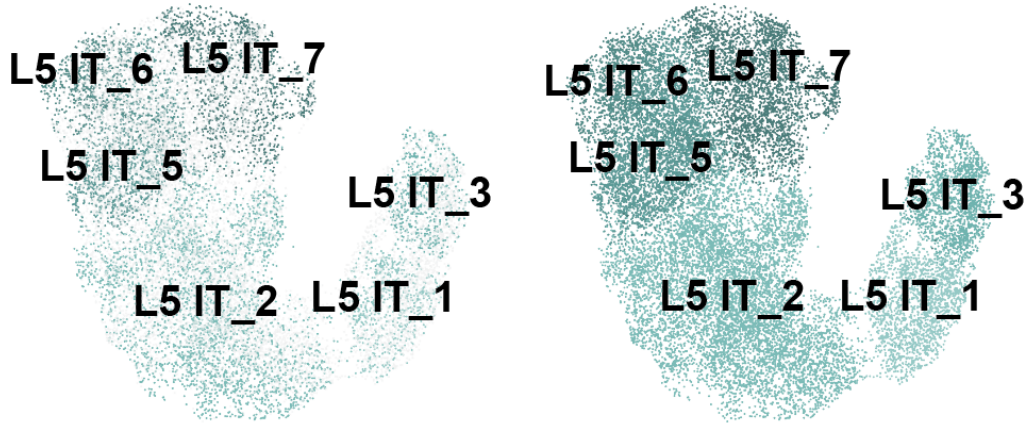
(2) Split cells by class, hold out 1 donor at a time, predict cell subclass



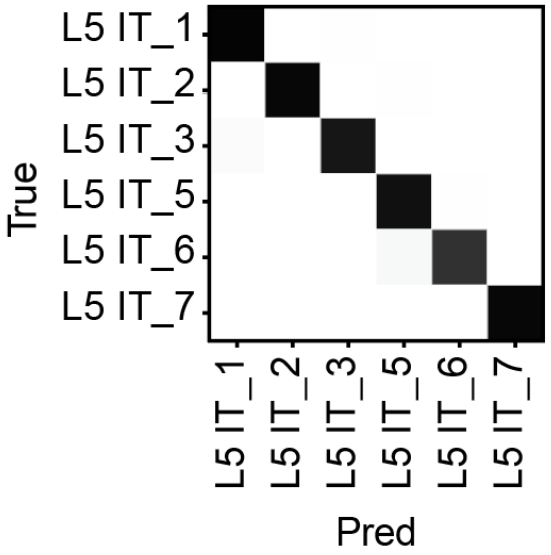
Gabitto, Travaglini et al (2024)

Benchmark the reference taxonomy using K-fold validation to determine which types are mappable

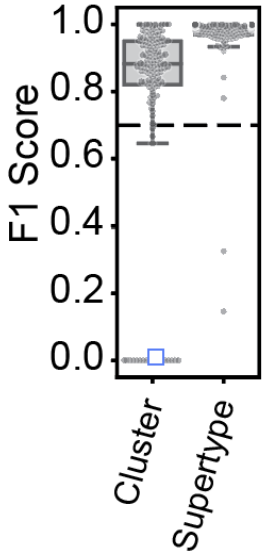
(5) Prune difficult to predict labels with <3 differentially expressed genes and re-run prediction



Held out supertypes | Predicted supertypes

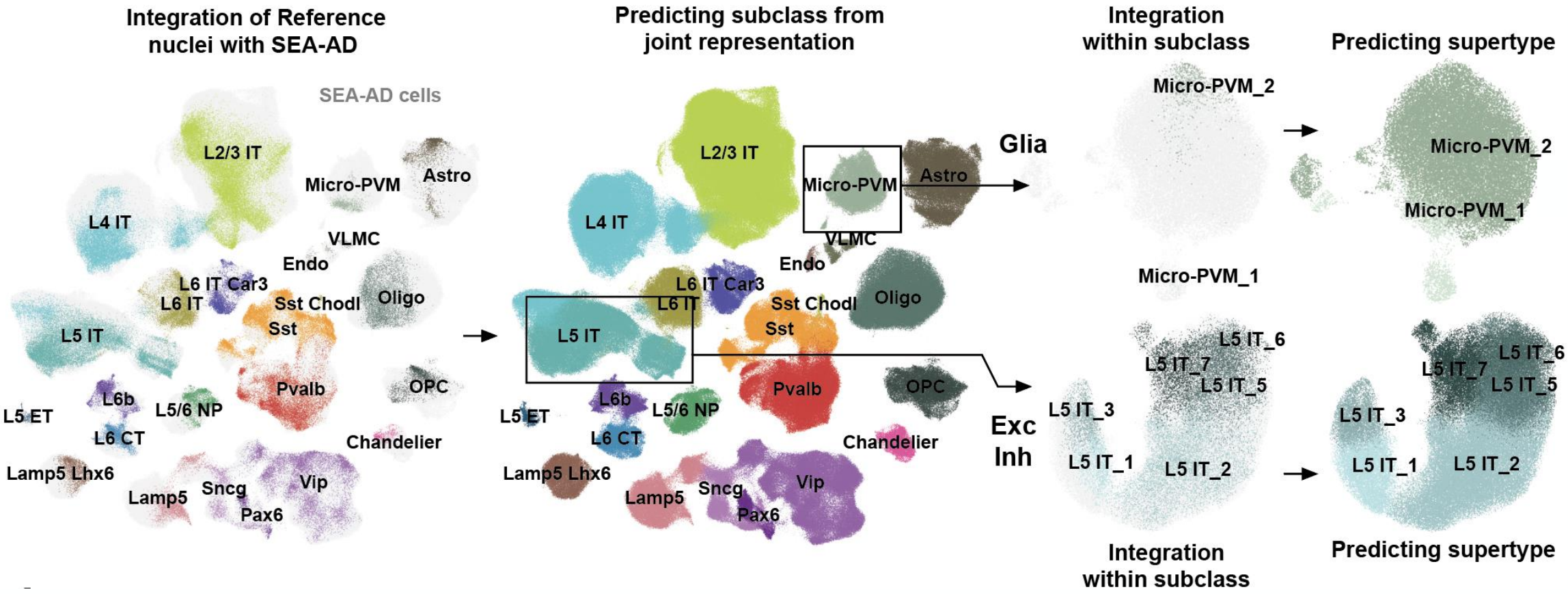


High (> 0.9) F1 scores for nearly all supertypes



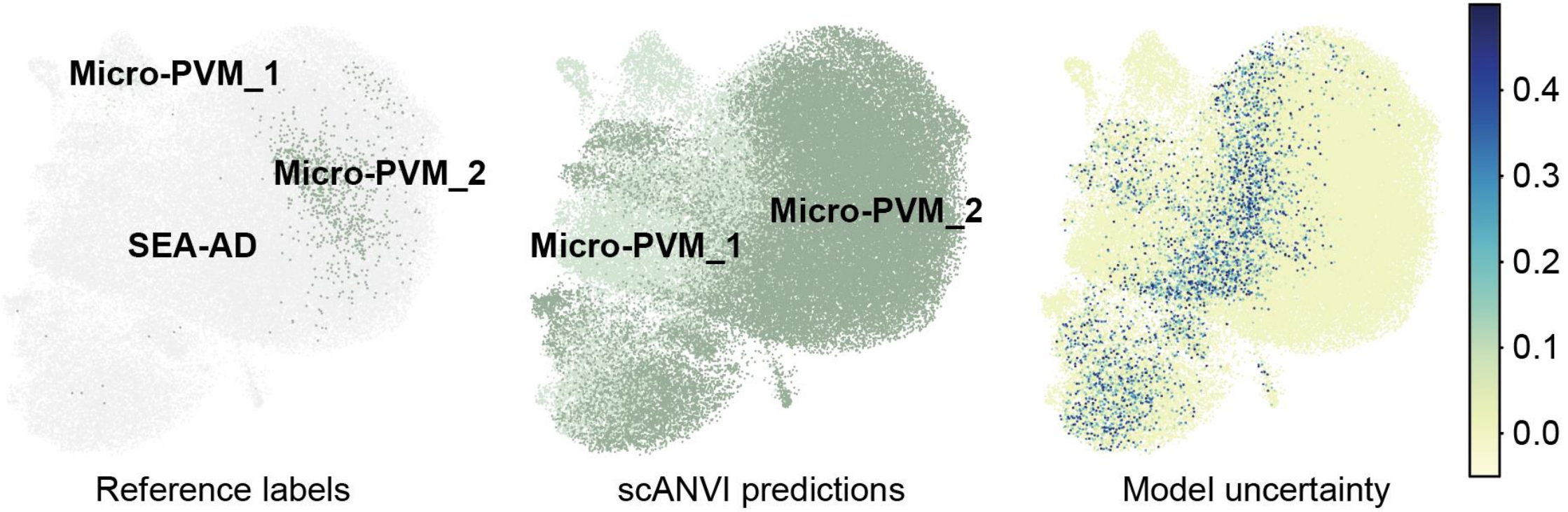
Gabitto, Travaglini et al (2024)

Hierarchically map SEA-AD data to each cell type taxonomic level



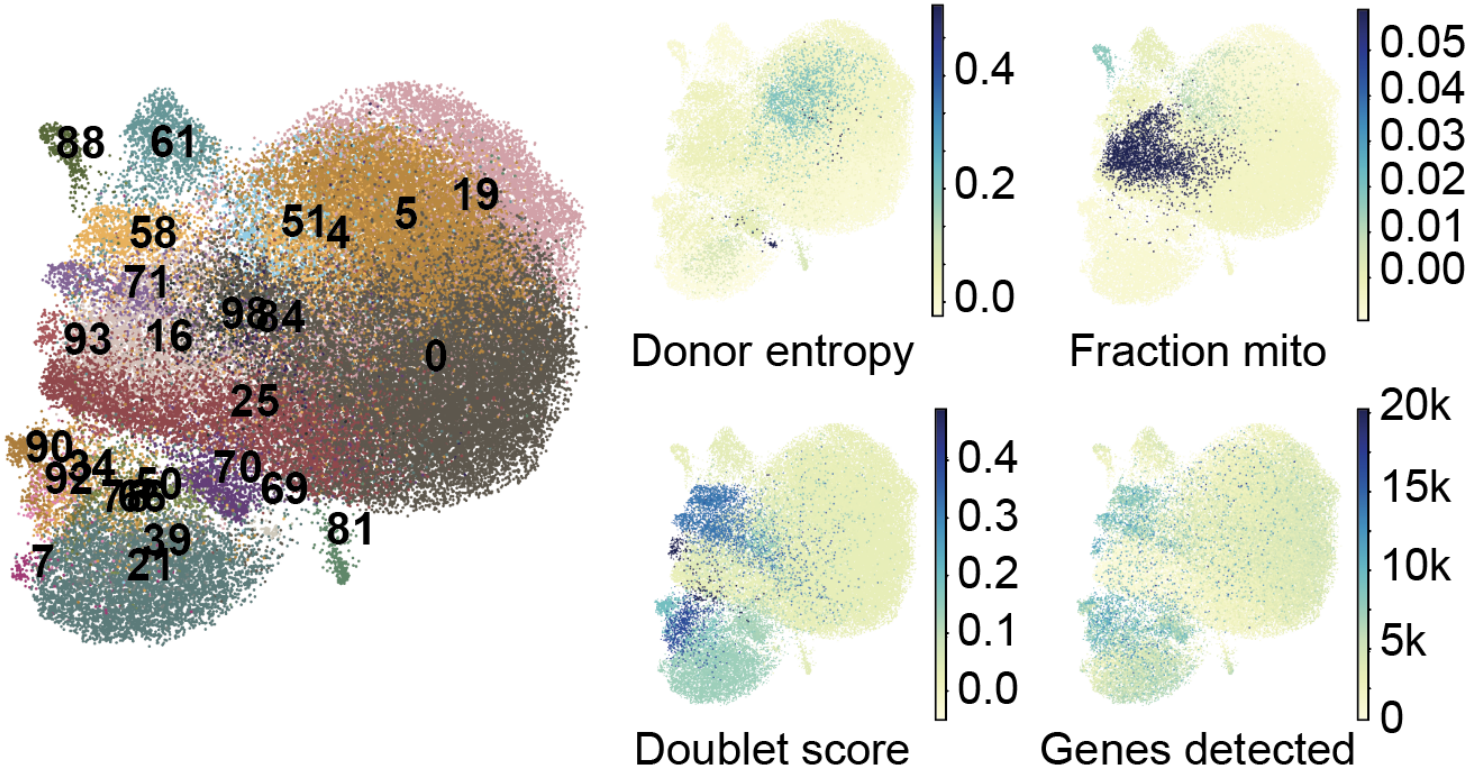
Identify low quality nuclei in the SEA-AD data using defined metrics

(1) Parts of the nearest neighbor graph contain few reference cells and have poor confidence from the model

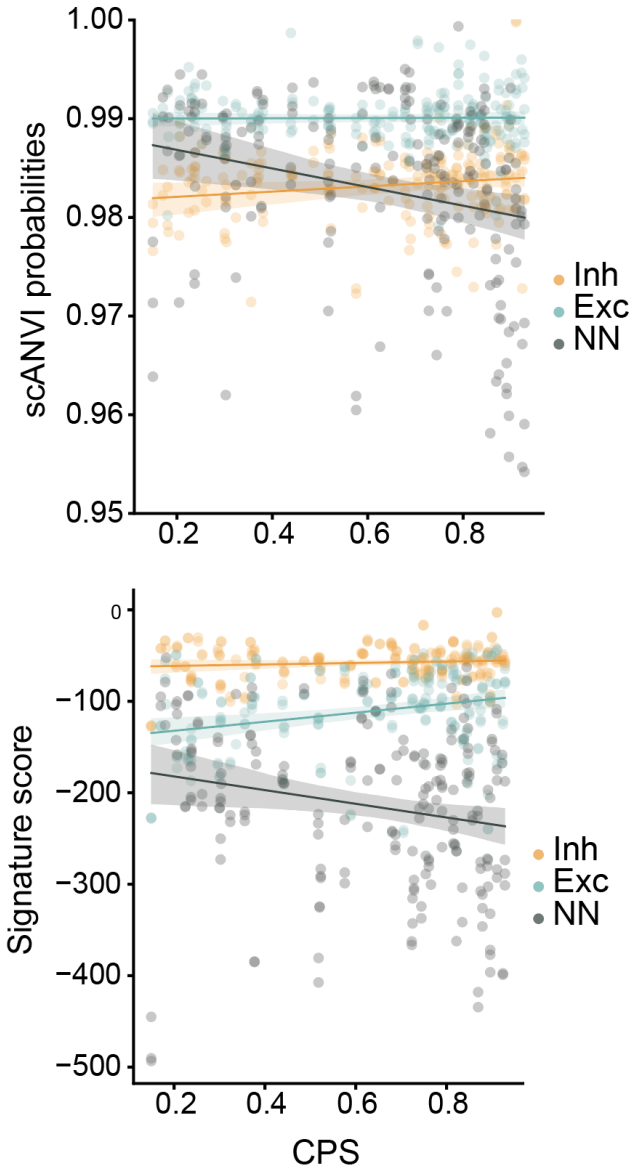


Identify low quality nuclei in the SEA-AD data using defined metrics

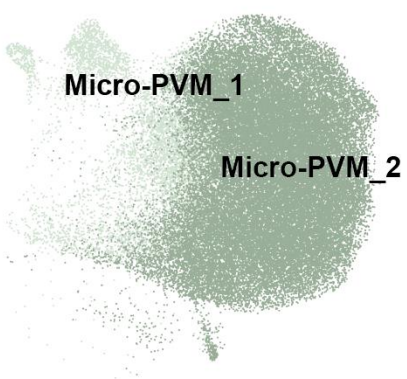
(2) Perform high resolution leiden clustering, merge clusters without differentially expressed genes, calculcate per-cluster QC metrics



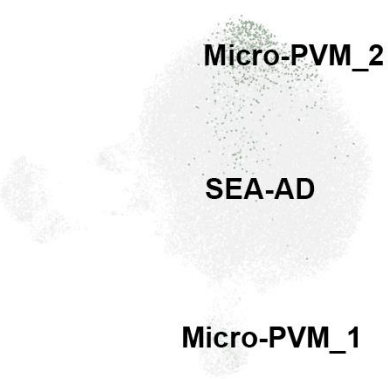
Identify new cellular states/types not present in the reference



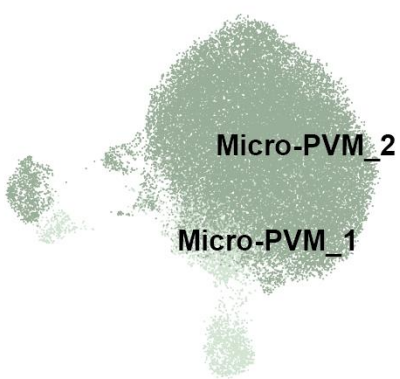
(3) After removing flagged cells, there is still transcriptional heterogeneity within predicted supertypes, so we re-compute the latent representation/nearest neighbor graph and cluster as above

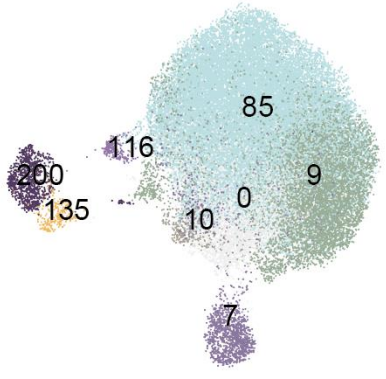


scANVI predictions



Reference labels

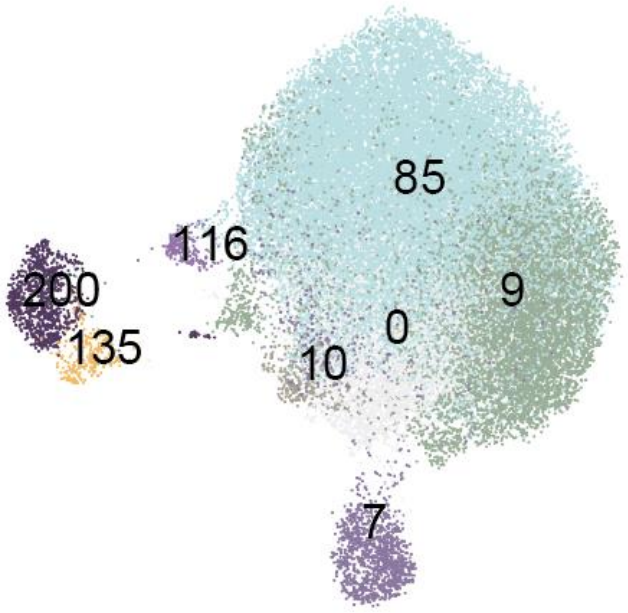


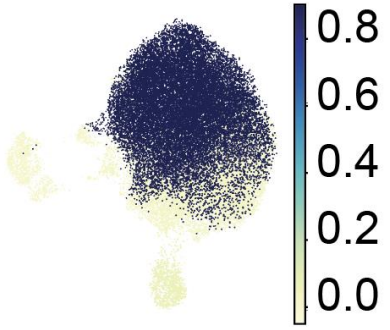


scANVI predictions

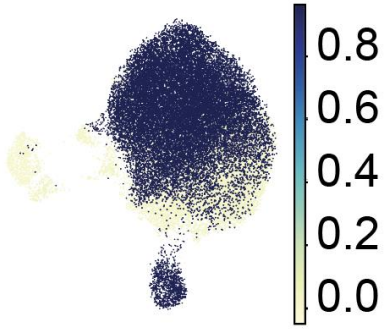
Identify new cellular states/types not present in the reference

(4) Add clusters with poor reference support to the SEA-AD taxonomy

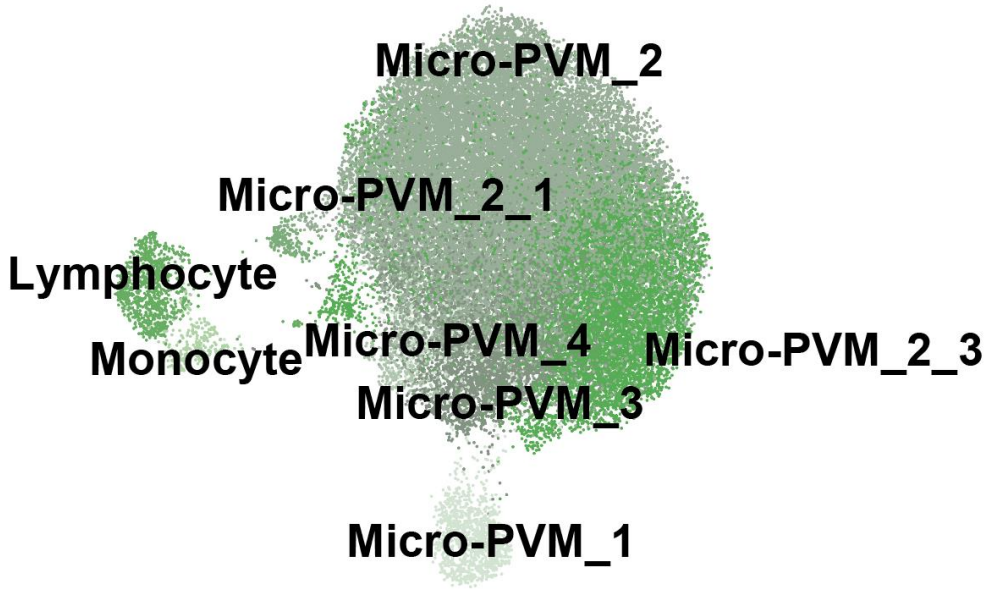




Fraction of reference

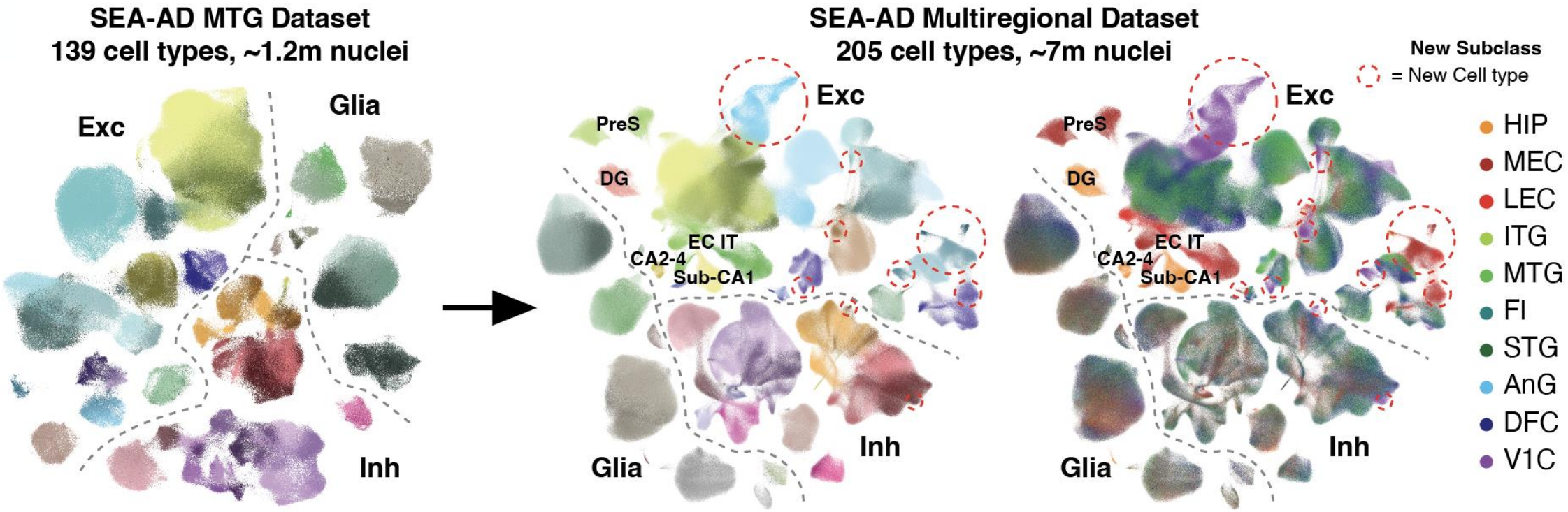


Max supertype fraction

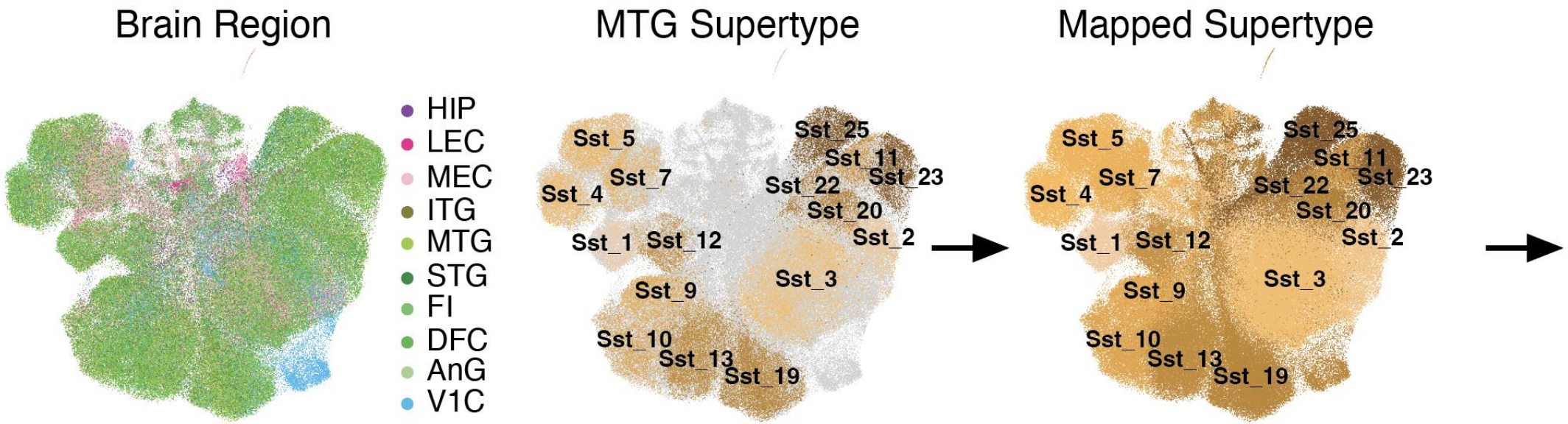


Expanded taxonomy

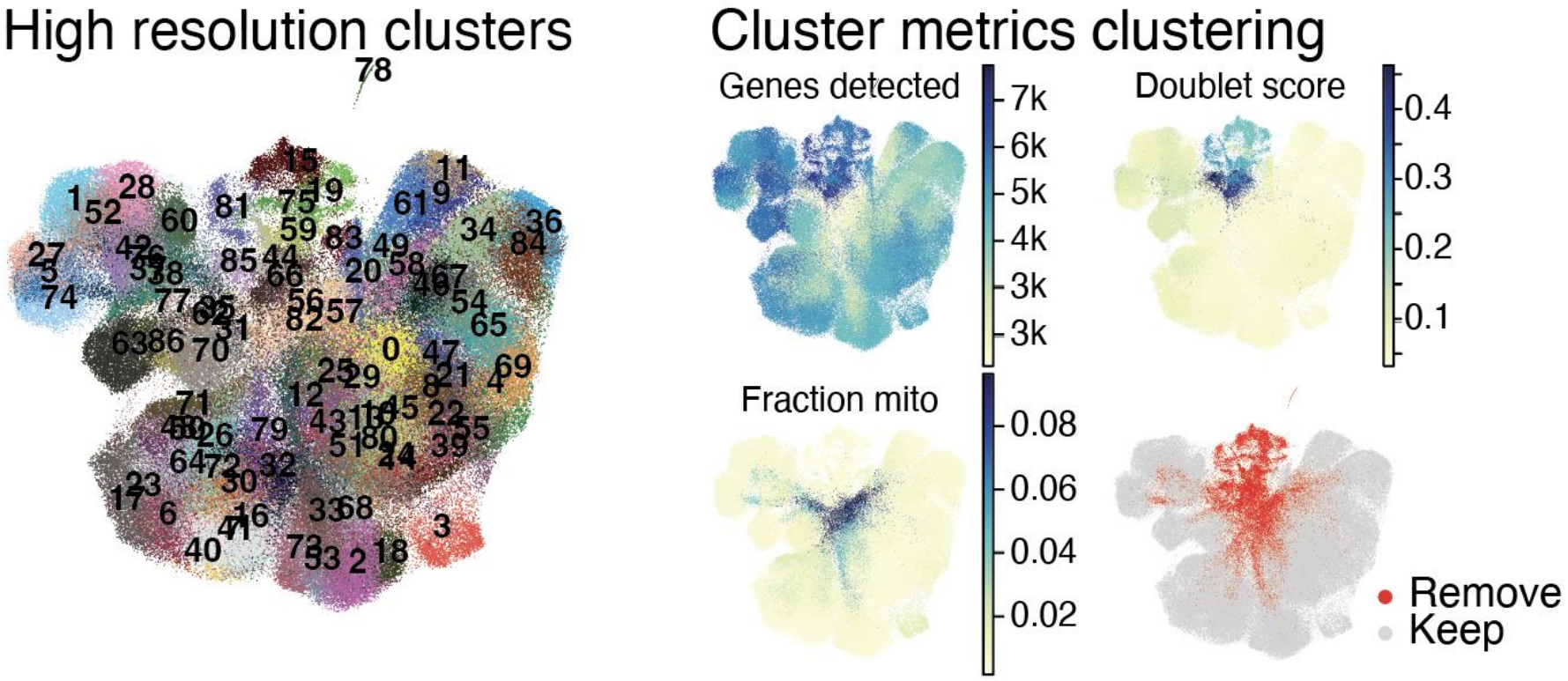
All -omics data mapped to an expanded BICAN taxonomy to account for regional cell types and AD-specific states



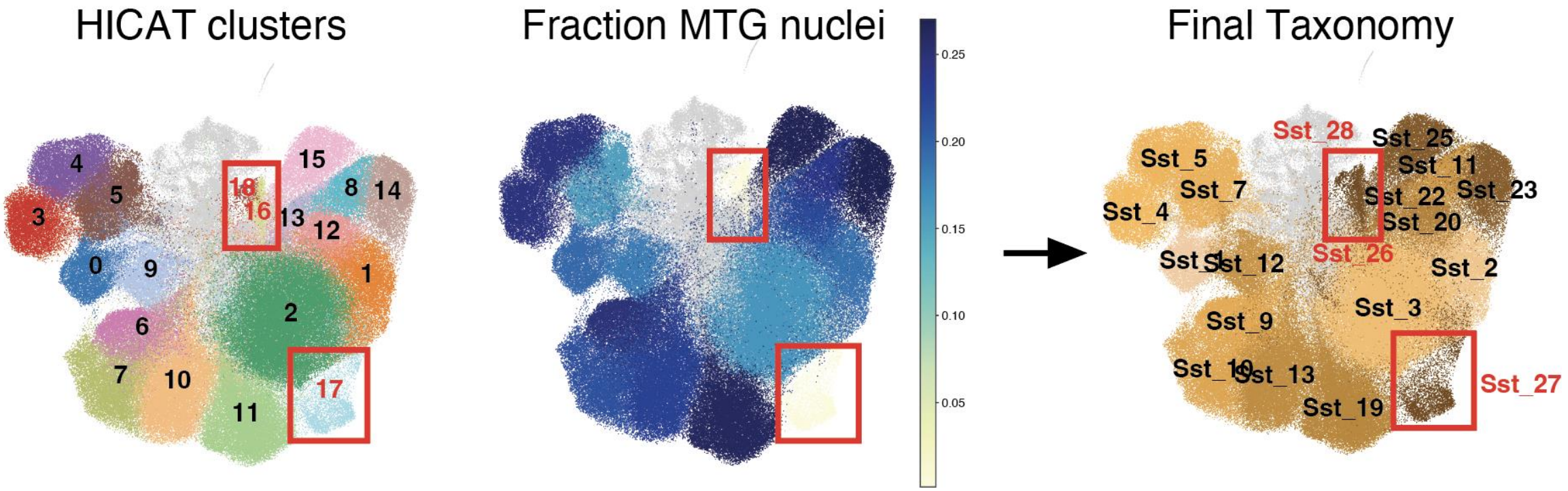
Repeated the same procedure in updating the taxonomy for the multiregional preprint so that it is consistent



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Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD)



The Seattle Alzheimer's Disease Brain Cell Atlas (SEA-AD) consortium is supported by the National Institutes on Aging (NIA) grant U19AG060909.

agenda/afternoon **– breakout rooms**

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Room 1165 – 1185

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3:40-4:00 | feedback survey

4:00 – 5:00 | additional time for user exploration

cross-species alignment and comparative analyses/

Morgan Wirthlin, Scientist II

allen institute/

allen institute







ethology/

1973 Nobel Prize in Physiology or Medicine

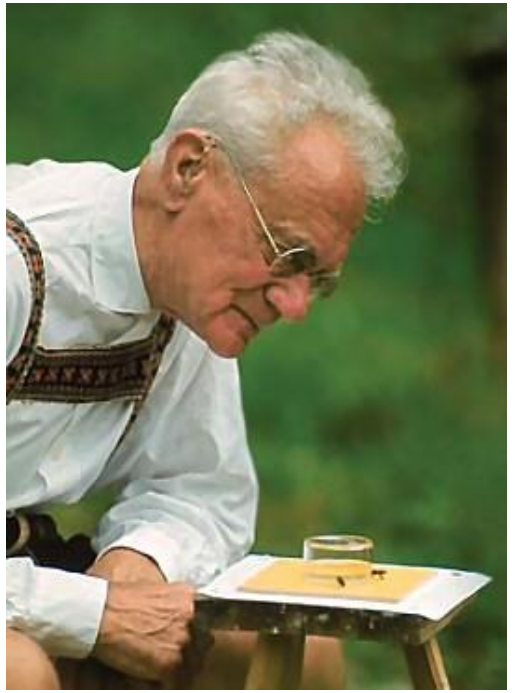
"for their discoveries concerning organization and elicitation of individual and social behaviour patterns"



Nikolaas Tinbergen



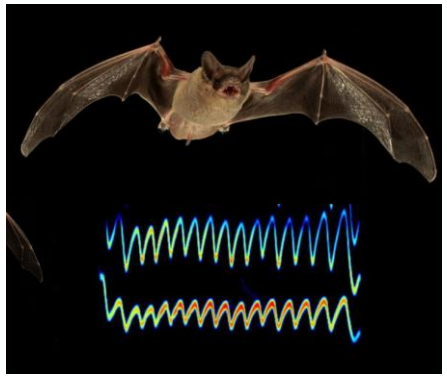
Konrad Lorenz

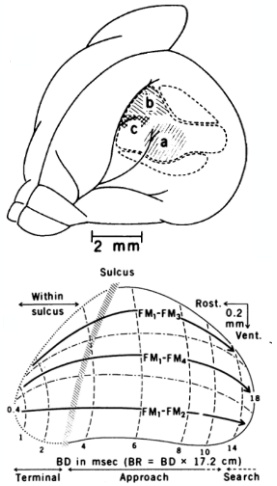


Karl von Frisch

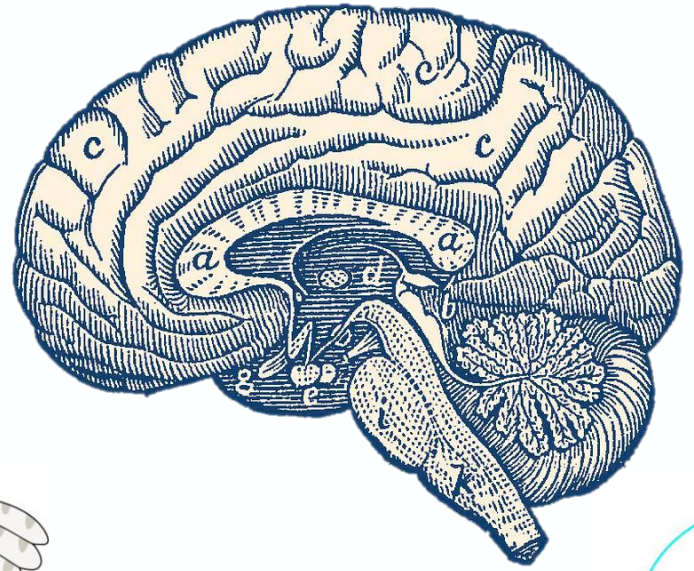
neuroethology/

The investigation of neural mechanisms through naturalistic behavior

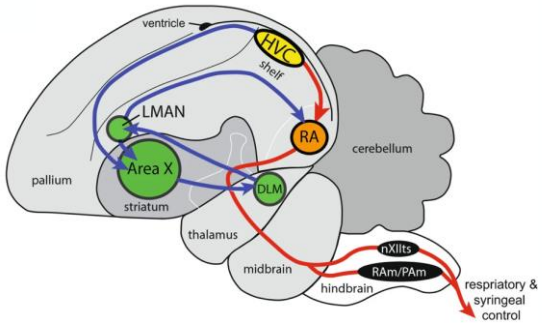




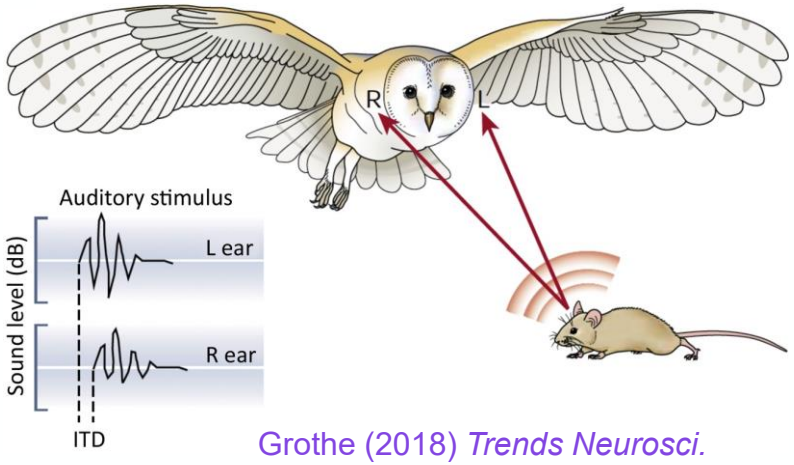
Suga & O'Neill (1979) Science



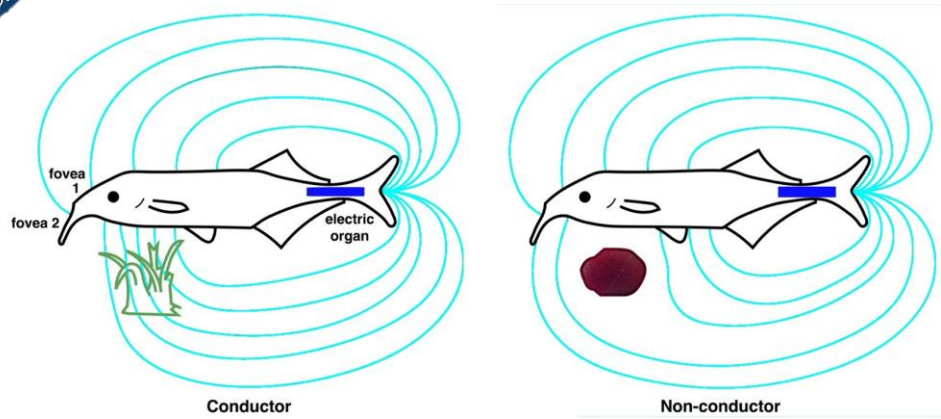




Doupe & Kuhl (1999) Ann. Rev Neuro.



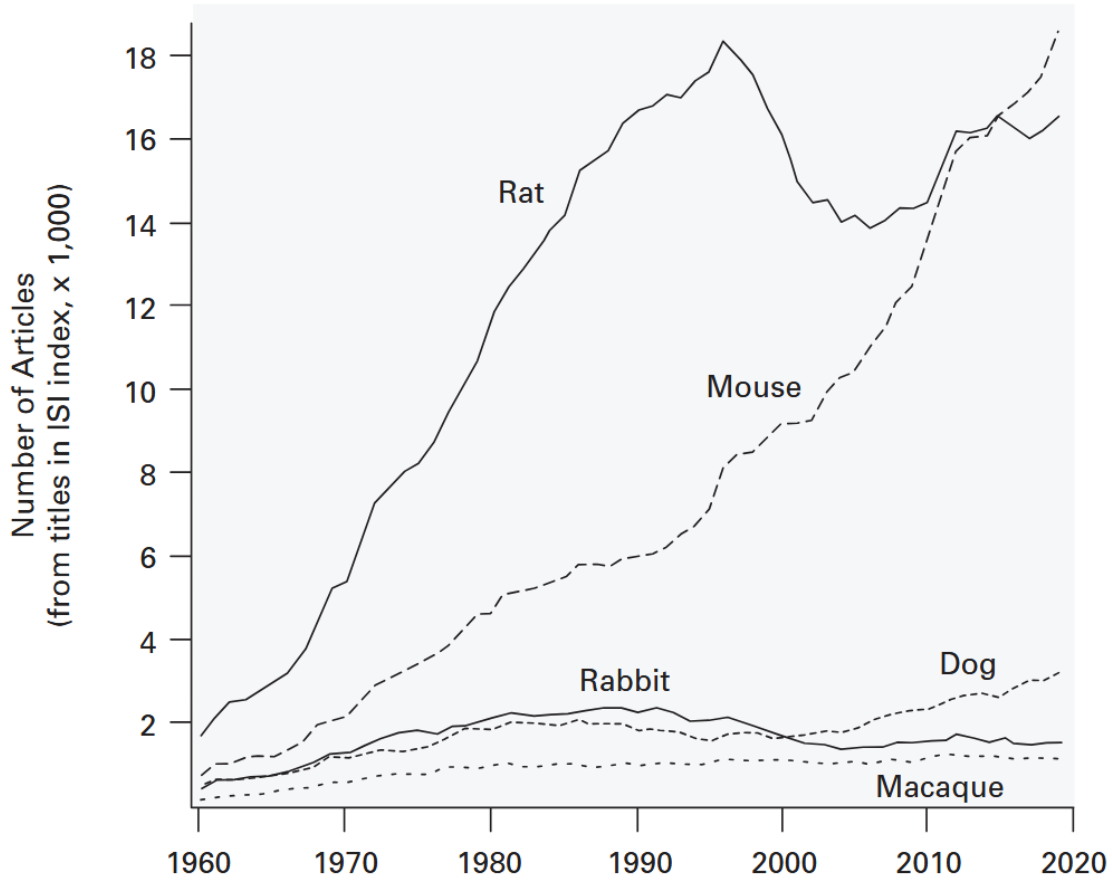
Grothe (2018) *Trends Neurosci.*



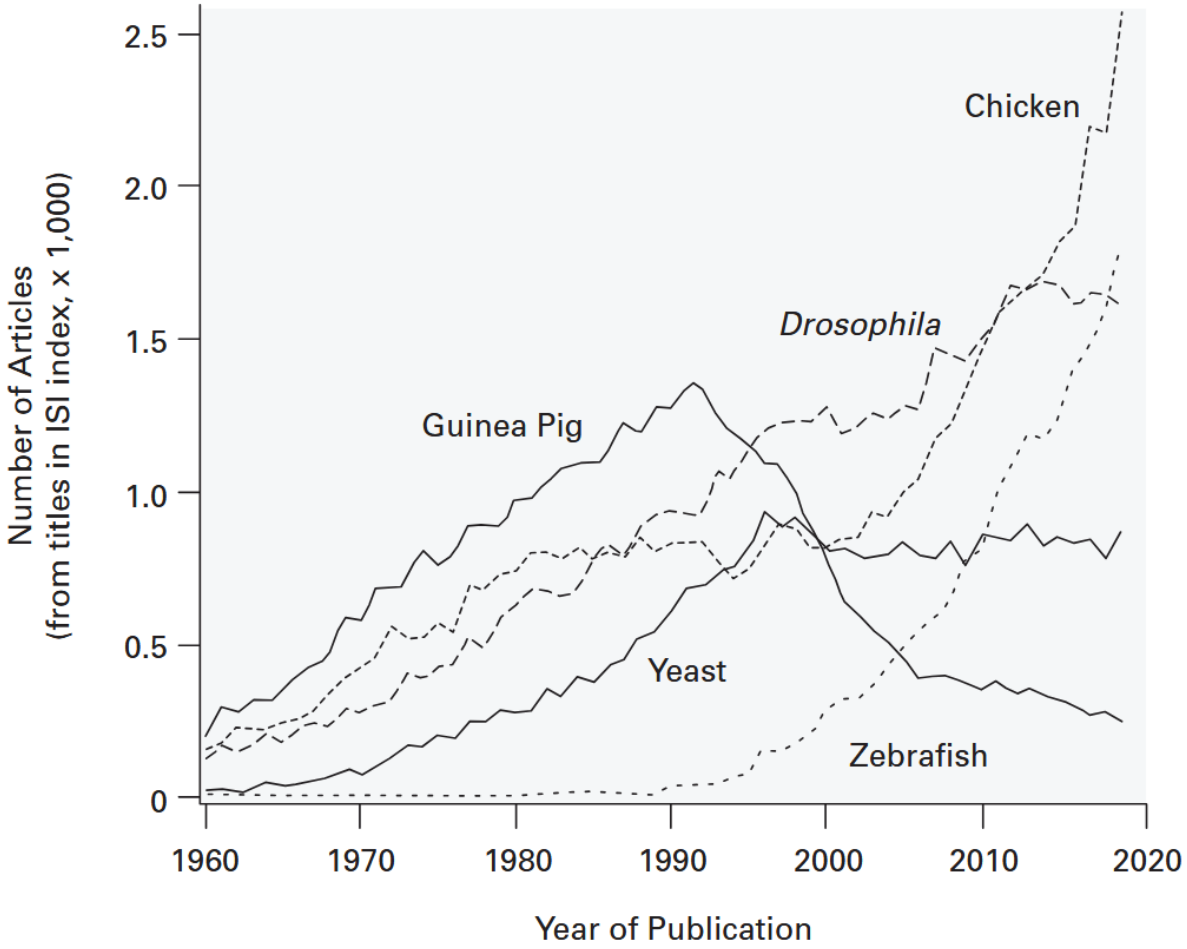
von der Emde (1999) *J Exp Biol*

model species impoverishment over time/

Mammals



Non-Mammals



Striedter (2022) *Model Systems in Biology*, MIT Press

rapidly expanding *omics resources/

22,000+ animal genomes…

EARTH BIOGENOME PROJECT

GENOME 10K

VGP VERTEBRATE GENOMES PROJECT

i5K



B10K

BAT 1K

4 million+ transcriptomic datasets…

NCBI National Center for Biotechnology Information

国家生物信息中心 CHINA NATIONAL CENTER FOR BIOINFORMATION

Sequence Read Archive

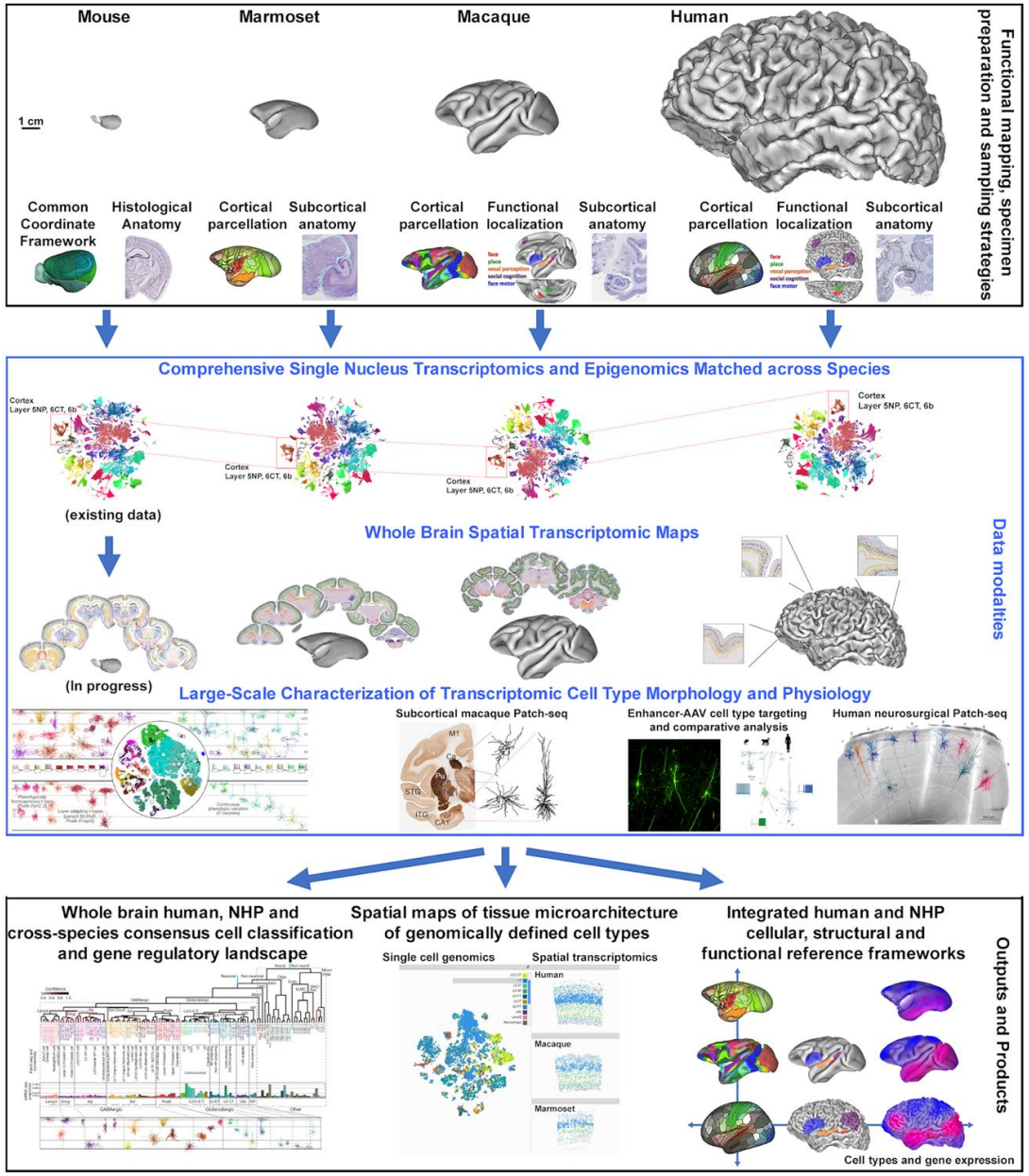
EMBL

400,000+ epigenomic datasets…



Methyl-seq, ChIP-seq, ATAC-seq, DNAse-seq, Ribo-seq

human and mammalian brain atlas (HMBA) /



Standardized anatomy

Functional Imaging

Single cell genomics

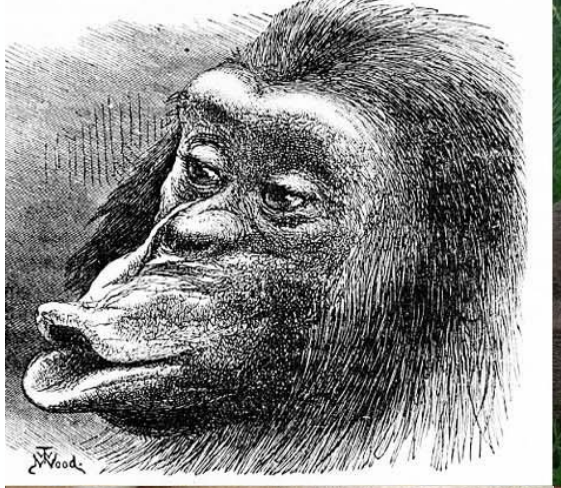
Spatial transcriptomics

Patch-seq

Products

Key driving concepts:

- **Whole brain atlases**
- **Comparative human, NHP and mouse**
- **Multimodal**
- **Link function (fMRI) to cell types**
- **Individual variation**
- **Species conservation and divergence**













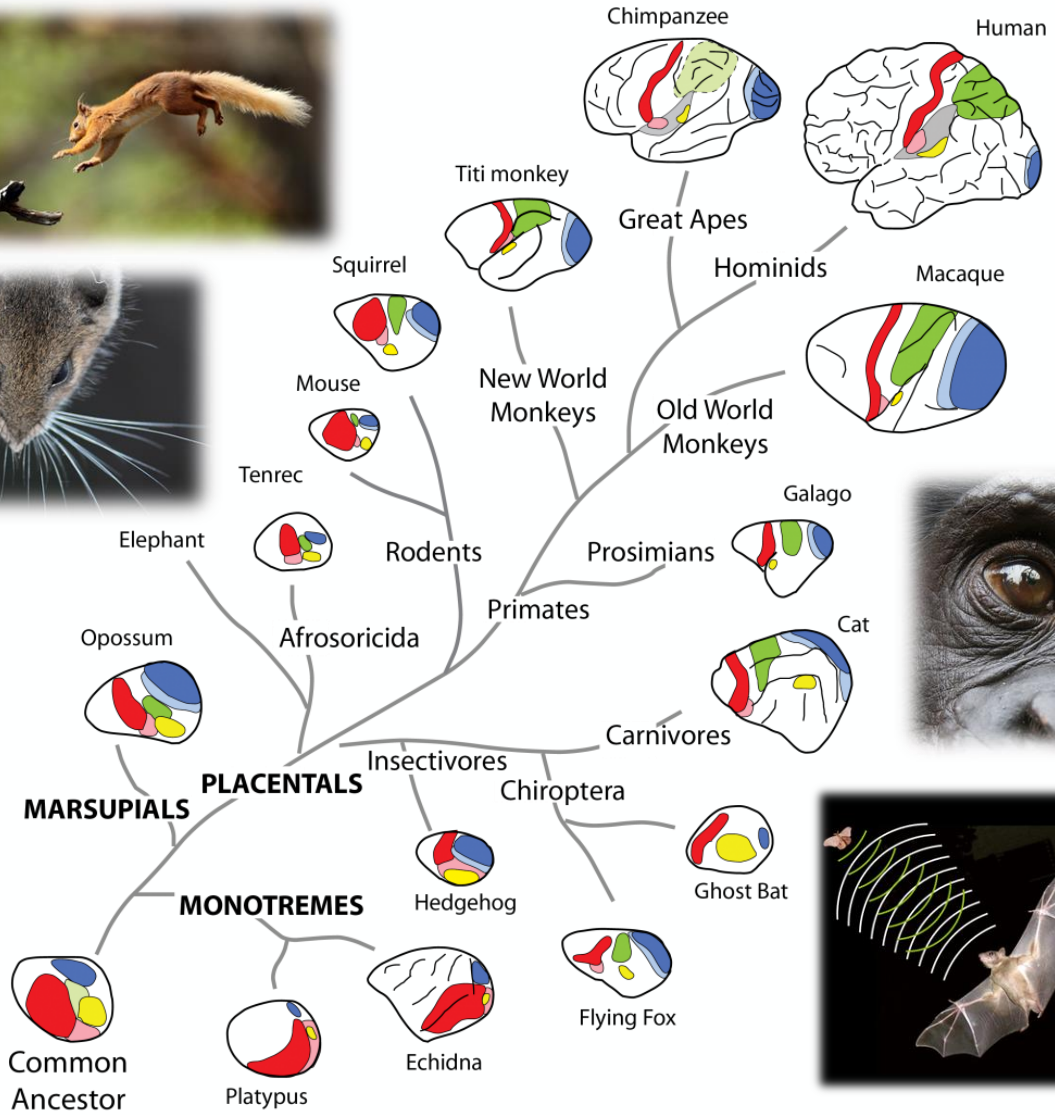

brain organization differences reflect sensory ecology/

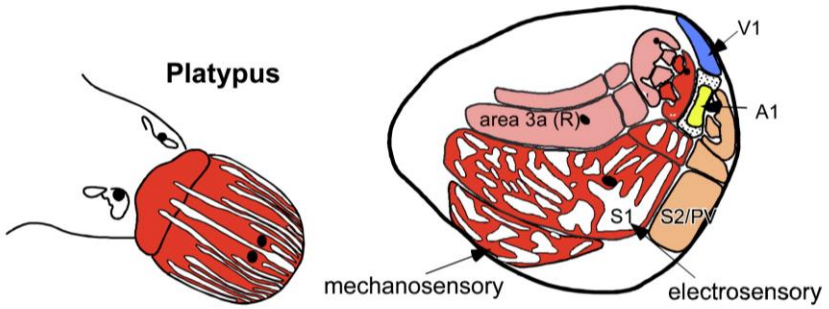


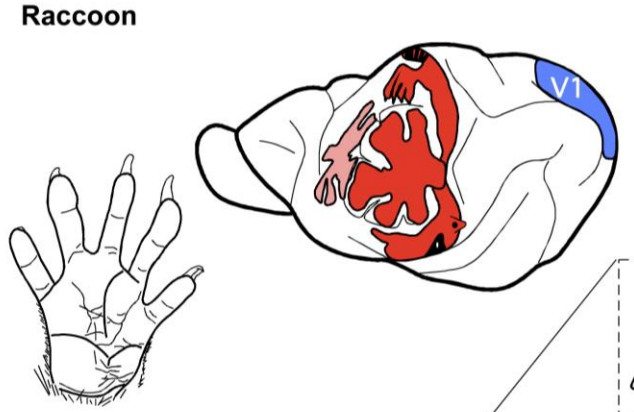


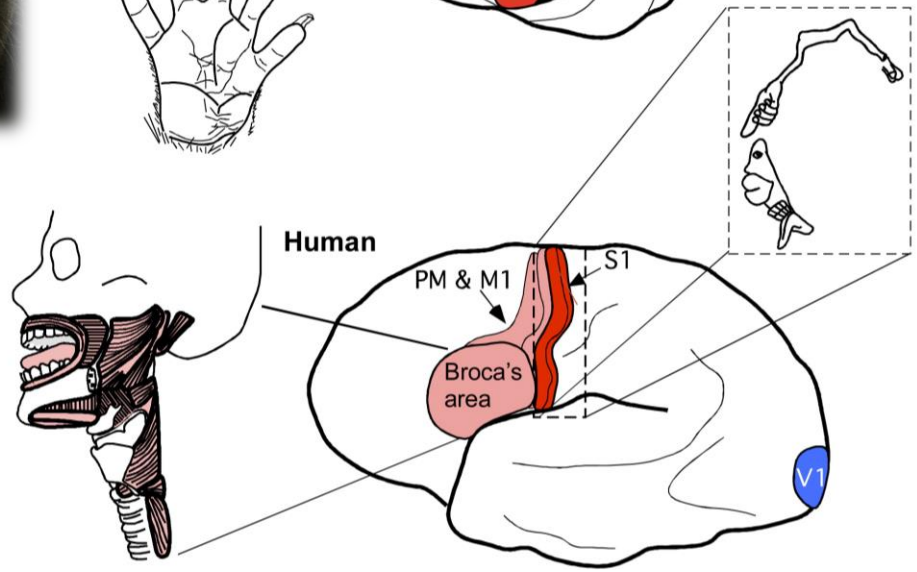












Krubitzer (2007) *Neuron*

at what level does neural evolutionary innovation occur?

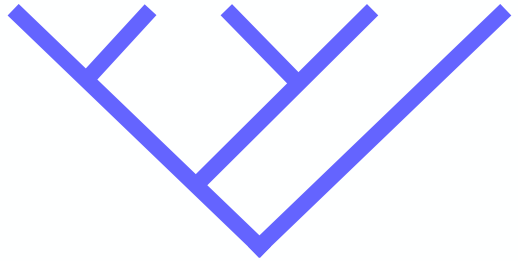
Genomic
- Single-nucleotide variation
- Protein-coding sequence changes
- Novel gene evolution

Epigenomic
- Gene regulatory elements
- DNA methylation
- Histone modification

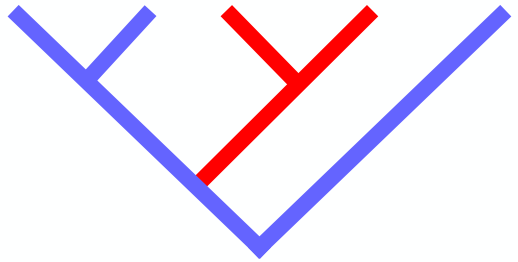
Transcriptomic
- Differential expression
- Module composition

Cellular
- Cytoarchitectonics
- Cell type abundance
- Novel cell types
- Circuit connectivity
- Electrophysiology

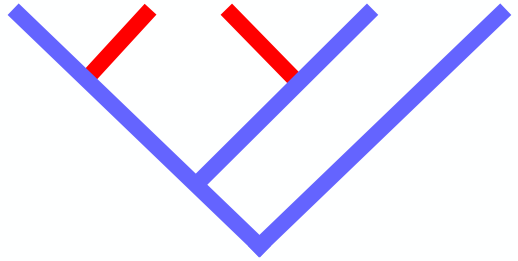
Conservation



Divergence



Convergence



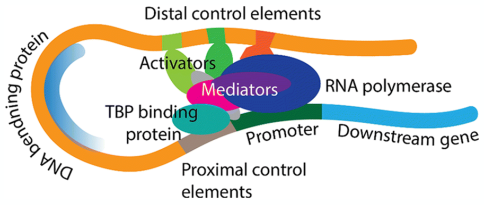
at what level does neural evolutionary innovation occur?

Genomic

- Single-nucleotide variation
- Protein-coding sequence changes
- Novel gene evolution

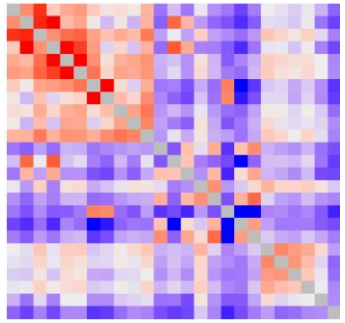
Epigenomic

- **Gene regulatory elements**
- DNA methylation
- Histone modification



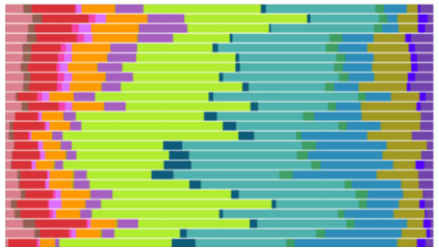
Transcriptomic

- **Differential expression**
- Module composition

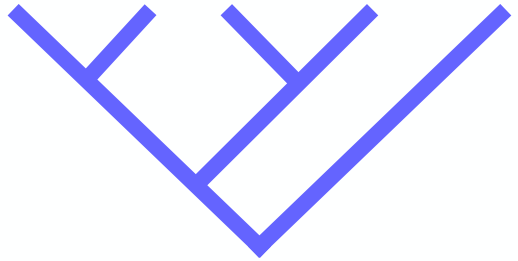


Cellular

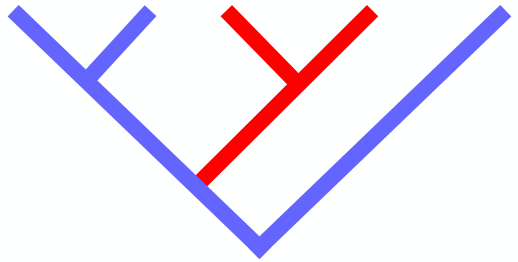
- Cytoarchitectonics
- **Cell type abundance**
- Novel cell types
- Circuit connectivity
- Electrophysiology



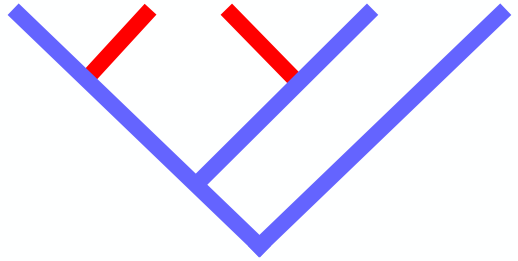
Conservation



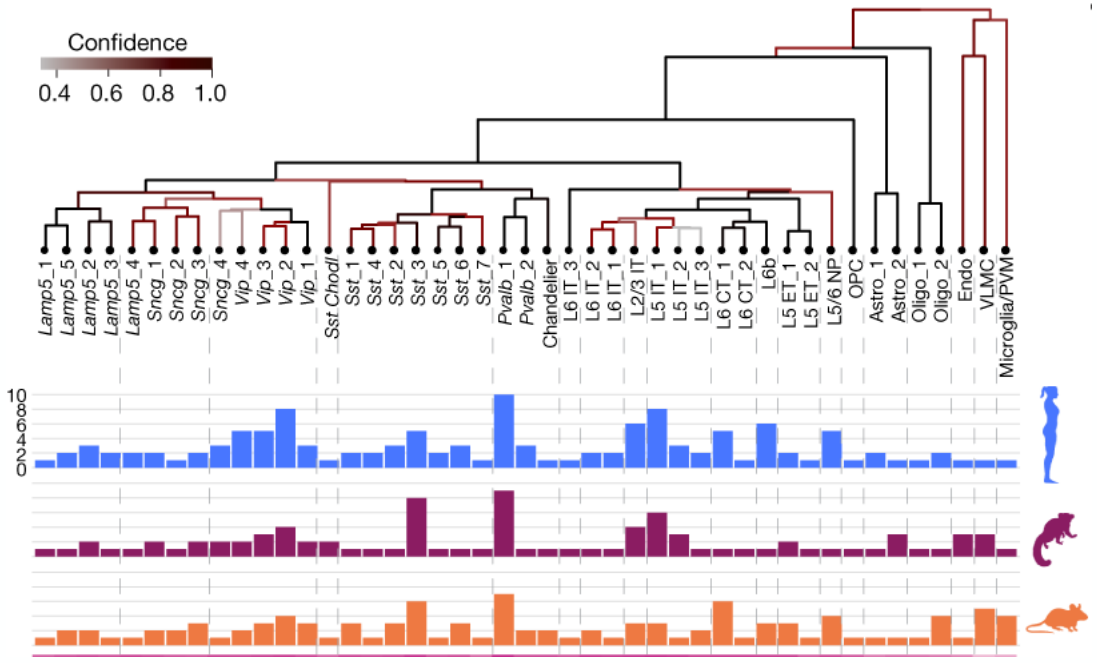
Divergence



Convergence



a cellular taxonomy of mammalian M1/

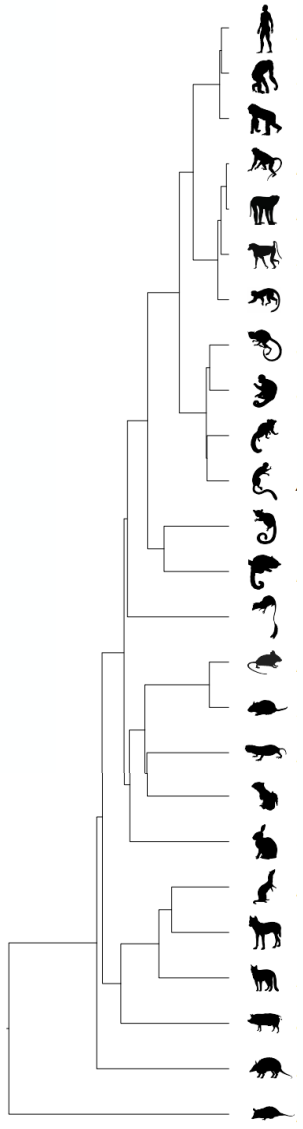


3-Species Mammalian M1 Taxonomy
Bakken *et al.* (2021) *Nature*
BICCN Consortium (2021) *Nature*

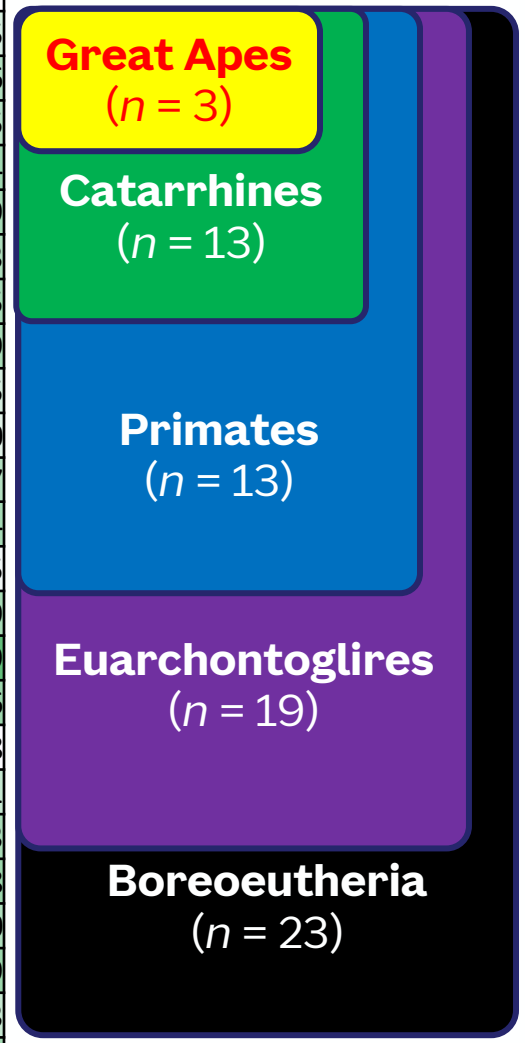


25-Species Mammalian M1 Taxonomy
This study

multiomic sampling across mammalian M1 diversity/



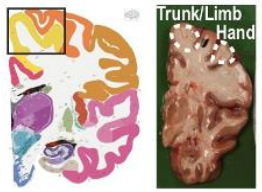
Common name	Species	Donors	RNA libraries	ATAC libraries	Total Cells (post-QC)
Human	*Homo sapiens*	5	12	Nemo	73,805
Chimpanzee	*Pan troglodytes*	6	12	0	101,056
Gorilla	*Gorilla gorilla gorilla*	5	14	0	97,935
Rhesus macaque	*Macaca mulatta*	6	13	2	113,554
Pig-tailed macaque	*Macaca nemestrina*	5	15	8	110,209
Olive baboon	*Papio anubis*	3	12	0	103,913
Green monkey	*Chlorocebus sabaeus*	3	12	12	109,395
Squirrel monkey	*Saimiri boliviensis boliviensis*	3	12	12	108,539
Capuchin	*Sapajus apella*	2	10	0	74,485
Marmoset	*Callithrix jacchus*	5	10	0	66,310
Owl monkey	*Aotus nancymaae*	3	12	0	100,087
Galago	*Otolemur garnettii*	2	12	0	109,824
Gray mouse lemur	*Microcebus murinus*	3	9	0	53,955
Northern treeshrew	*Tupaia belangeri*	5	13	12	108,779
Mouse	*Mus musculus*	16	20	Nemo	159,019
Rat	*Rattus norvegicus*	4	12	12	116,786
Naked mole rat	*Heterocephalus glaber*	5	8	0	44,428
Arctic ground squirrel	*Urocitellus parryii*	4	11	0	86,631
European rabbit	*Oryctolagus cuniculus*	5	13	11	108,073
Ferret	*Mustela putorius furo*	6	12	12	117,568
Coyote	*Canis latrans*	3	11	0	101,400
Cat	*Felis catus*	3	12	0	117,749
Pig	*Sus scrofa*	4	10	11	85,063
Nine-banded armadillo	*Dasypus novemcinctus*	4	12	9	108,866
Short-tailed opossum	*Monodelphis domestica*	5	12	12	82,516



Total mammals: 25

Collaborating Labs: Chet Sherwood, David Fitzpatrick, Don Bolser, Fenna Krienen, Gabriel Balmus, Greg Wilkerson, Guoping Feng, Jeffrey Padberg, Jon Kaas, Kelly Drew, Kimberley Phillips, Leah Krubitzer, Mark Krasnow, Sarah Freeman, Ted Barrett

generating a 25-species mammalian M1 cellular taxonomy/



Tissue Dissection



10x snRNAseq



Metadata Cleanup

	Cell1	Cell2	...	CellN
Gene1	3	2	.	13
Gene2	2	3	.	1
Gene3	1	14	.	18
...	.	.	.	.
...	.	.	.	.
...	.	.	.	.
GeneM	25	0	.	0

Count Matrix



Create Anndata File



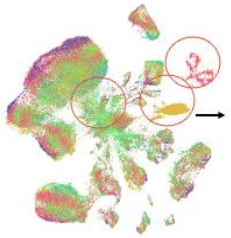
Visualize UMAPs in Cell x Gene



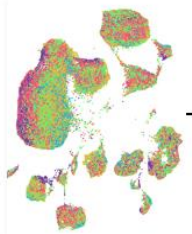
Donor Correction (scVI)

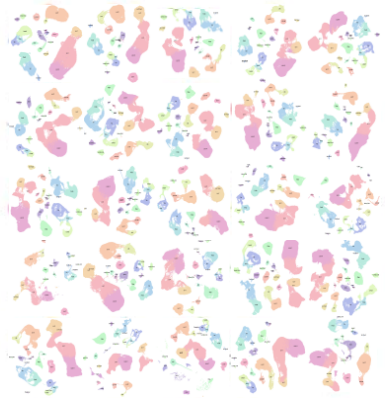


Cell type mapping
(scANVI and scratch)

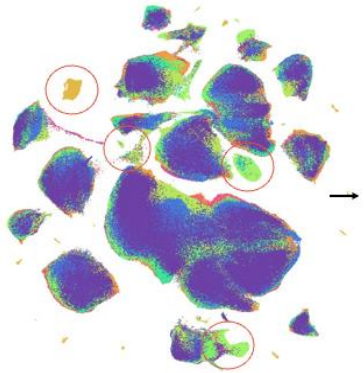


Per Species QC

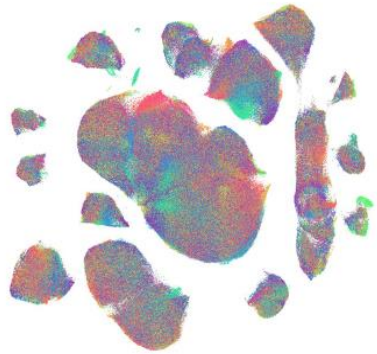




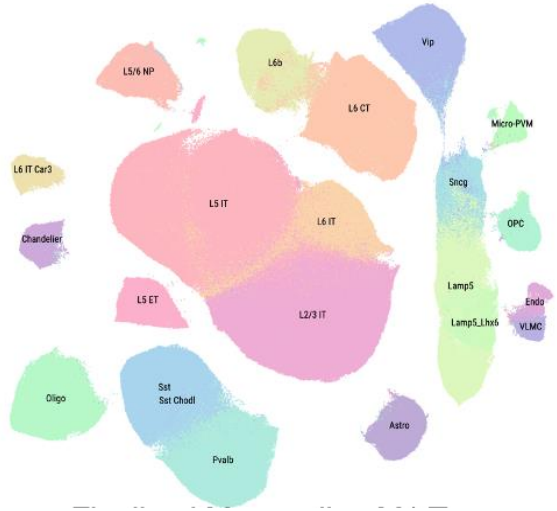
Merge and Integrate Species (scVI)



Merged QC



Finalize cell type labels



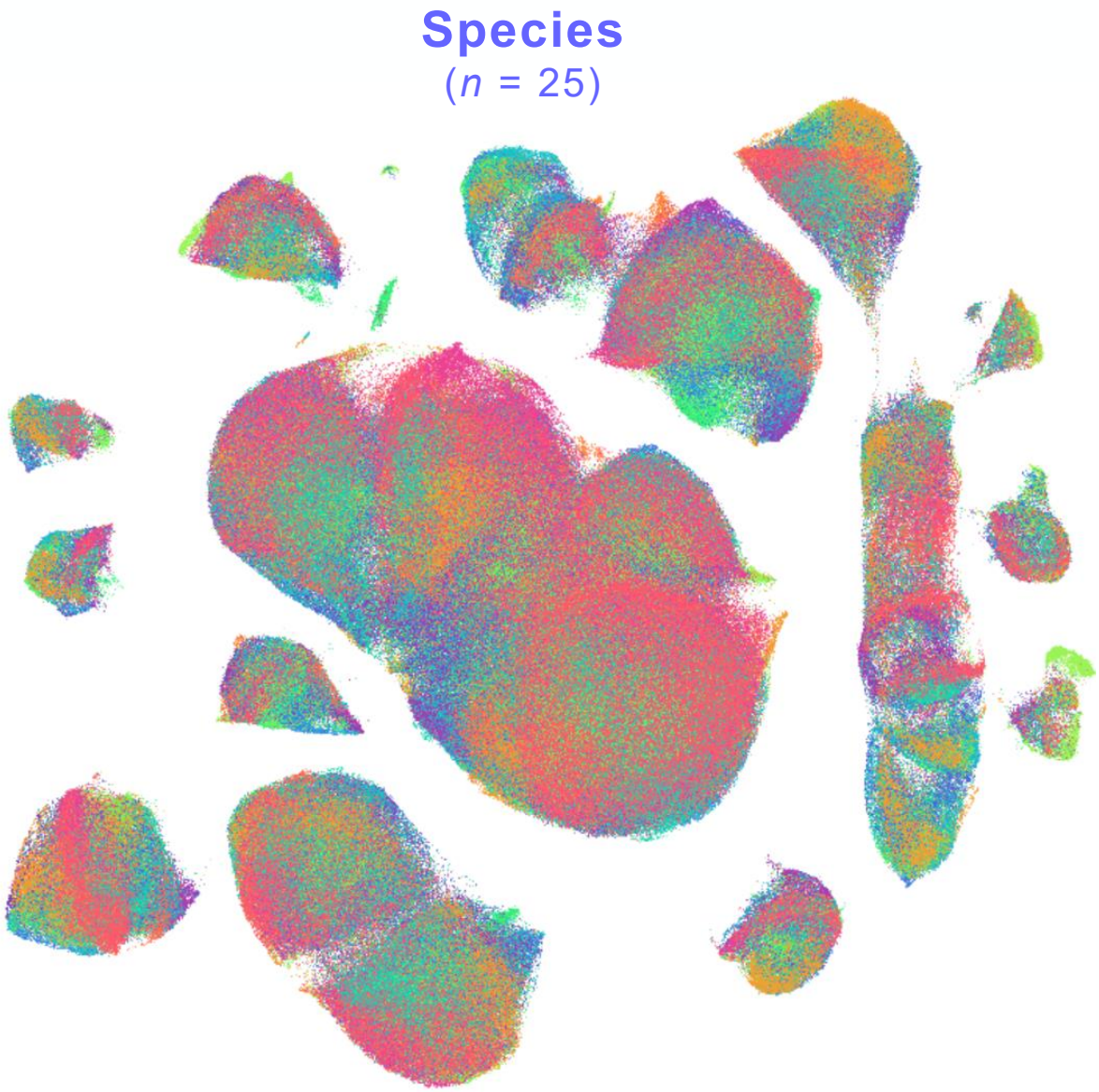


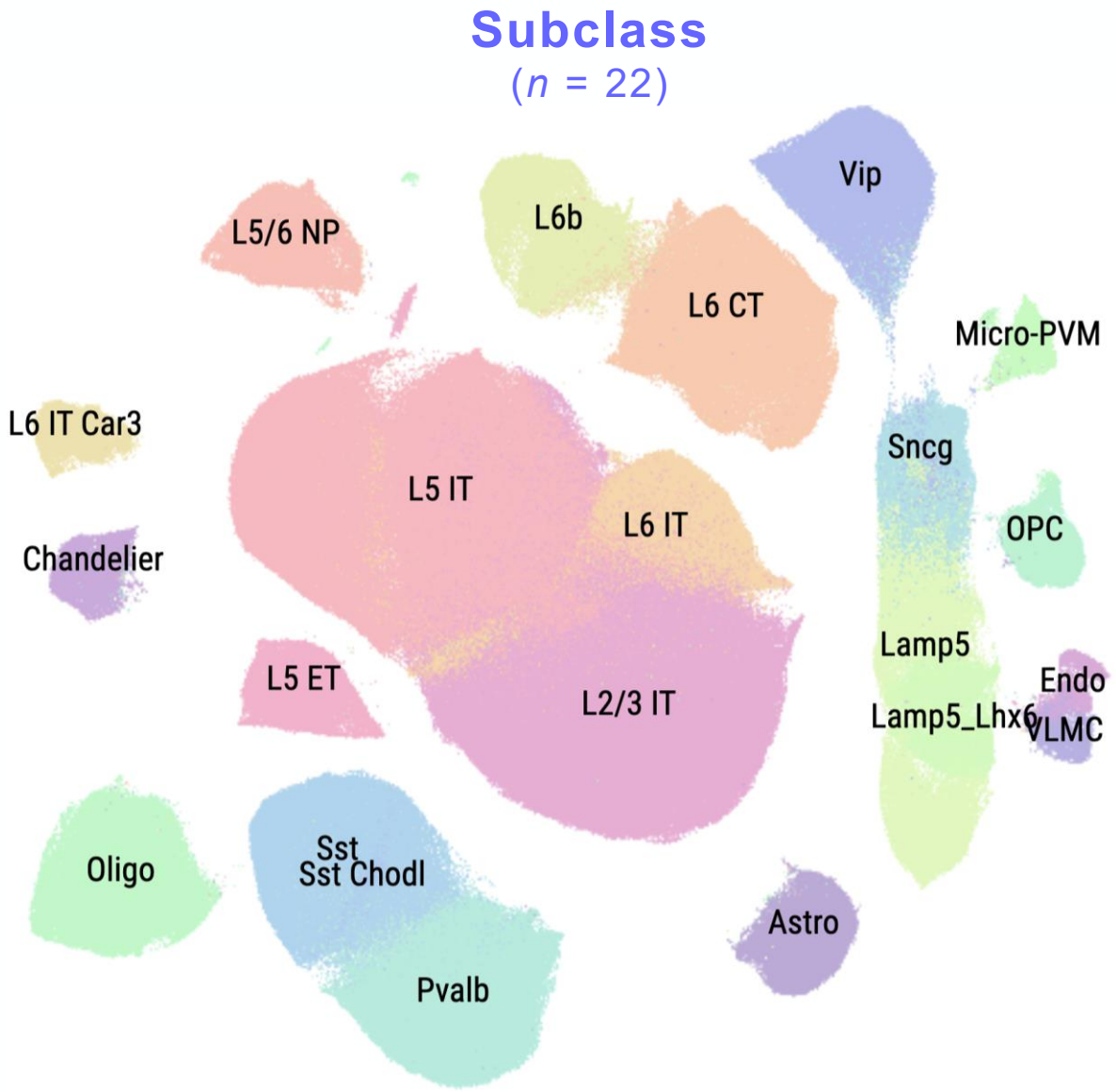
Jessie Puckett



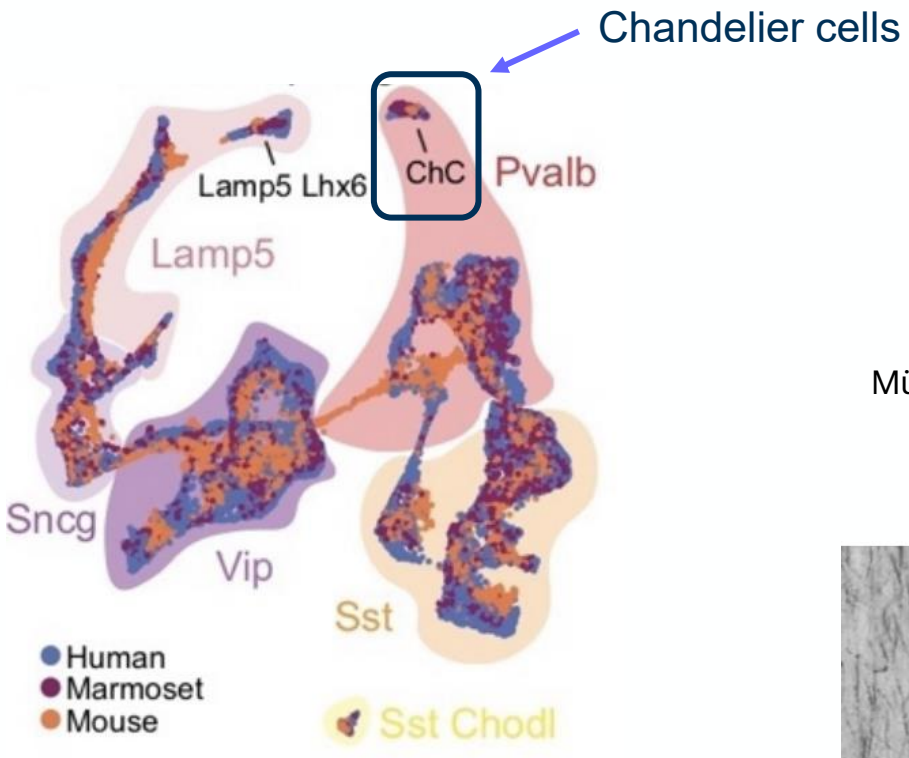
Nelson Johansen

an integrated mammalian M1 taxonomy/

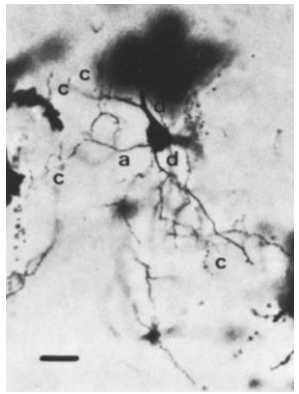




chandelier cell conservation across mammals?/

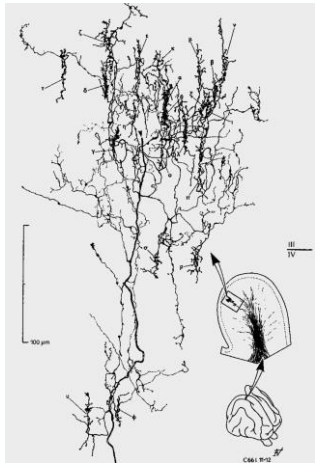


Bakken et al. 2021 *Nature*



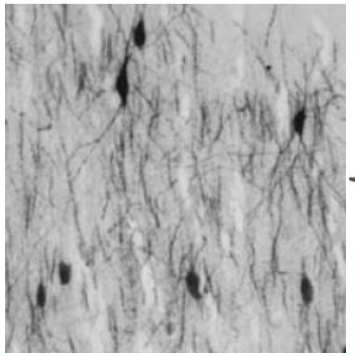


Müller-Paschinger et al. 1983



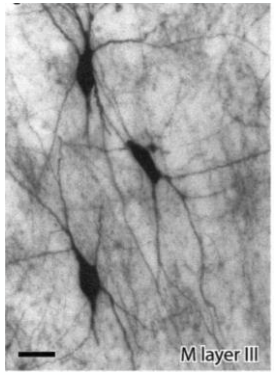


Fairén & Valverde 1980





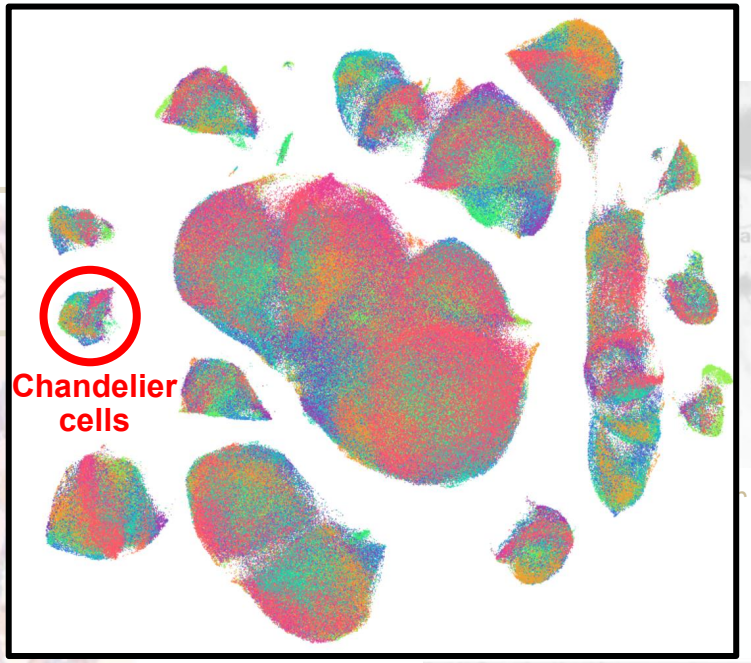
Sherwood et al. 2009

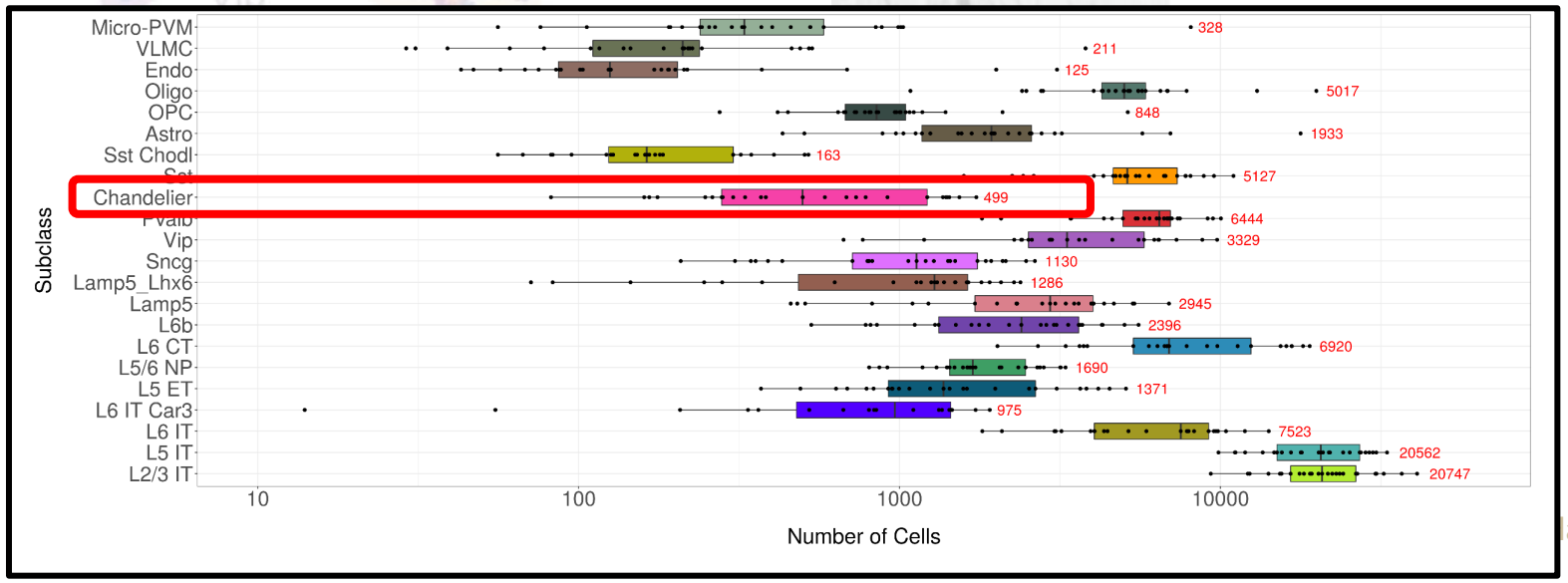




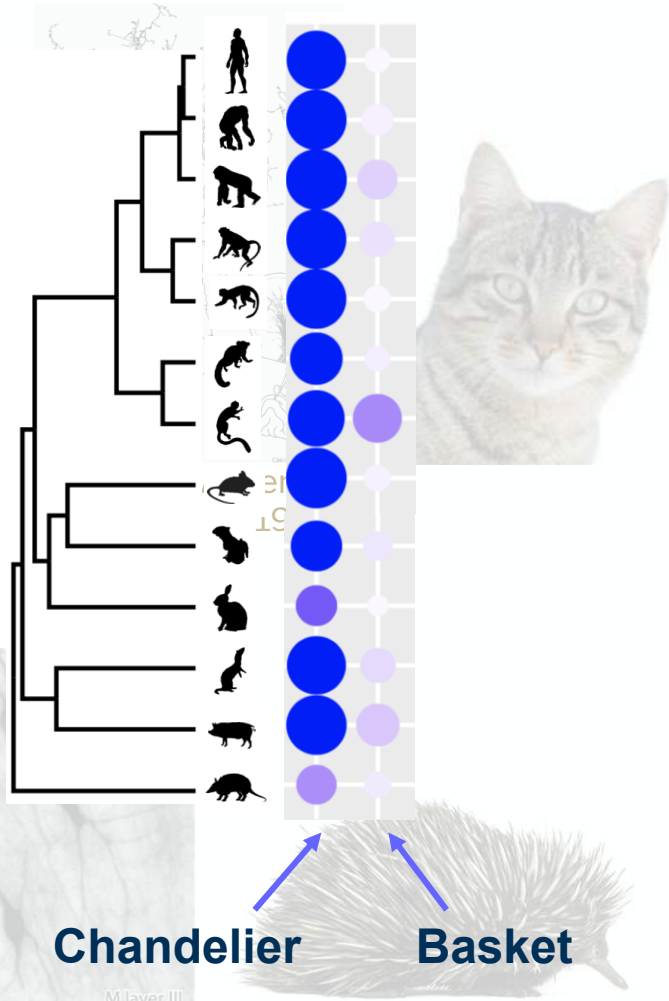
Hassiotis et al. 2005

chandelier cell conservation across mammals?/Yes!

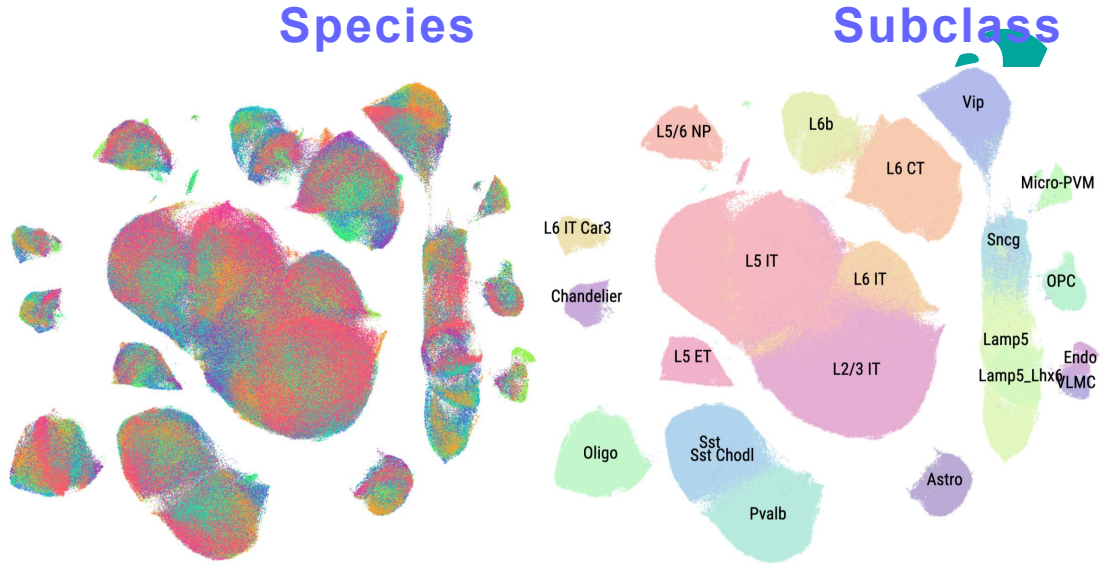




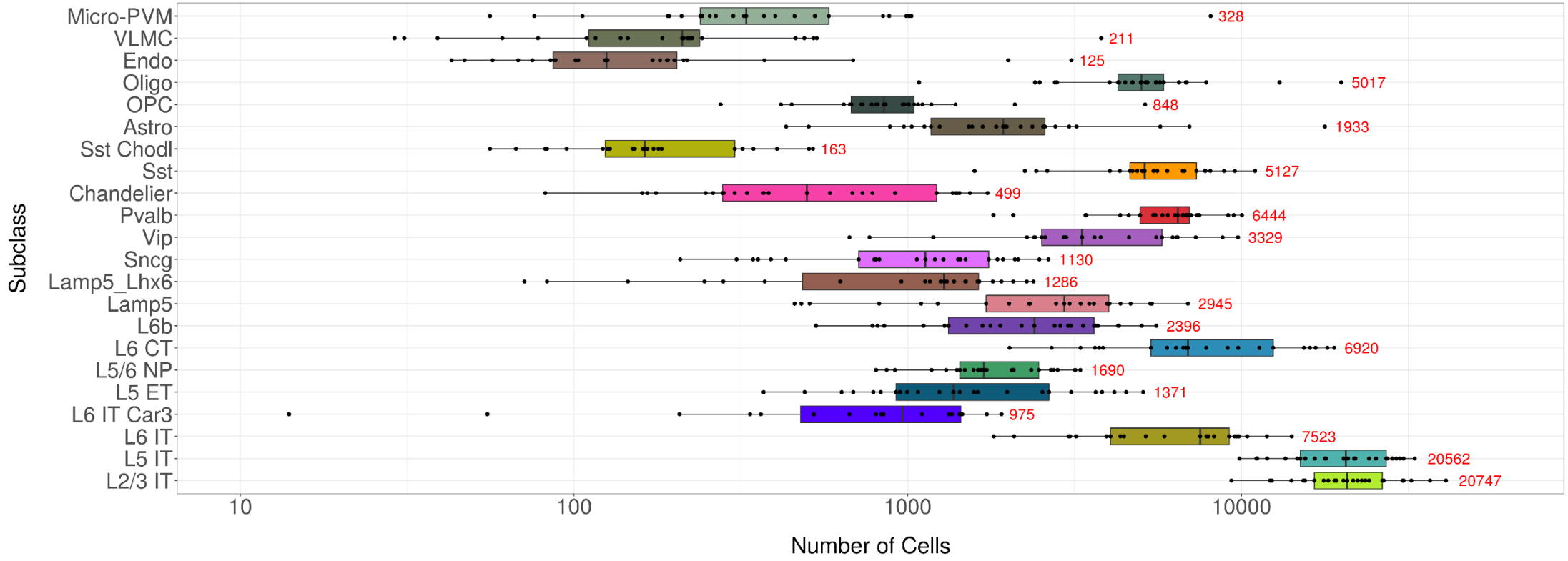
UNC5B expression



conserved cell types, variation in cell type abundance/



Cell counts per subclass

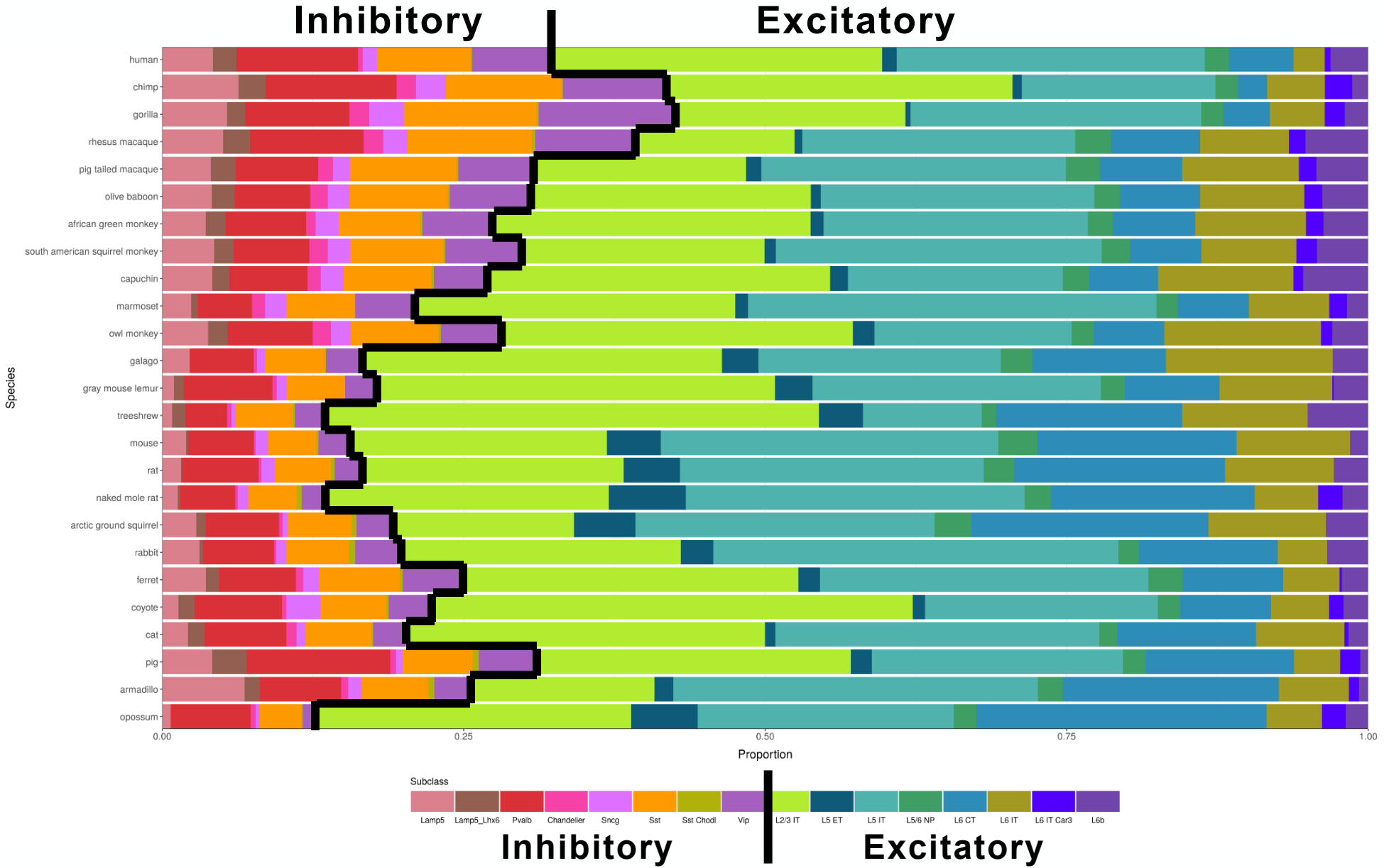


species differences in cell type abundance/

Neuronal Subclass Proportions Across Species



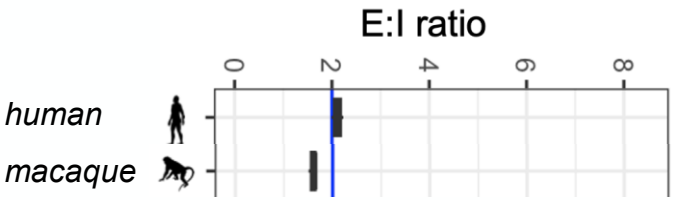
species differences in cell type abundance/

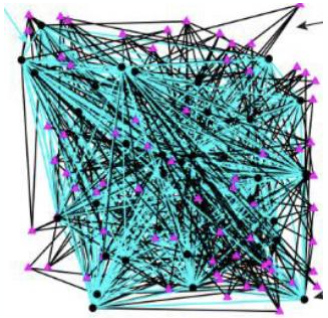


excitatory-to-Inhibitory (E:I) ratios vary across species/

~10-fold expansion of Interneuron-to-Interneuron connectivity:
Increased computational capacity?

Primates

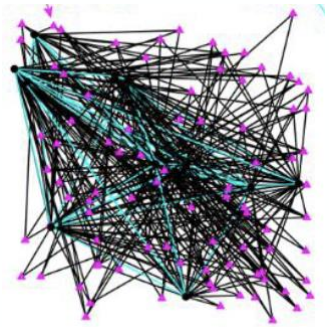




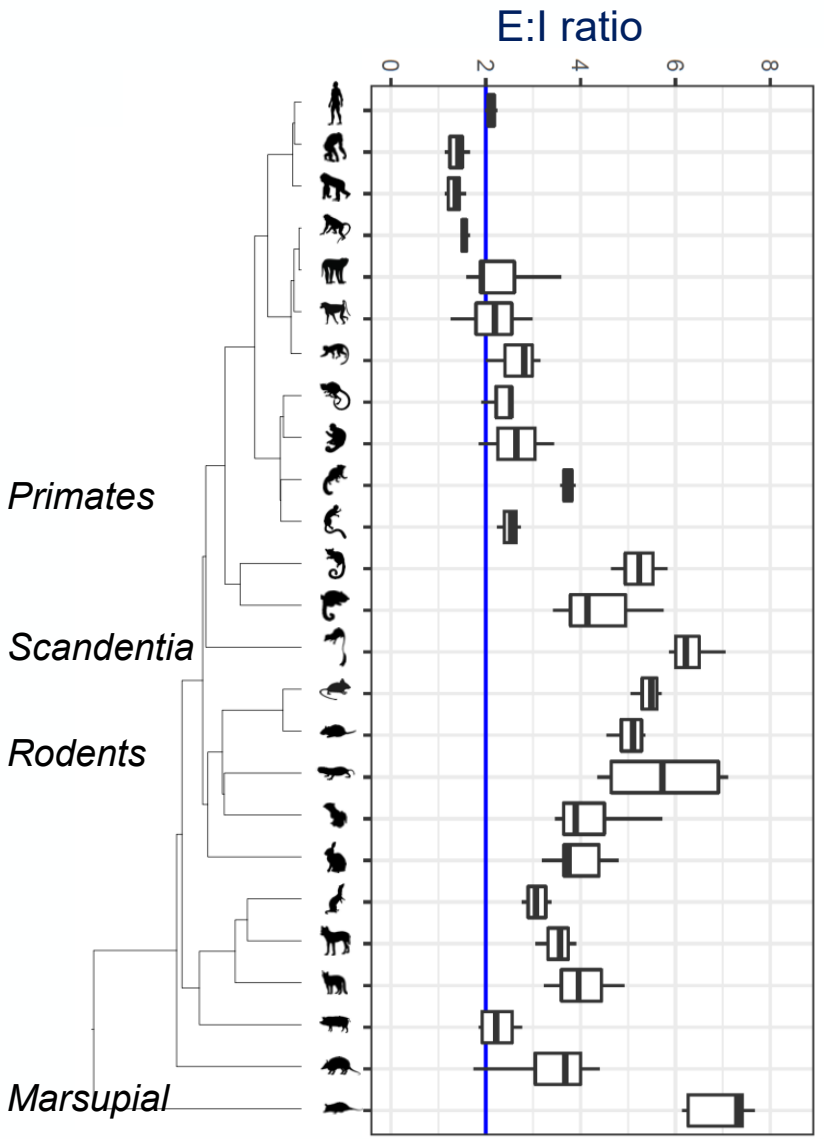
IN-to-ExN
connections

Rodents





Loomba et al. 2022 *Science*



Diverse species
with high E:I ratios

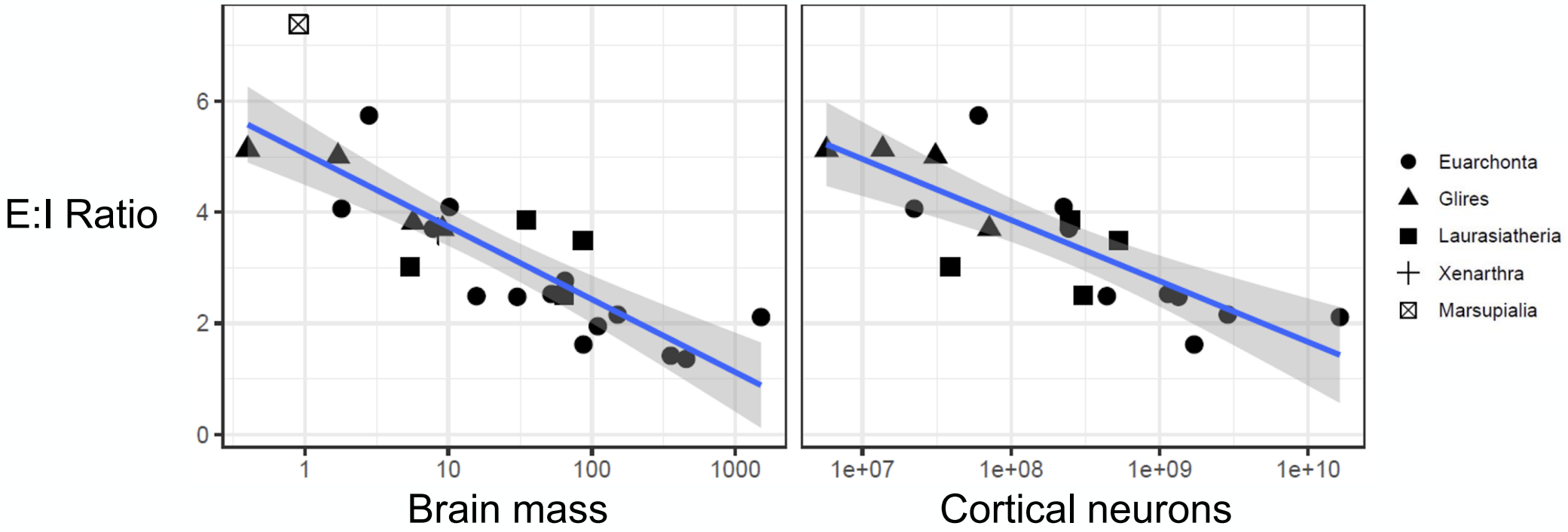






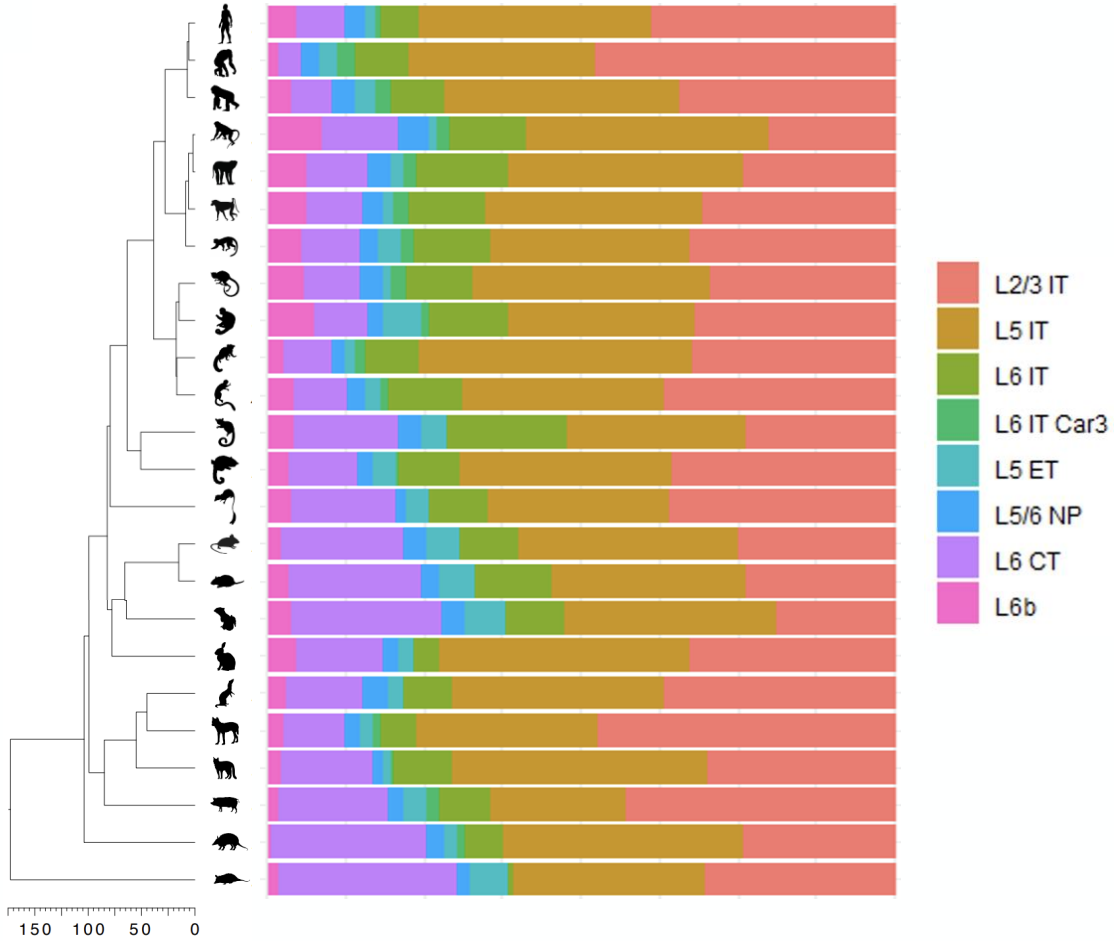


excitatory-to-inhibitory (E:I) neuron ratios scale *allometrically* as a function of brain size/

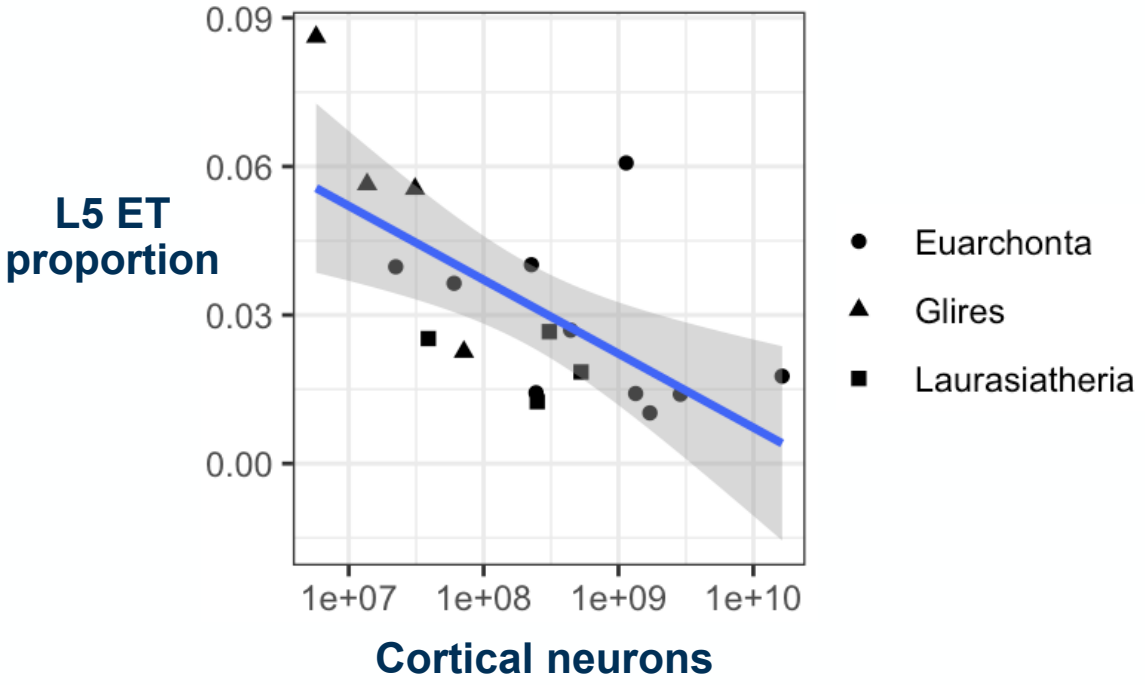


glutamatergic neuron abundances are divergent/

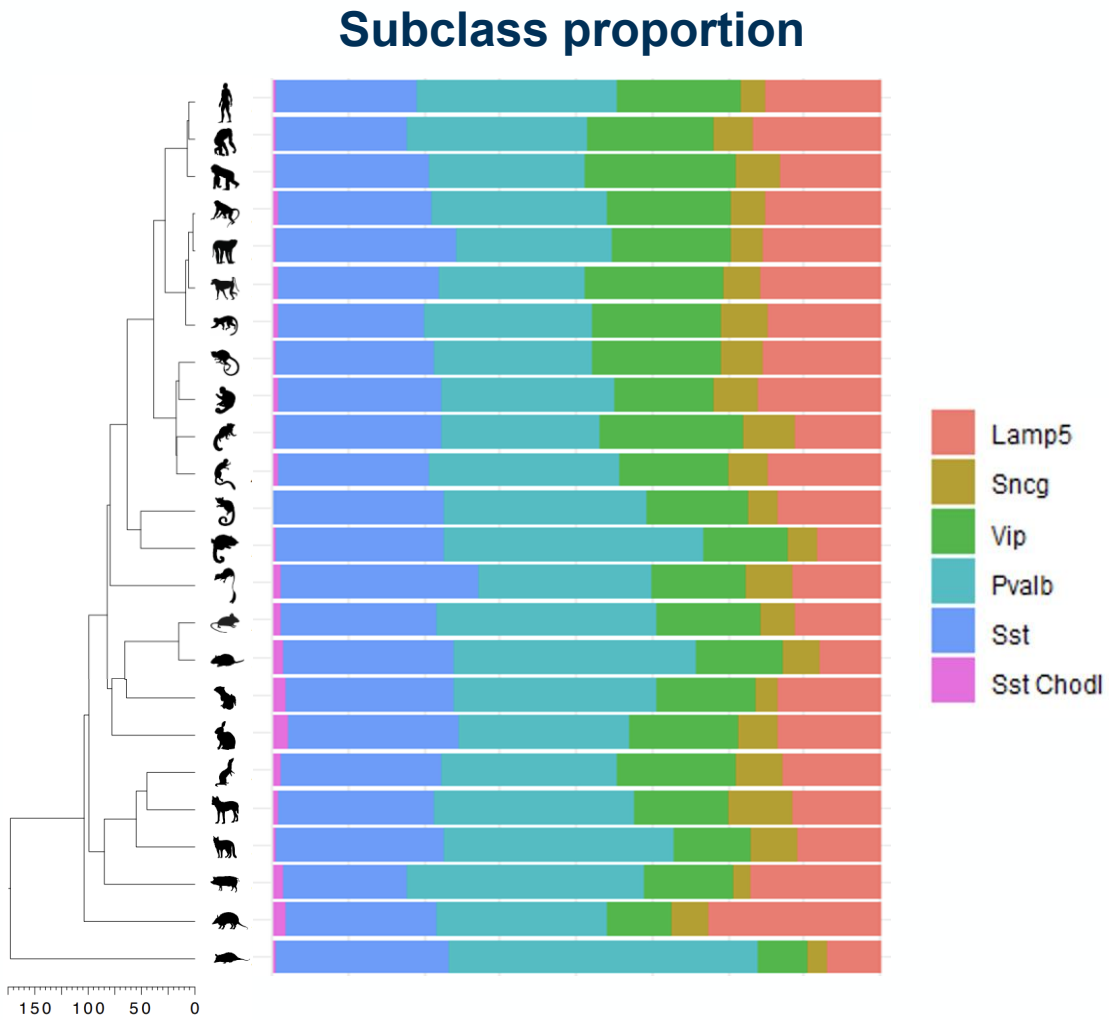
Subclass proportion



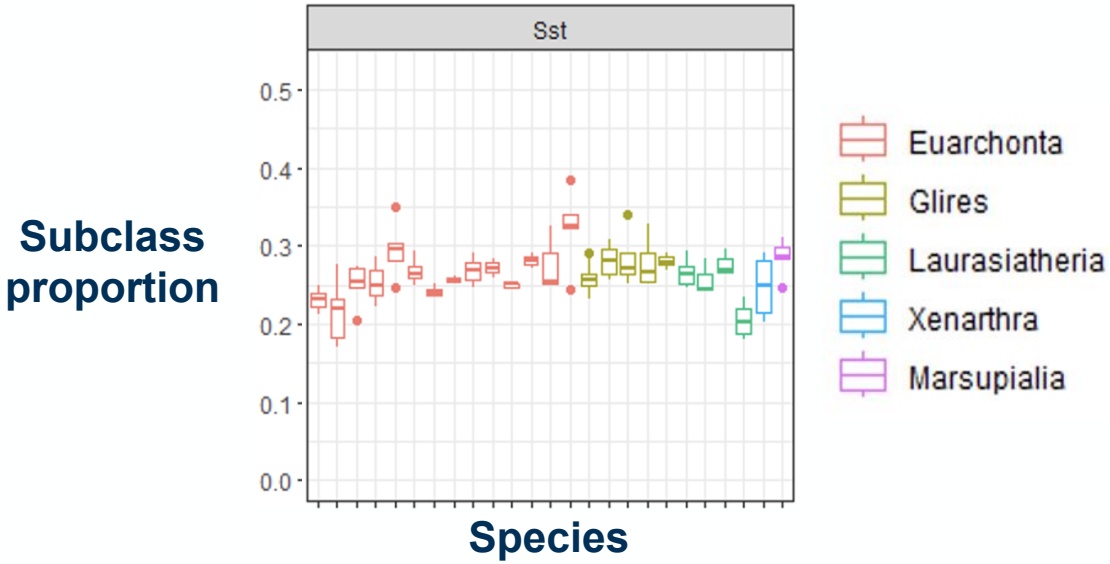
Relatively few L5 ET neurons in large cortices



interneuron abundances more conserved/

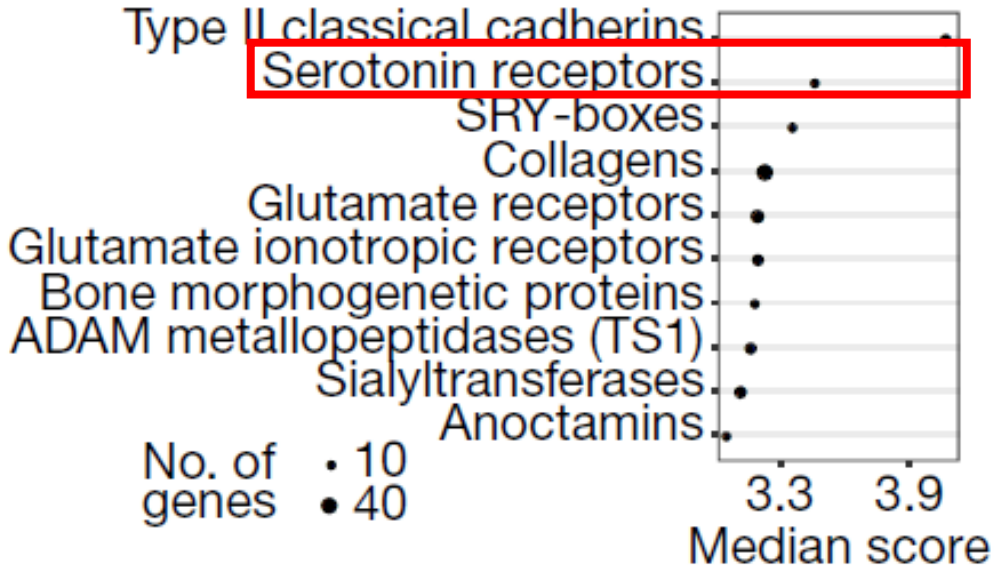


~25% SST interneurons in all species



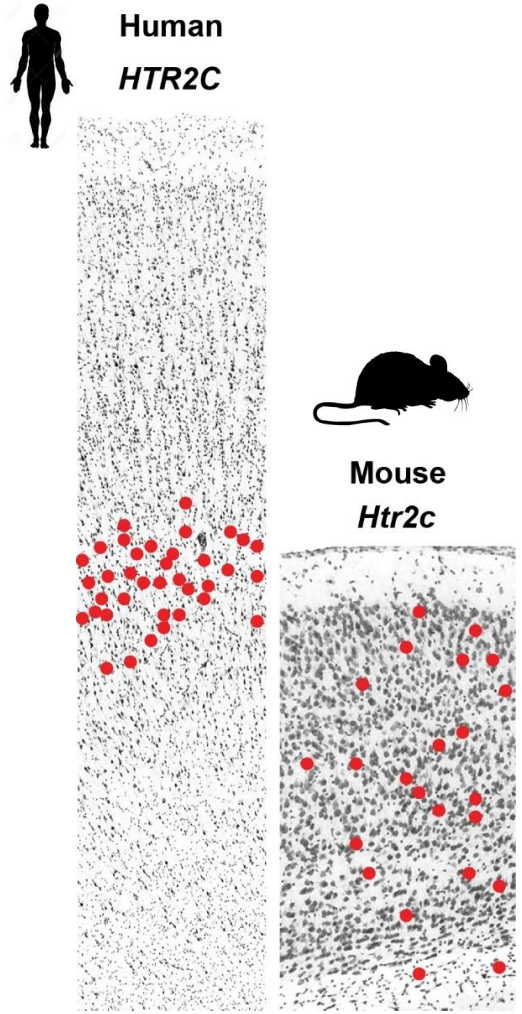
divergent expression of genes central to neuronal function/

Top 10 divergent gene families



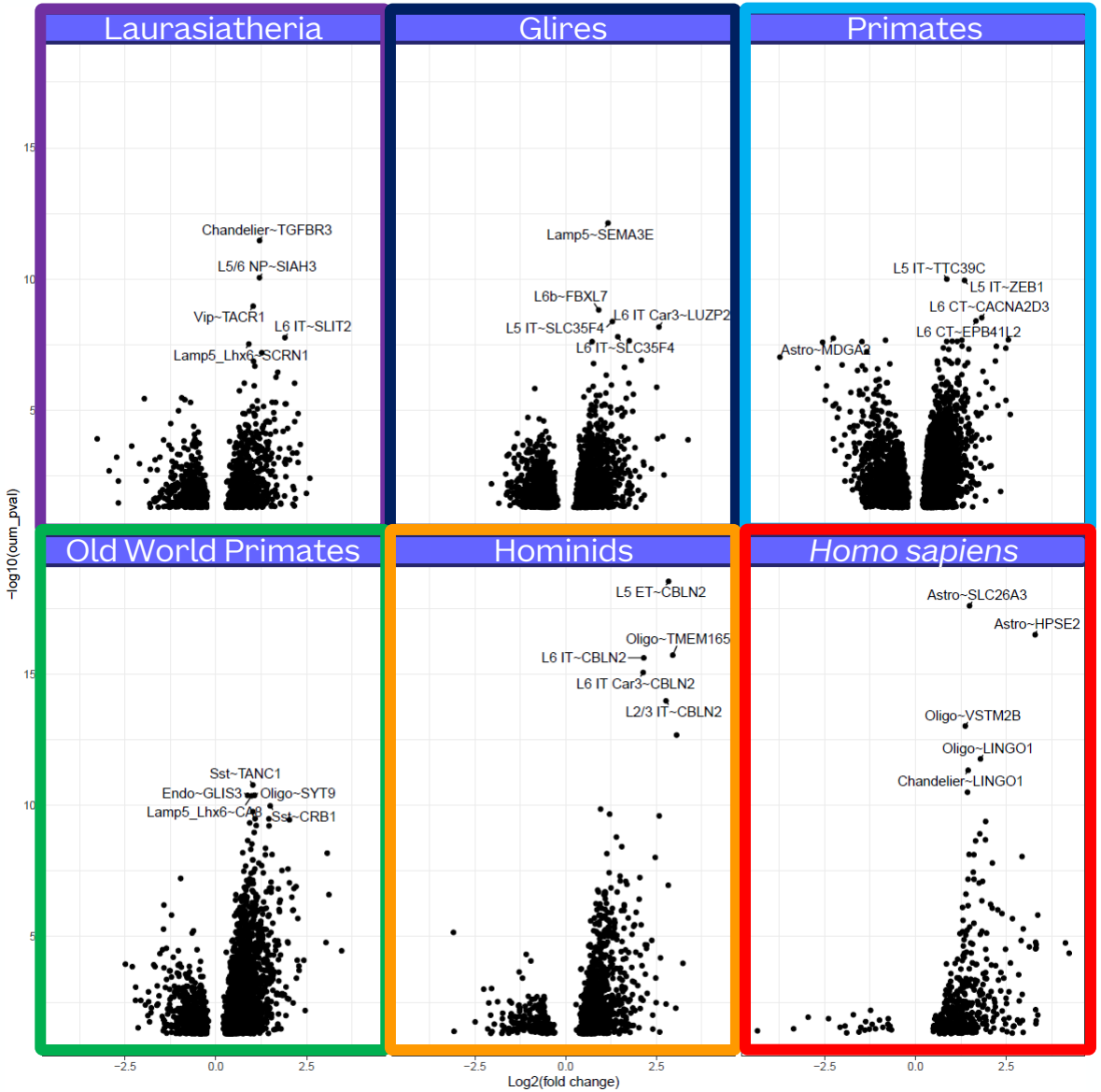
Hodge, Bakken *et al.* 2019

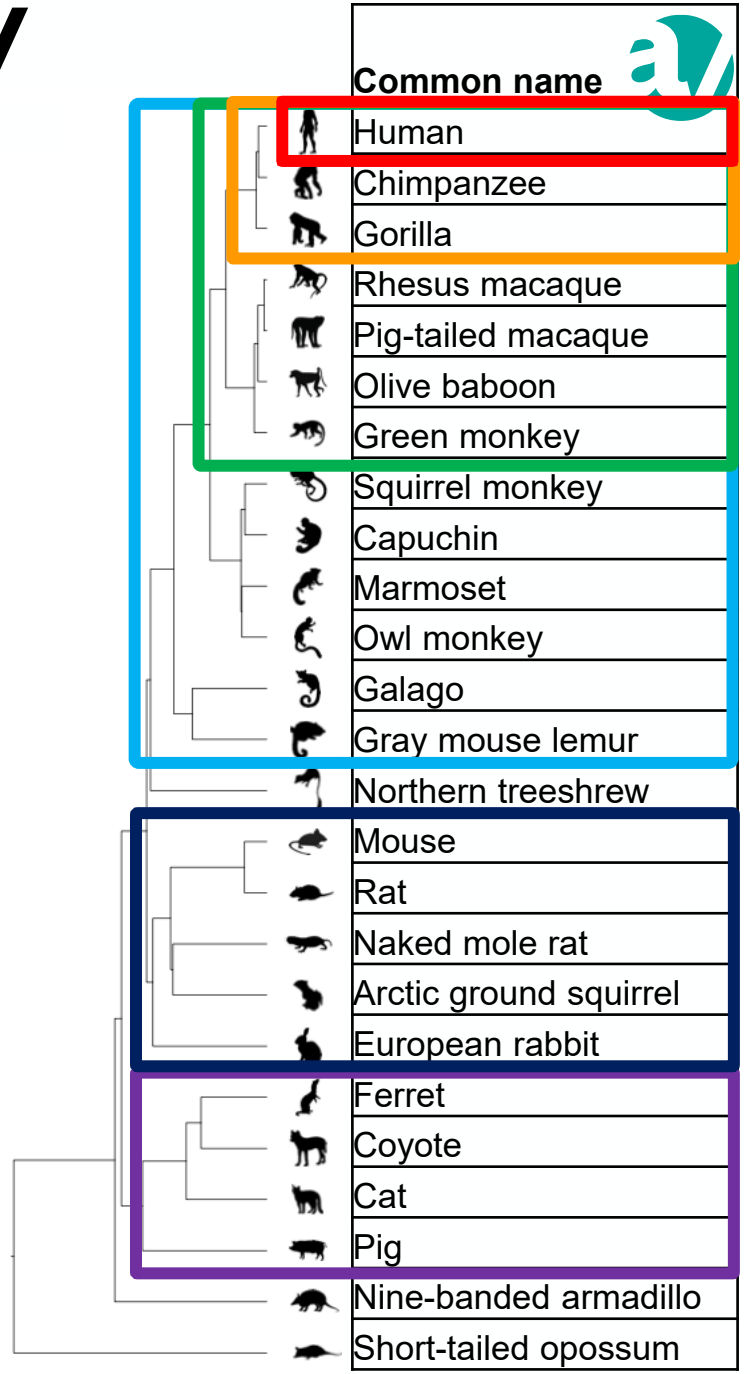
Serotonin receptor subunit *HTR2C* is expressed in human excitatory vs. mouse inhibitory neurons



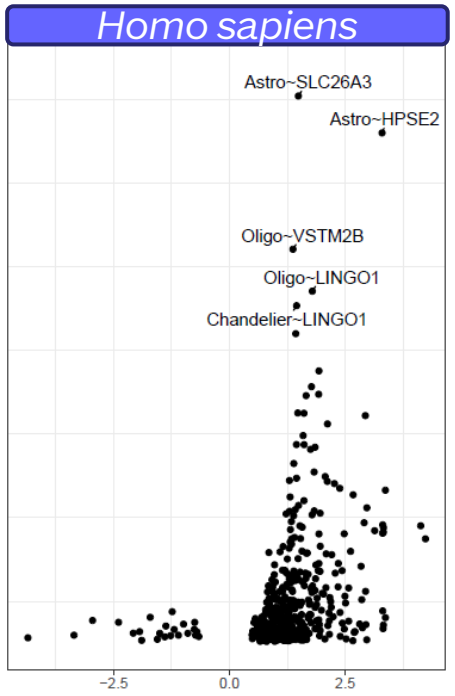
lineage-specific expression divergence/

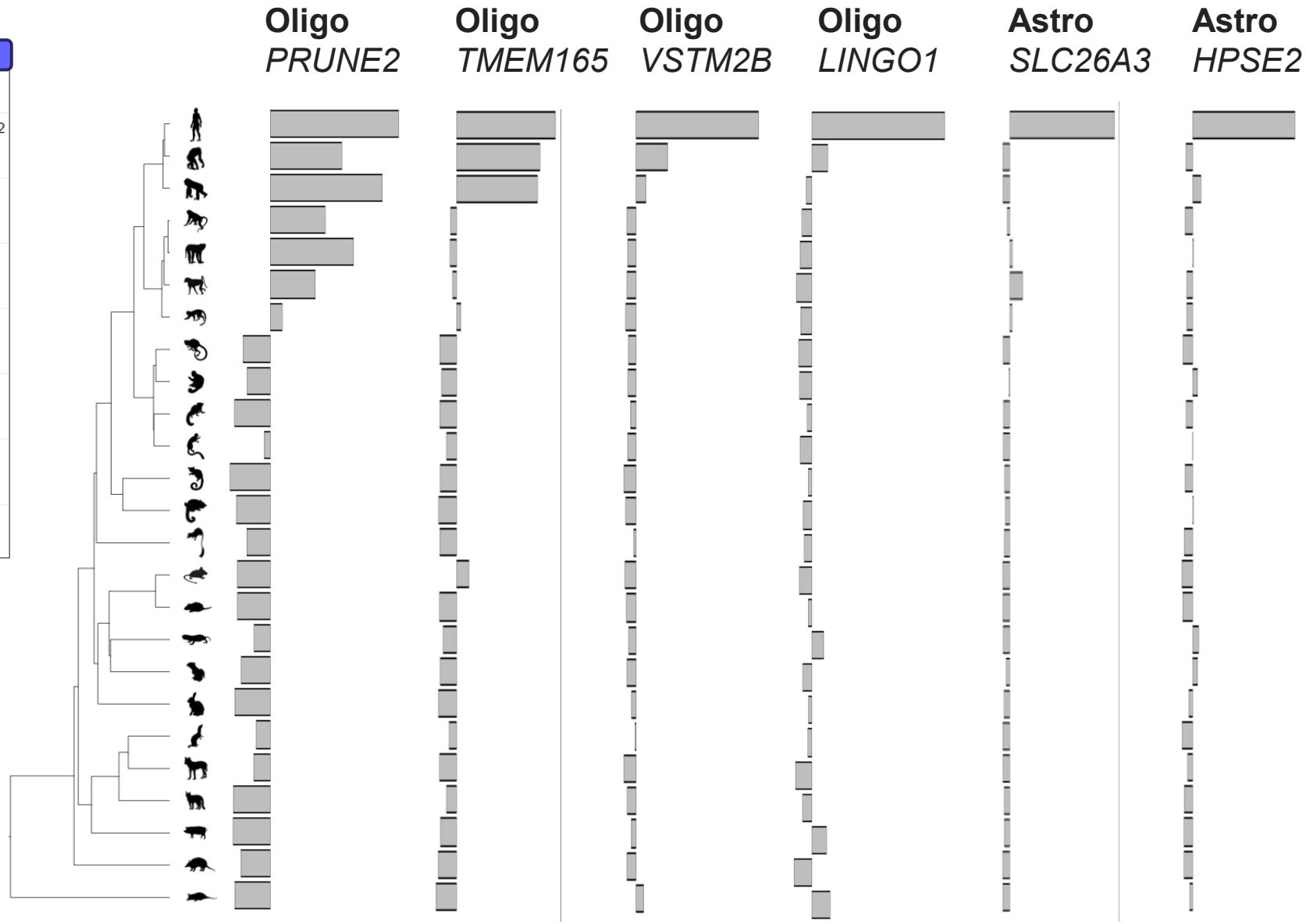
OU model with variance and selection strength parameters constant over tree



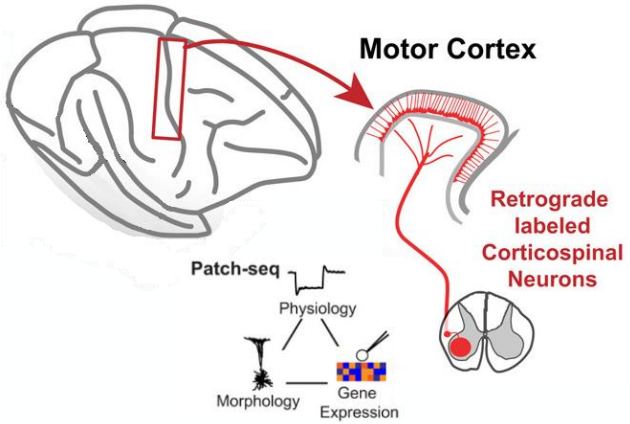


hominid-specific expression divergence in glial types/



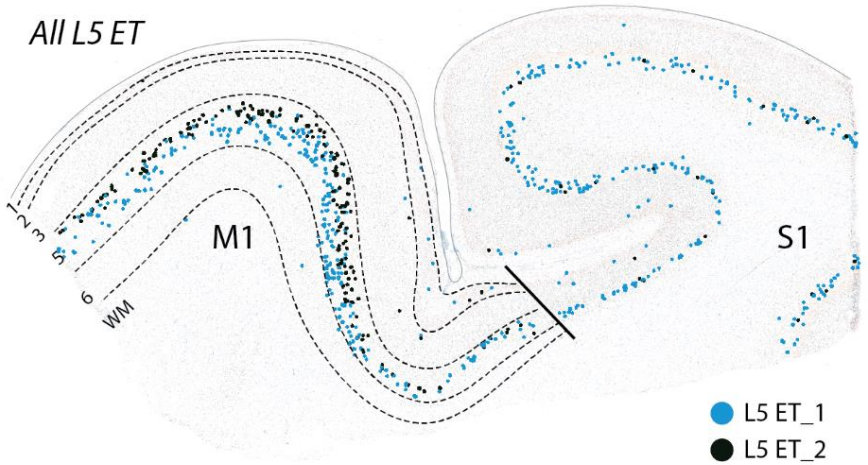


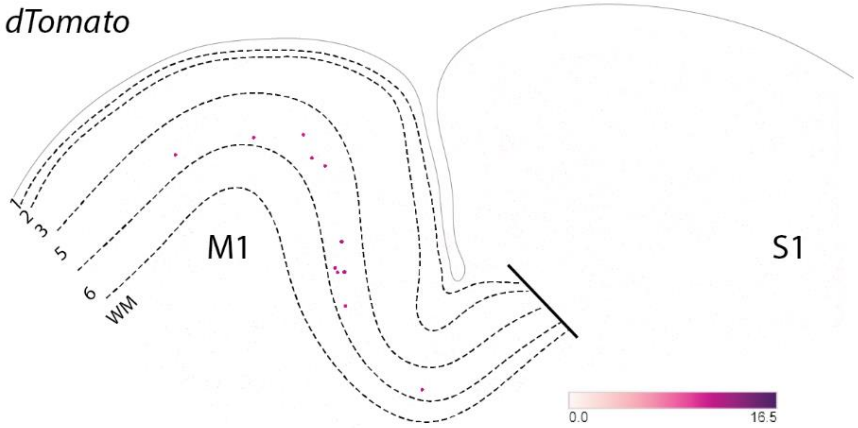
spatial transcriptomics on retrolabeled macaque M1/

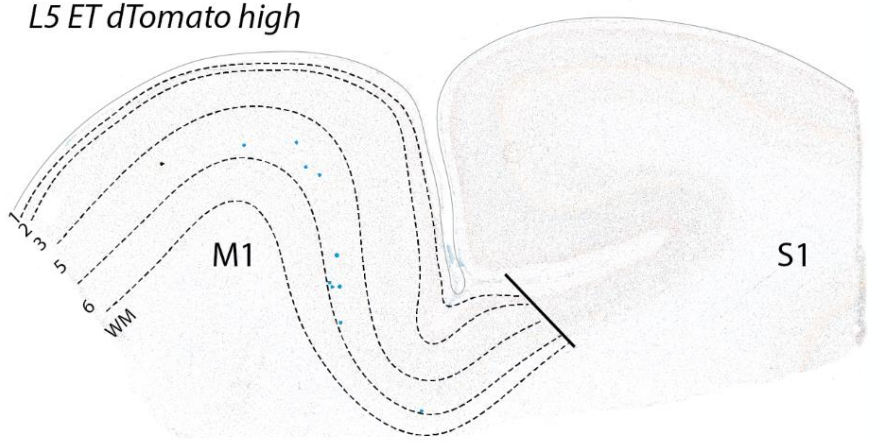


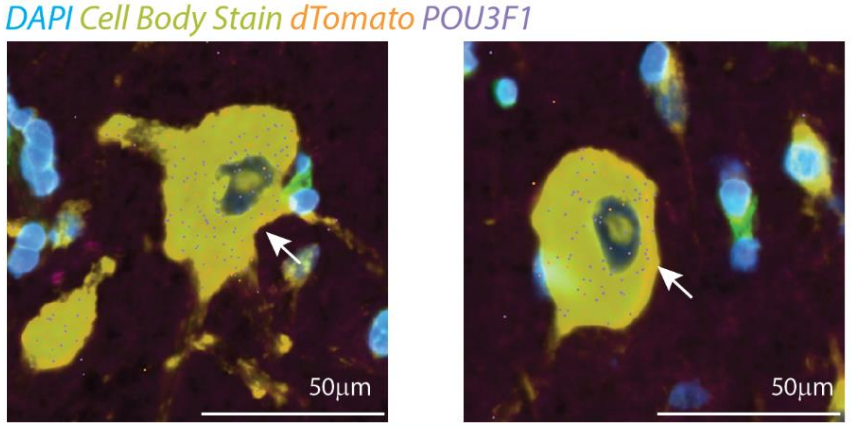
Dembrow *et al.* (2024)
Cell Reports

Spatial organization & distribution of L5ET subtypes





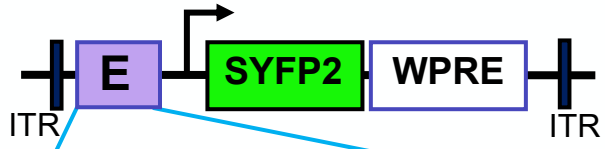




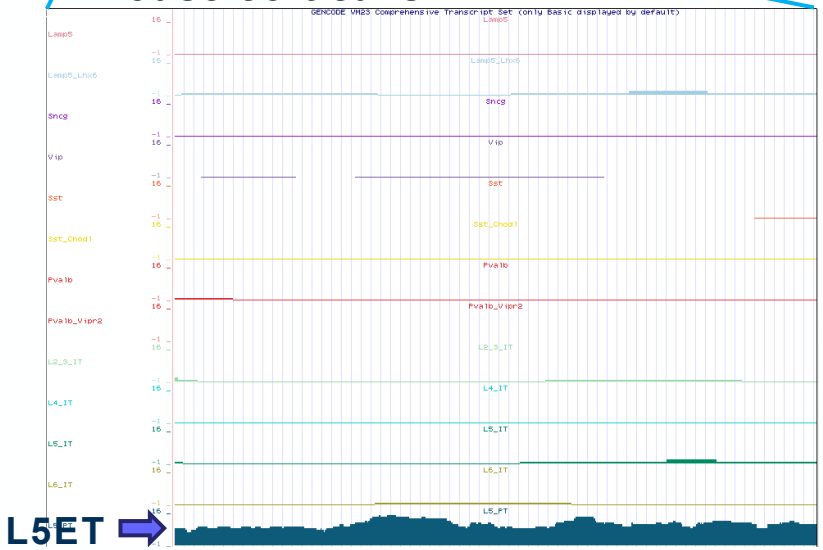
Rebecca Hodge, Steph Seeman, Brian Kalmbach

conservation of L5ET viral labeling specificity across species /

Viral plasmid: AiP12251



mouse cortical snATAC



Enhancer: AiE0453m

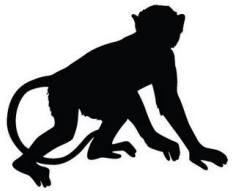
Mouse adult (RO)



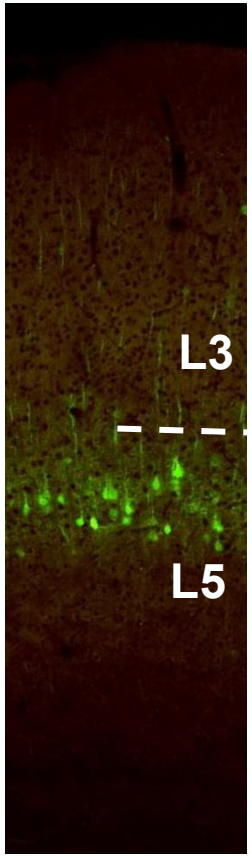
Rat P2 (ICV)

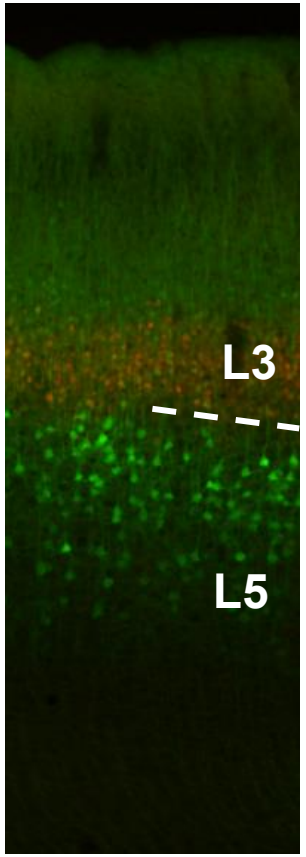


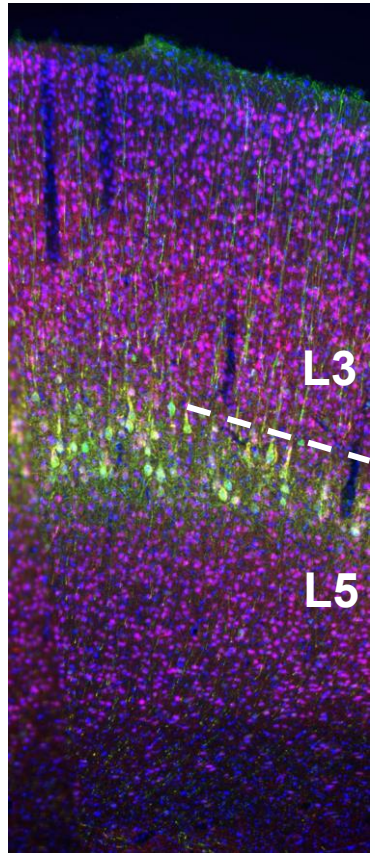
Treeshrew adult (stx)

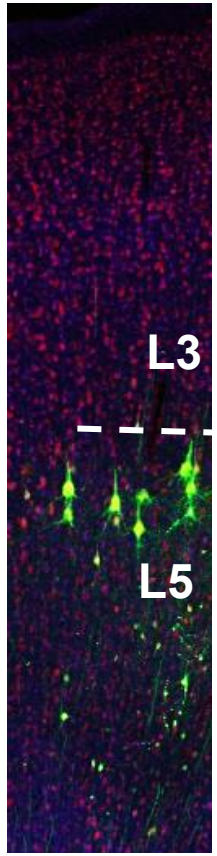


Macaque adult (stx)



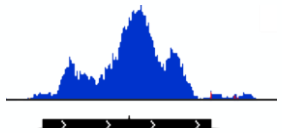




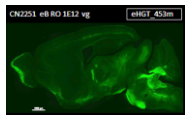


Jonathan Ting, Greg Horwitz, David Fitzpatrick & Teams

synthetic enhancers enable refined cell type-specific targeting/



31,994 L5ET ATAC peaks



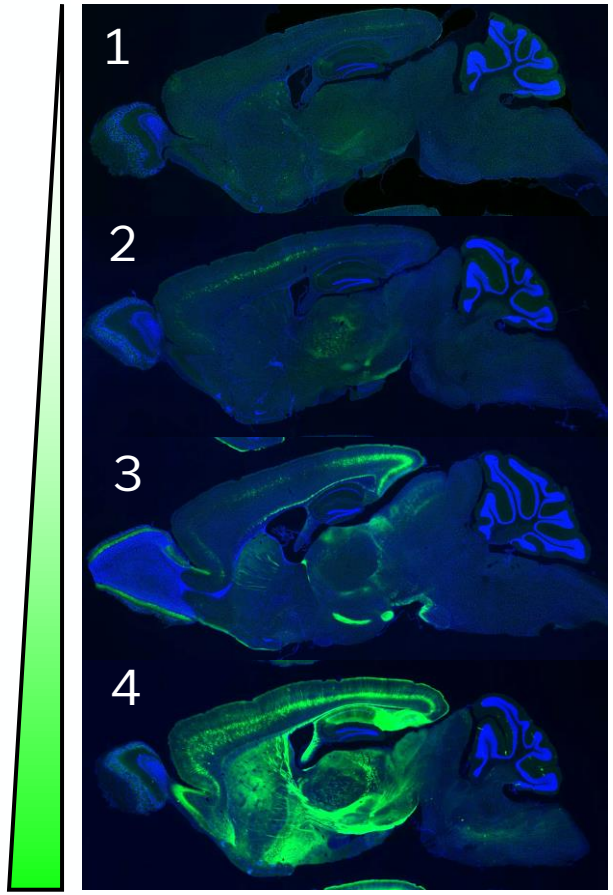
52 screened on-target enhancers



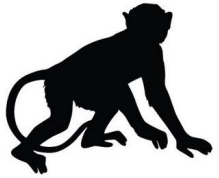
Enriched TF motifs & seq. features



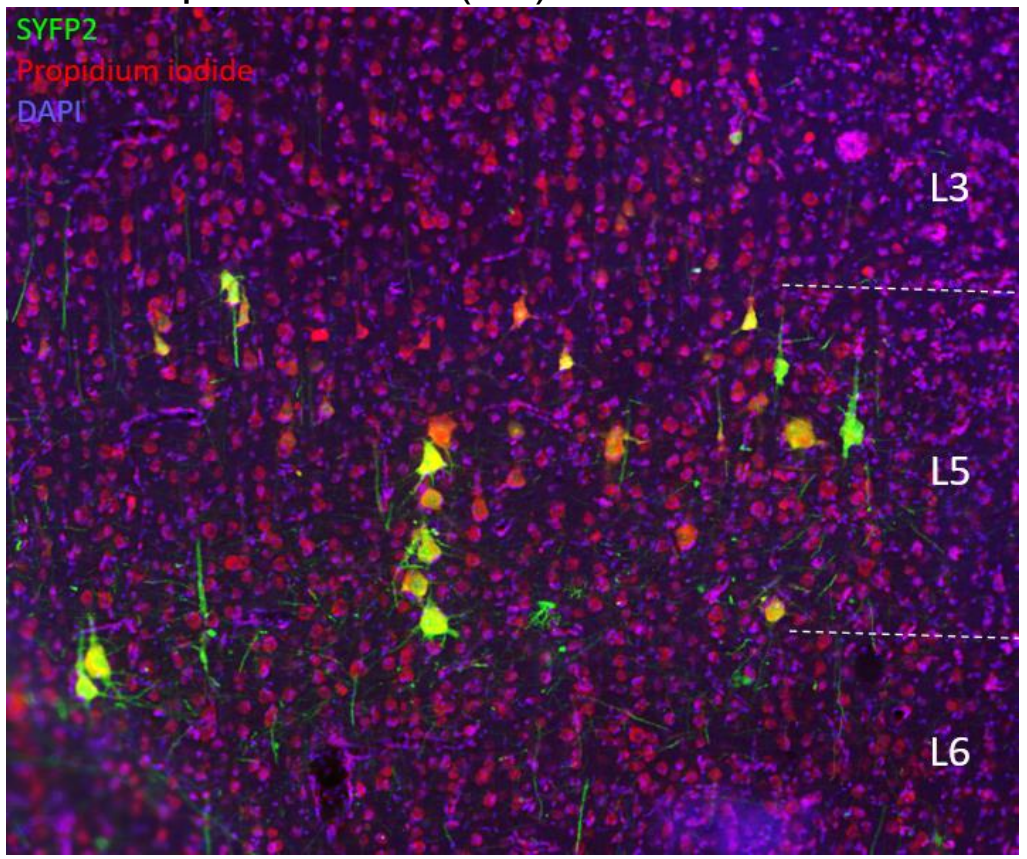
Mouse adult (RO)



Cross-species testing



Macaque adult M1 (stx)

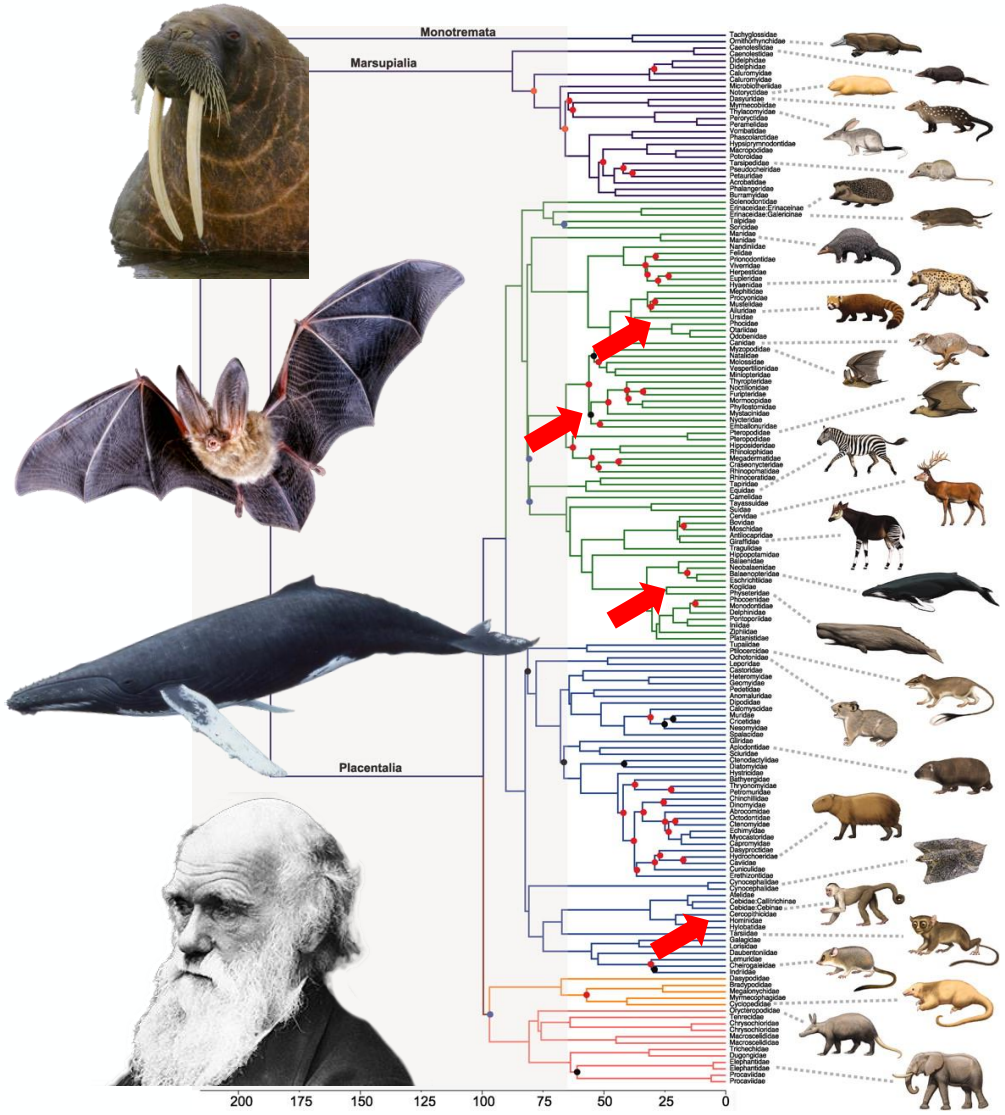


4 synthetic L5ET enhancers with divergent areal specificity

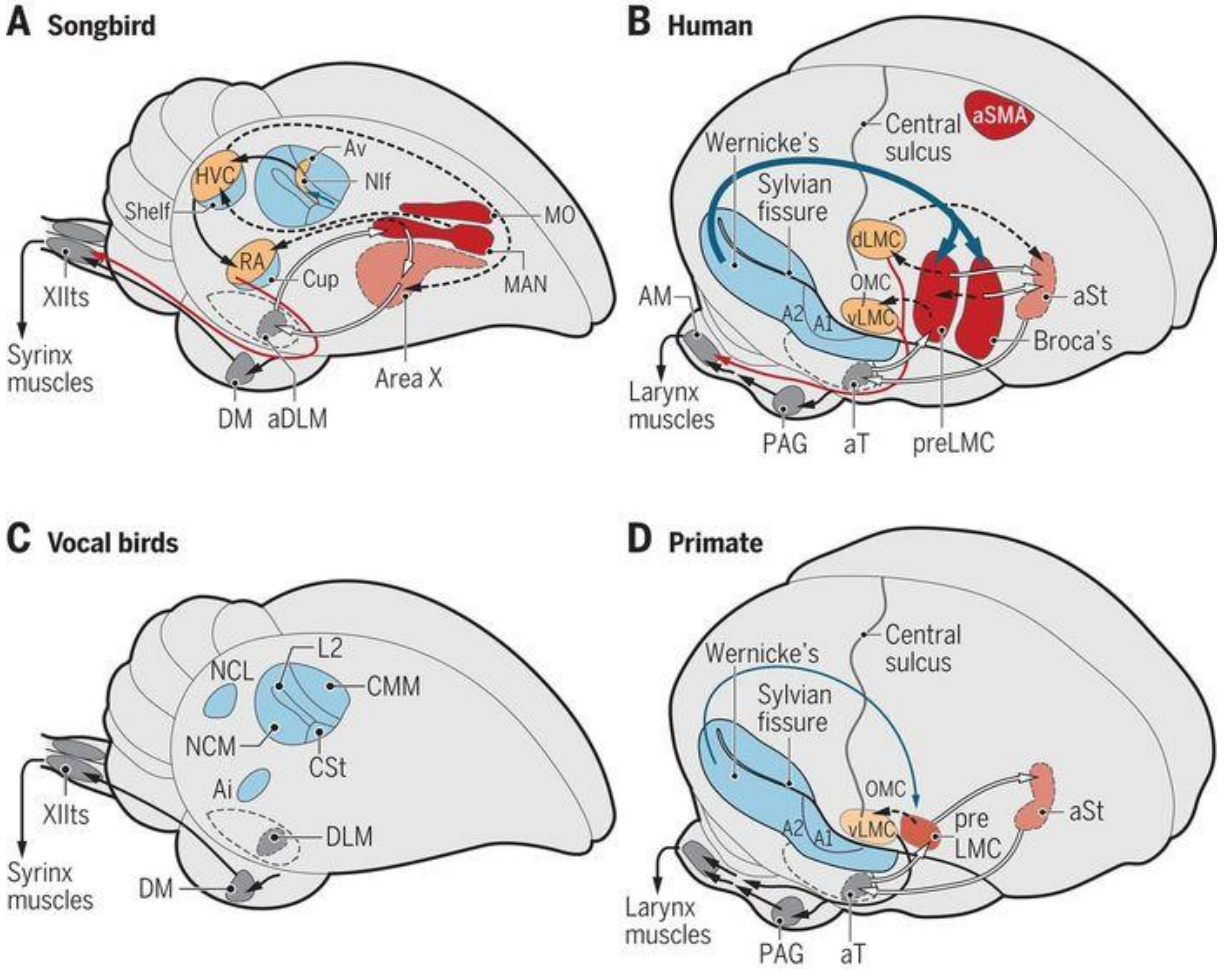
Enrichment of expression in L5ET Betz cells

Jonathan Ting, Greg Horwitz & Teams

next phase: the evolution of speech and vocal learning/

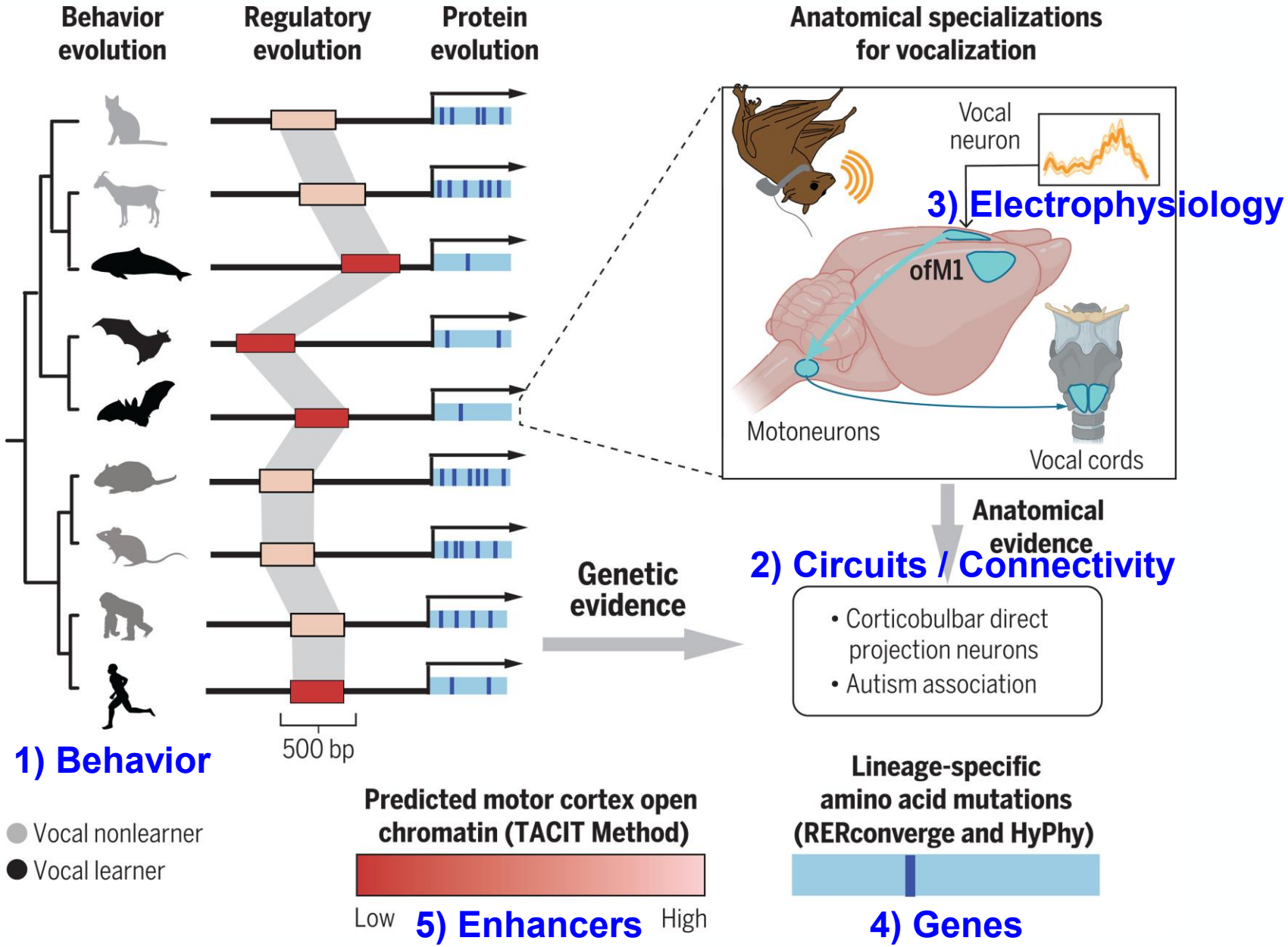


Meredith *et al.* (2011) *Science*



Jarvis (2019) *Science*

convergence at multiple levels in vocal learning mammals/





Egyptian Fruit Bat
Rousettus aegyptiacus

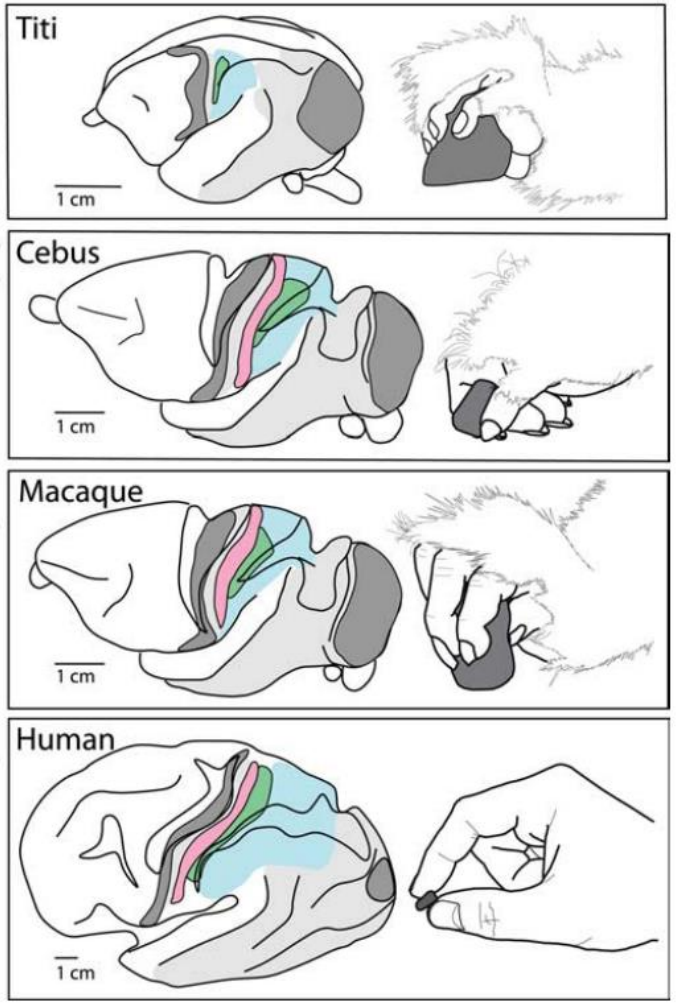
ZOONOMIA
Diverse genomes reveal mammalian

Genomes of
240 Mammals
26 Vocal Learners

Wirthlin et al. (2024) *Science*

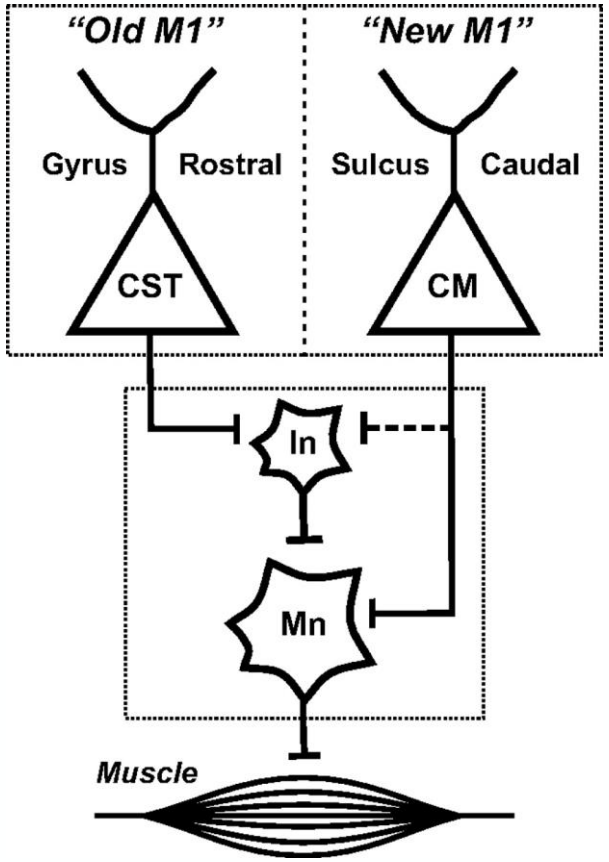
next phase: the evolution of dexterity/

Precision grip in primates

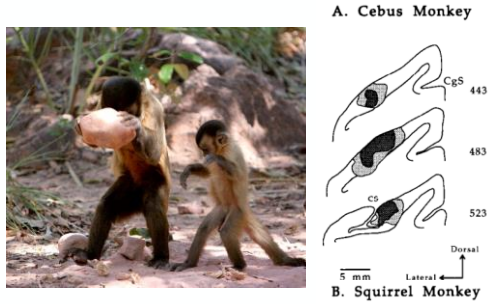


Padberg et al. (2007)

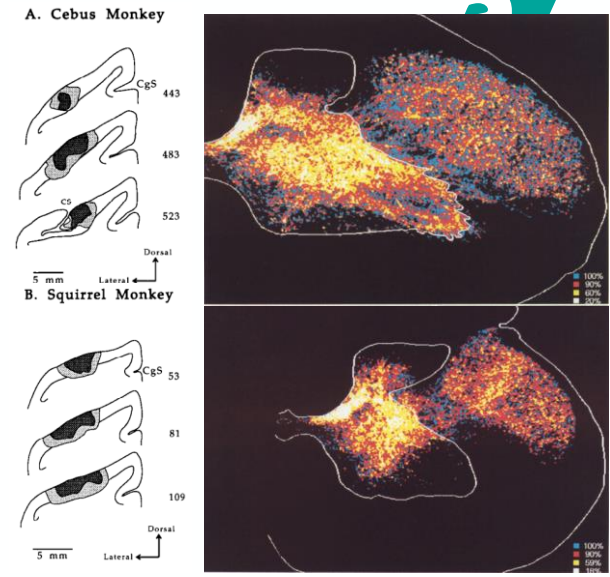
Convergent evolution of fine-motor control



Rathelot & Strick (2008)

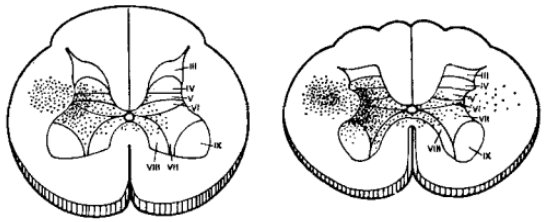






Bortoff and Strick (1993) *J Neuro*





Buxton & Goodman (1967)

the necessity of deep evolutionary comparisons/



Primates
State 2

Vs.



Rodents
State 1

Common assumption:
"Wow, State 2 is a human- / primate-specific specialization!"



Primates
State 2



Rodents
State 1



Bats
State 1



Carnivores
State 1



Marsupials
State 1

Primates: derived trait
Rodents: ancestral trait



Primates
State 2



Rodents
State 1



Bats
State 2



Carnivores
State 2



Marsupials
State 2

Primates: ancestral trait
Rodents: derived trait



Primates
State 2



Rodents
State 1



Bats
State 0.8

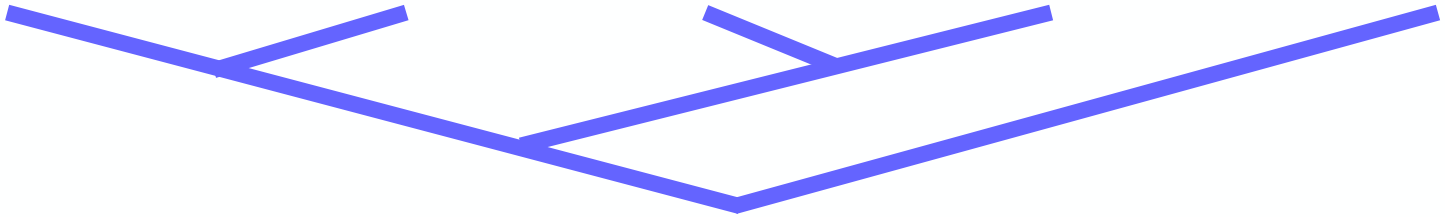


Carnivores
State 5



Marsupials
State 2.5

Trait varies, follows a rule



thank you/

EvoGen Team:
Trygve Bakken
Jessie Puckett
Nelson Johansen
Alexandra de Sousa
Matthew Schmitz
Yuanyuan Fu
Inkar Kapen

Collaborating Labs:
Chet Sherwood, David Fitzpatrick, Don Bolser, Fenna Krienen, Gabriel Balmus, Greg Wilkerson, Guoping Feng, Janet Song, Jeffrey Padberg, Jon Kaas, Kelly Drew, Kimberley Phillips, Leah Krubitzer, Mark Krasnow, Sarah Freeman, Shixuan Liu, Ted Barrett

We wish to thank the Allen Institute founder, Paul G. Allen, for his vision, encouragement, and support.



questions?

allen institute/

agenda/afternoon **– breakout rooms**

Session 2: analysis of your own sequencing data using allen institute resources.

Room 1165 – 1185

1:10-1:35 | Using a reference cell type taxonomy to understand Alzheimer's disease

1:35-2:00 | cross-species alignment and comparative analyses

2:00-2:20 | other interactive tools for -omics data exploration (CellxGene, Cytosplore, etc.)

2:20-2:30 | break

2:30-2:40 | ABC Atlas demo & external use case examples

2:40-3:40 | user exploration of allen institute tools

3:40-4:00 | feedback survey

4:00 – 5:00 | additional time for user exploration

other interactive tools for -omics data exploration/

Meghan Turner, Scientist II

allen institute/

allen institute

omics data exploration/

- The BKP offers several tools for omics data exploration, including the **ABC Atlas** and **MapMyCells**

- These include tools for exploration of:
  - Single cell/nucleus RNA-seq data
  - Single cell/nucleus ATAC-seq data
  - Spatial transcriptomics data
  - Patch-seq data

- These additional tools offer different advantages and shortfalls

- We note that this is not an exhaustive list—thousands of tools exist for analysis and exploration of –omics data!

types of tools for exploring –omics data/

- Single cell/nucleus transcriptomics
  - Exploring 2D/3D latent spaces (UMAP, tSNE, etc.)
  - Other (better) ways to explore transcriptomics data
  - Mapping and label transfer
- Cell-level spatial transcriptomics data viewers
- Patch-seq data exploration
- Viewing and selecting enhancers from genome-level data
- Custom (mostly R-shiny) tools for complex use cases
  - Trajectory analysis
  - Taxonomy comparison
  - Summarizing gene knowledge across atlases

single cell transcriptomics

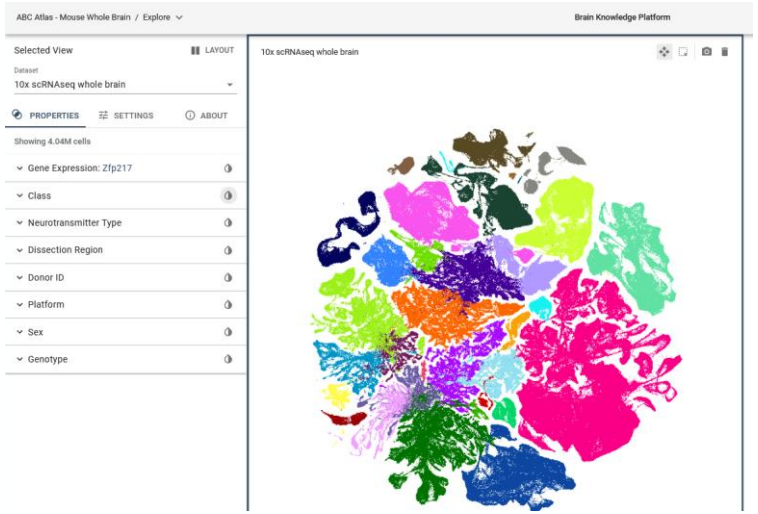
Exploring cells and cell types



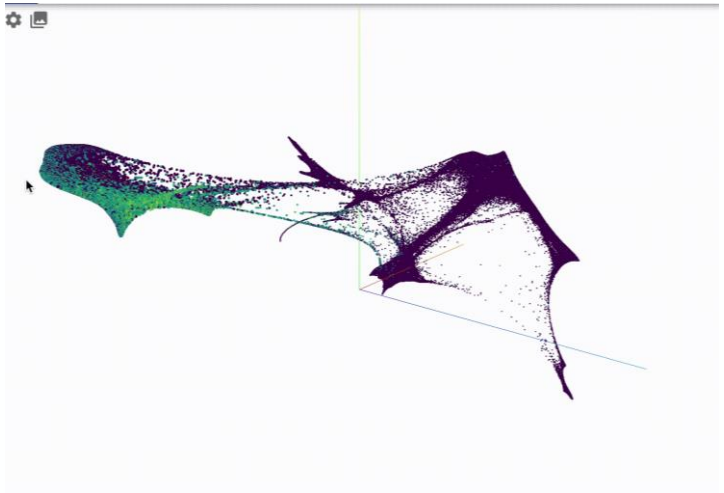
alleninstitute/

many tools for latent space exploration/

ABC Atlas



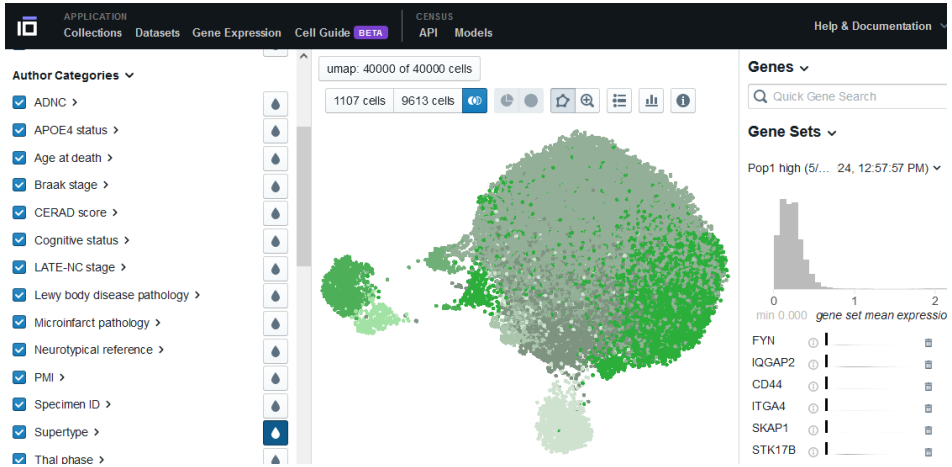
Cirrocumulus



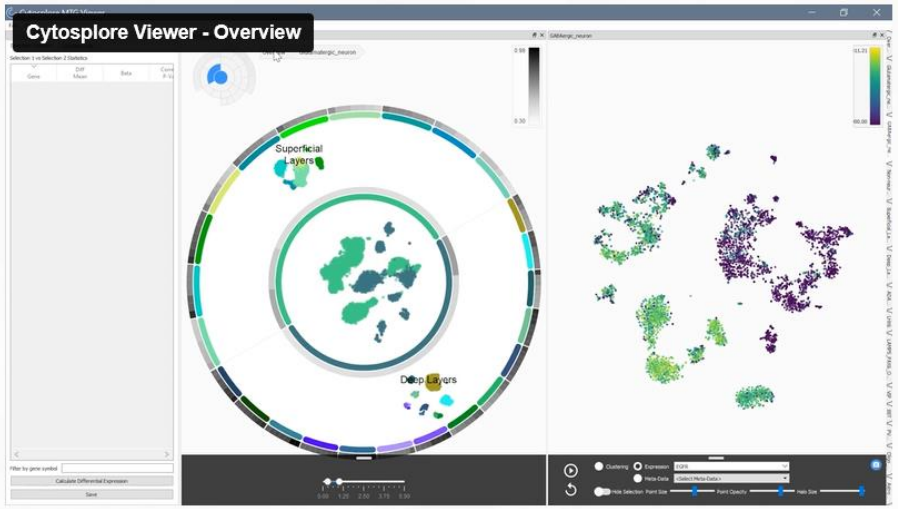
(Several other Allen Brain Map viewers)

- Transcriptomics Explorer
- RNA-seq Data Browser
- SEA-AD Comparative Viewer

CELLxGENE



Cytosplore



(Note: many of these tools have other functionalities, which I'll return to later as relevant.)

choosing a latent space exploration tool/

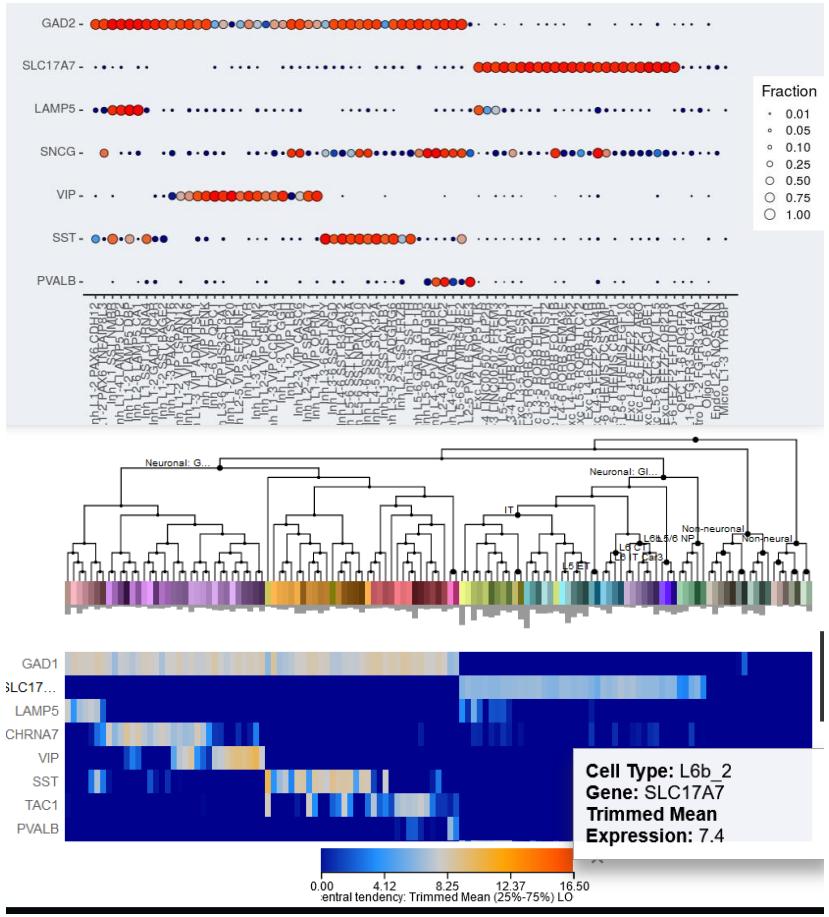
- All of these tools show cells from a sc/snRNA-seq study in 2D coordinates and allow you to color-code them by gene expression or cell metadata (e.g., cell type, age, sex, QC)
  - *If this is what you want to do, choose any tool that has the data set you'd like*

Tool	Ease of use	Function-ality	Support	Spatial Data?	Public	Comments
ABC Atlas	XXX	X	XXXX	Yes	Web	Can view a growing list of existing, published Institute datasets. Fully owned by the Allen Institute's D&T team and functionality expected to improve. SPOCTT team has built vignettes for use.
Cirrocumulus	X	XXX	X	Yes	Local host	Can view 3D latent spaces; works on pretty much any data set. Hosted internally. Used for WMB annotation.
Cytosplore	X	XXXX	XXX	Yes	Desktop App	Many versions of tool that do (or can do) just about anything. Collaborator group (Lelieveldt) longtime/continuing collaborators. Fast computations!
CELLxGENE	XX	XX	XX	No	Web or local host	Does computations; connects to entire cell typing community; lots of CZI folks working on it (but not us).
Transcriptomics Explorer	XXX	X	X	No	Web	I'd used this tool for the heatmap view rather than the UMAP view; limited cortical data sets.
RNA-seq Data Browser	XX	XX	XX	No	Web	Can show up to 3 genes at once; for human MTG (Hodge) and mouse VISp/ALM only; does computations (outside UMAP).
SEA-AD Comparative Viewer	XXX	XX	X	No	Web	Shows UMAP for reference and SEA-AD data side-by-side; only for MTG data set

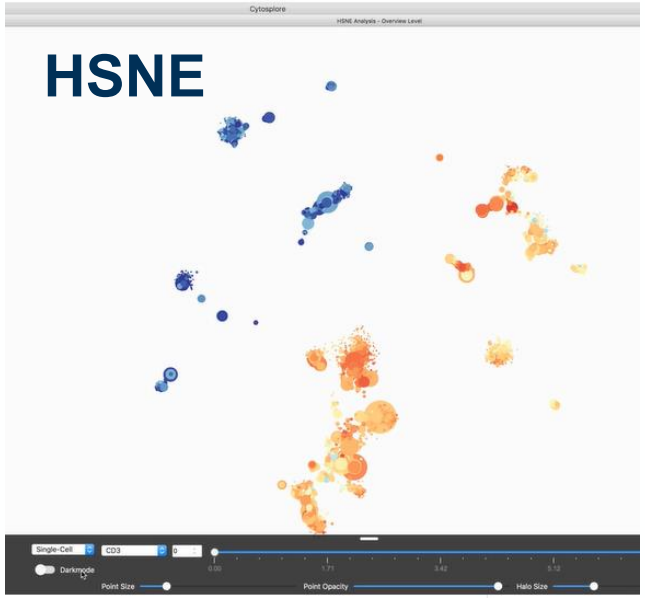
alternatives to UMAP data viewing/

- Sometimes it's more desirable to summarize gene expression per cell type (or other group)
- Here are a few alternatives :

Group plots

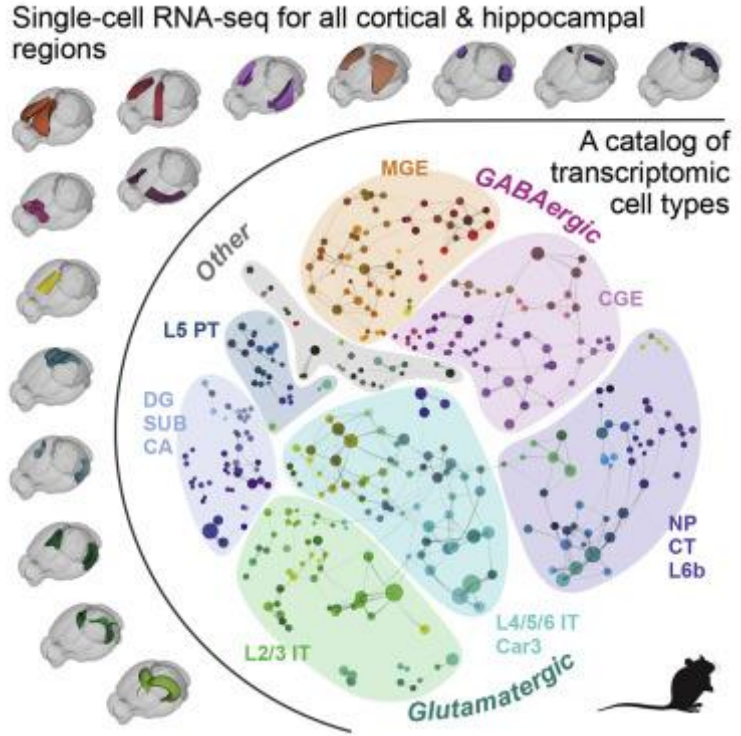


Accordion Hierarchies



van Unen et al, 2017, Nature Comm.

Constellation diagrams

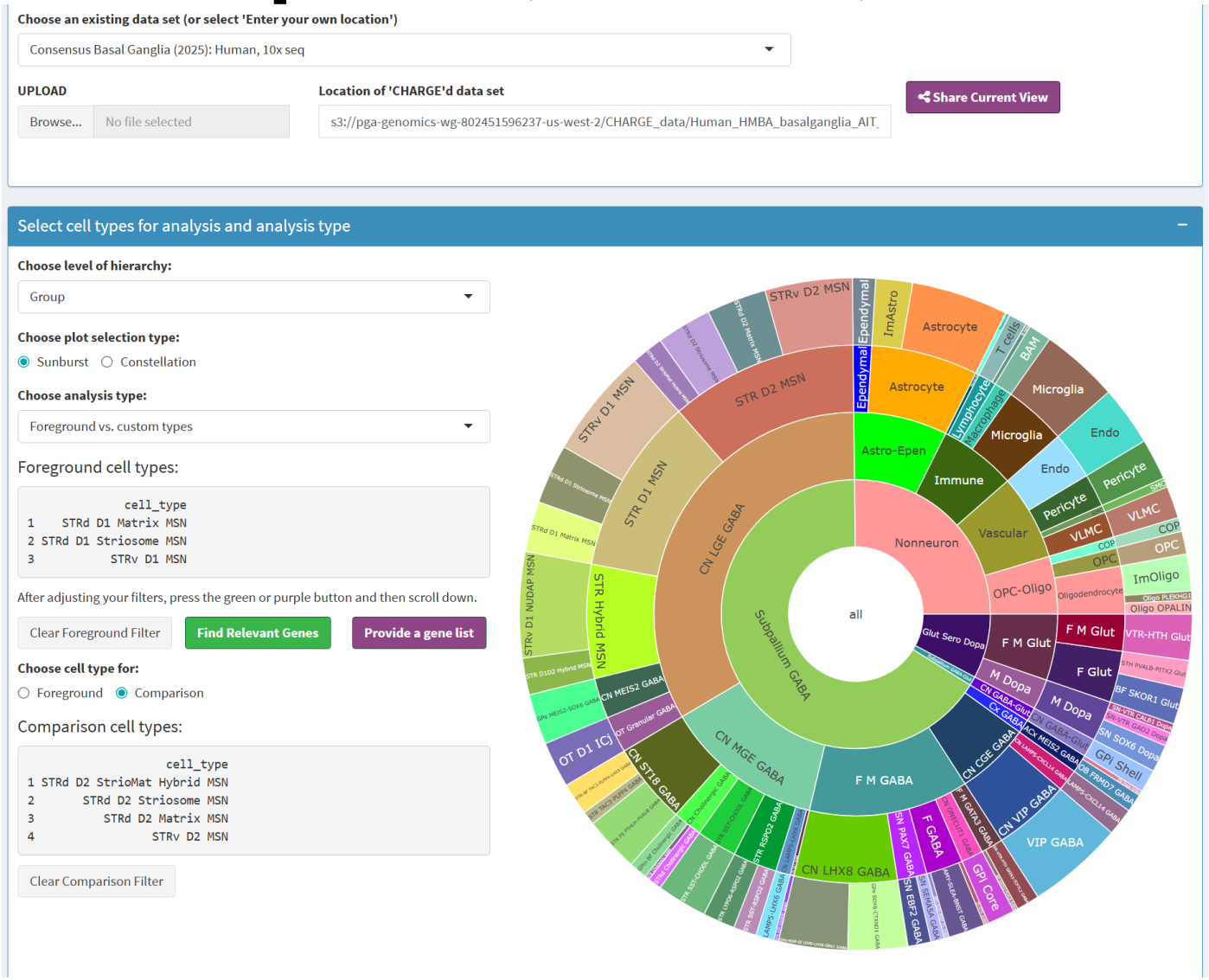


https://taxonomy.shinyapps.io/ctx_hip_browser/

Yao et al 2021, Cell

Cell HierARchy Gene Explorer (CHARGE)

- CHARGE is an interactive tool for quickly identifying genes of potential biological interest in a cell type taxonomy
- CHARGE takes a taxonomy as input and outputs visualizations and statistics for interesting genes and gene sets for viewing and download
  - *(Can run on any AIT file via [CHARGE_taxonomy]())*



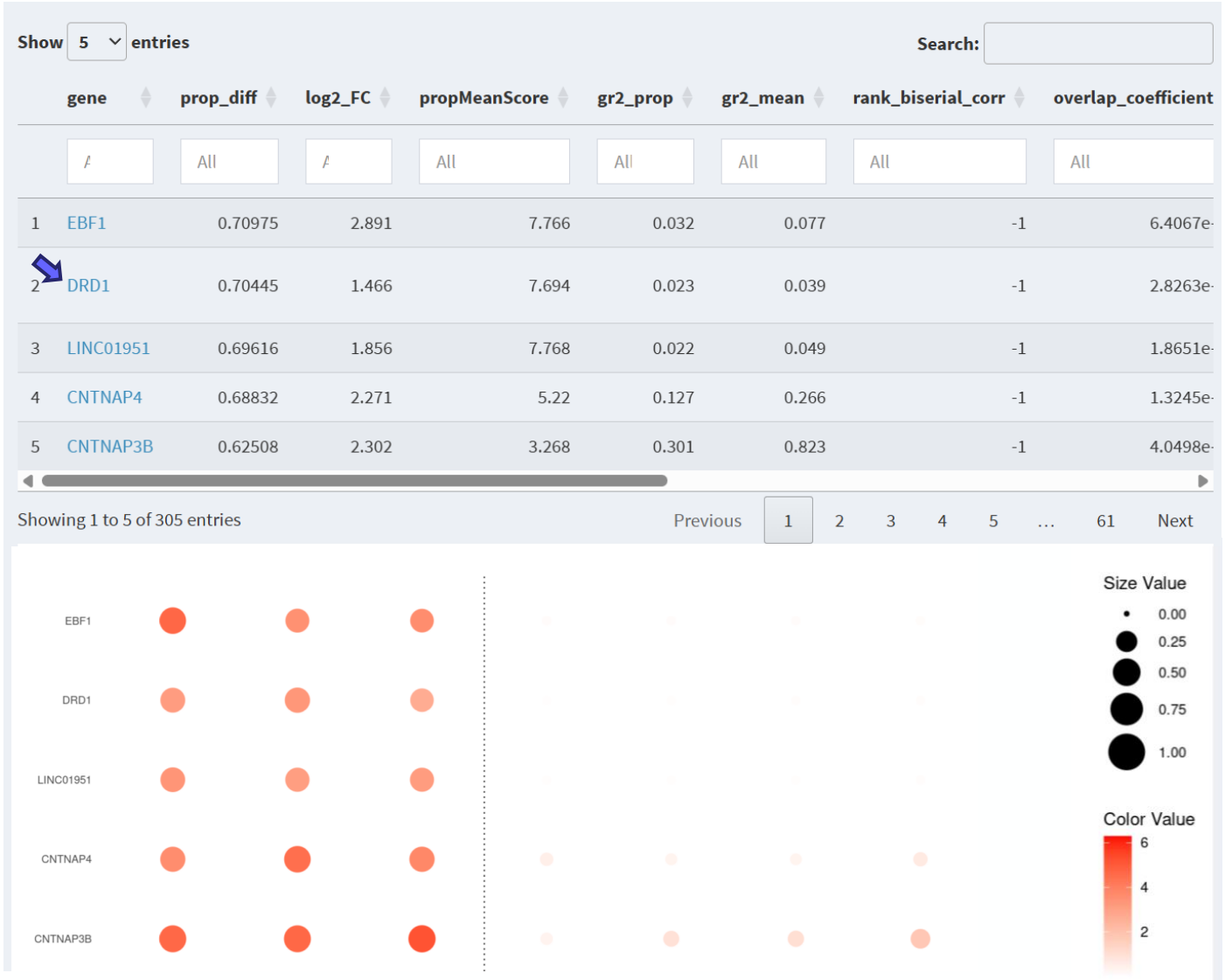
Select Human BG taxonomy

Select D1 MSN vs. D2 MSN

https://molgen-shiny.org/CHARGE/

Cell HierARchy Gene Explorer (CHARGE)

- CHARGE is an interactive tool for quickly identifying genes of potential biological interest in a cell type taxonomy
- CHARGE takes a taxonomy as input and outputs visualizations and statistics for interesting genes and gene sets for viewing and download
  - *(Can run on any AIT file via CHARGE_taxonomy)*



Filter and sort using basic metrics

(and link-out to ABC atlas)

See gene plots for top genes

https://molgen-shiny.org/CHARGE/

Cell HierARchy Gene Explorer (CHARGE)

- CHARGE is an interactive tool for quickly identifying genes of potential biological interest in a cell type taxonomy
- CHARGE takes a taxonomy as input and outputs visualizations and statistics for interesting genes and gene sets for viewing and download
  - *(Can run on any AIT file via [CHARGE_taxonomy]())*



Download plots or data

Calculate (basic) gene set enrichment for genes in table

https://molgen-shiny.org/CHARGE/

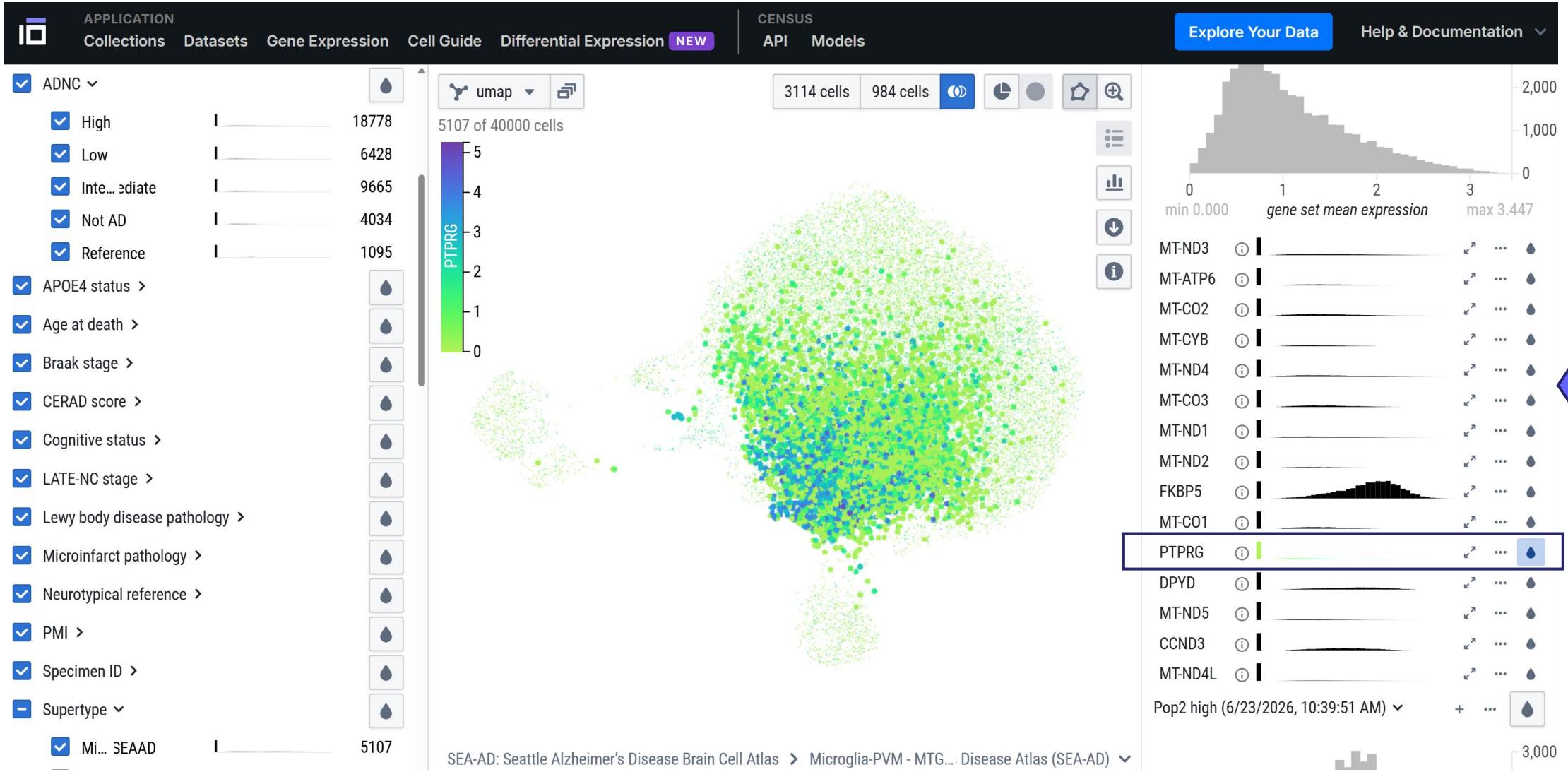
single cell transcriptomics

Exploring changes in disease



alleninstitute/

identify genes of interest using CELLxGENE/



Interactive tool can find some of the same microglial genes associated with AD as complex analyses...

but need to note covariates

find genes changing with AD pathology using the MTG gene trajectory viewer/

SEA-AD
Gene Trajectories

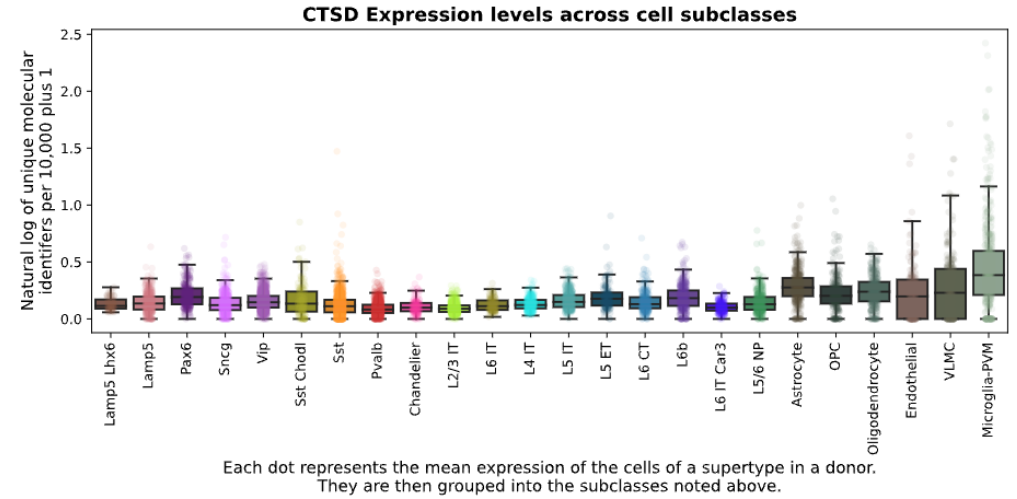
Please select an option and click open table to begin.

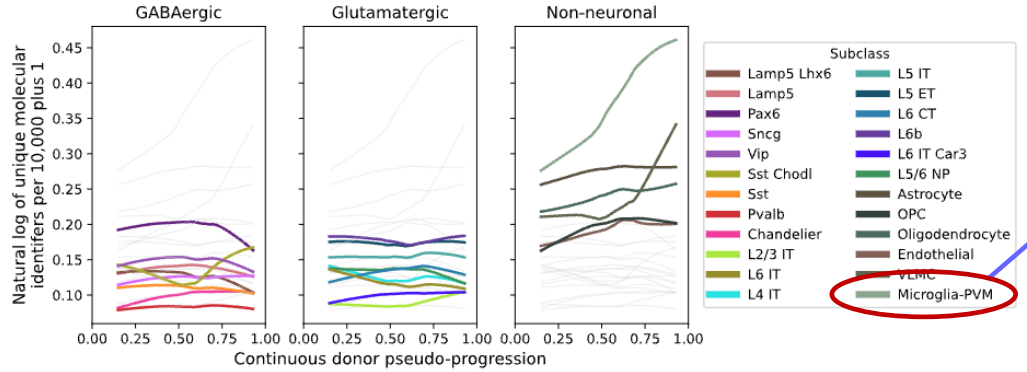
Taxonomy Level

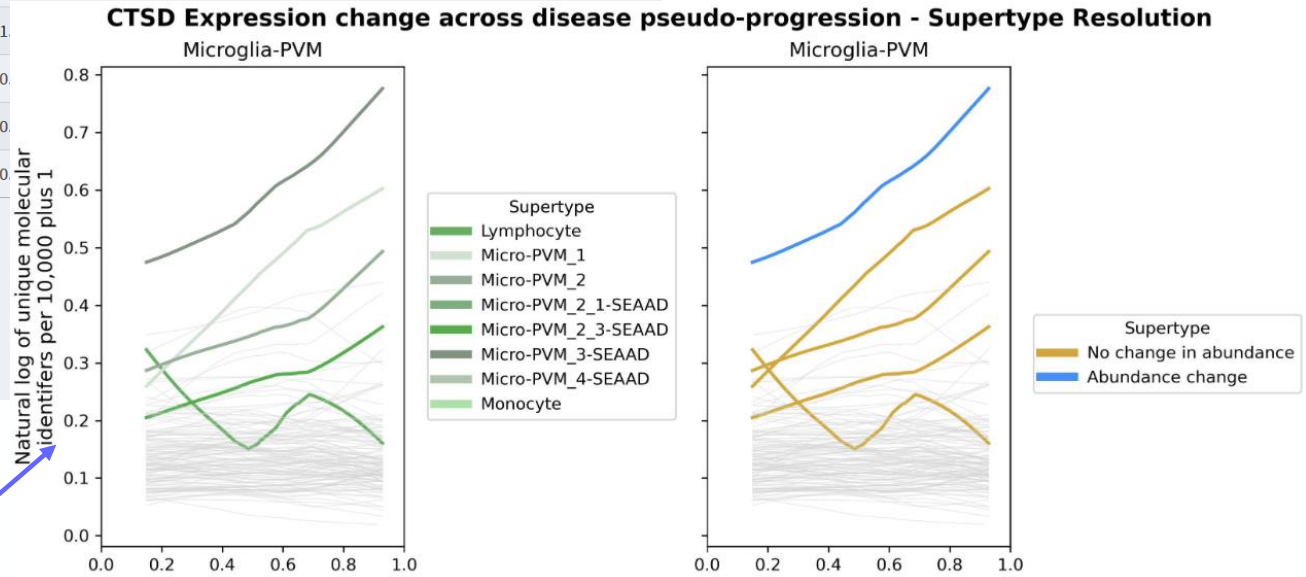
Subclass

Open beta coefficent table

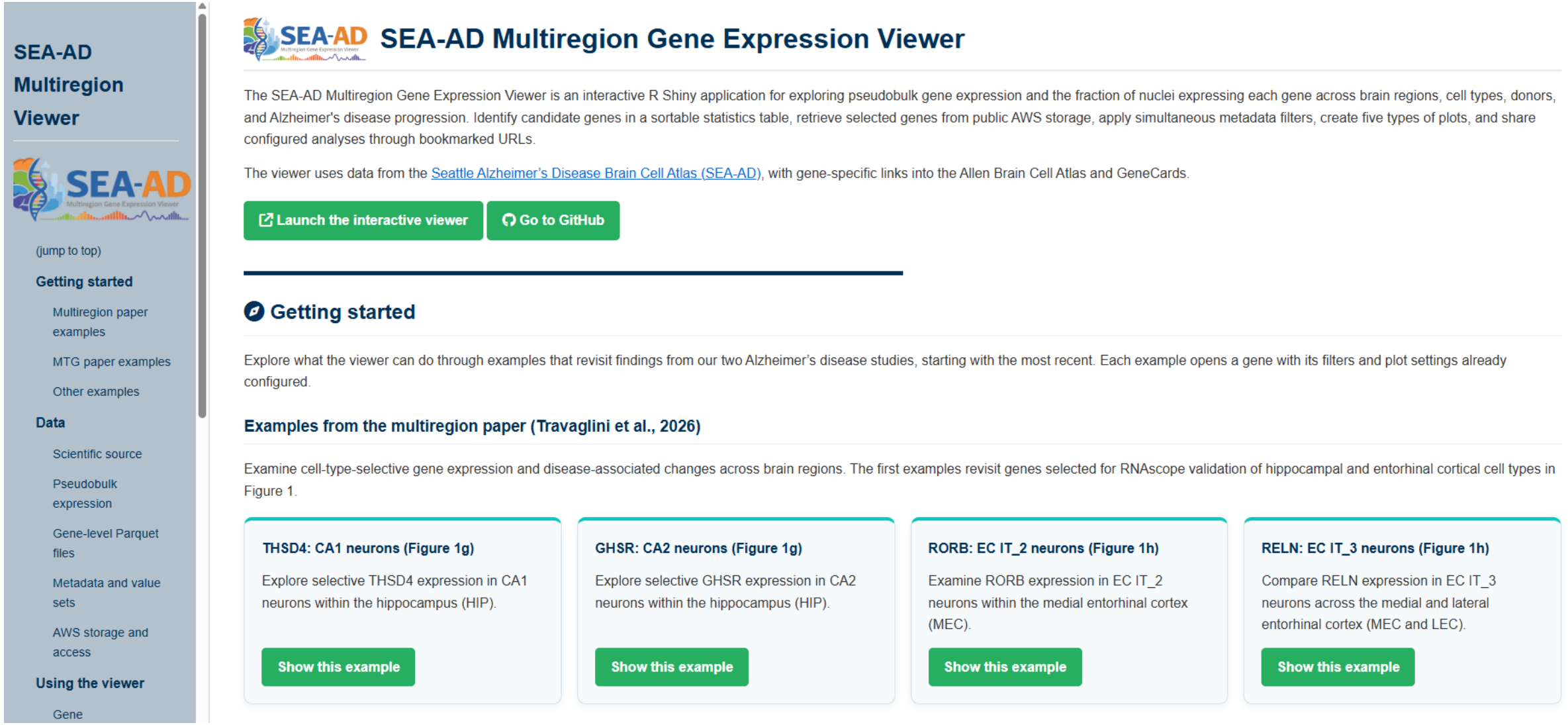
Row.number	Gene	Taxonomy.Level	Population	all	early	late	Mean.expression	Comparative.Viewer	Pseudoprogression.Plot
5615742	CTSD	Subclass	Microglia-PVM	3.46	1.1	1.42	0.49	Comparative Viewer	Pseudoprogression Plot
						1.36	0.3	Comparative Viewer	Pseudoprogression Plot
						-0.73	0.29	Comparative Viewer	Pseudoprogression Plot
						0.92	0.24	Comparative Viewer	Pseudoprogression Plot
						0.38	0.23	Comparative Viewer	Pseudoprogression Plot
						0.14	0.23	Comparative Viewer	Pseudoprogression Plot







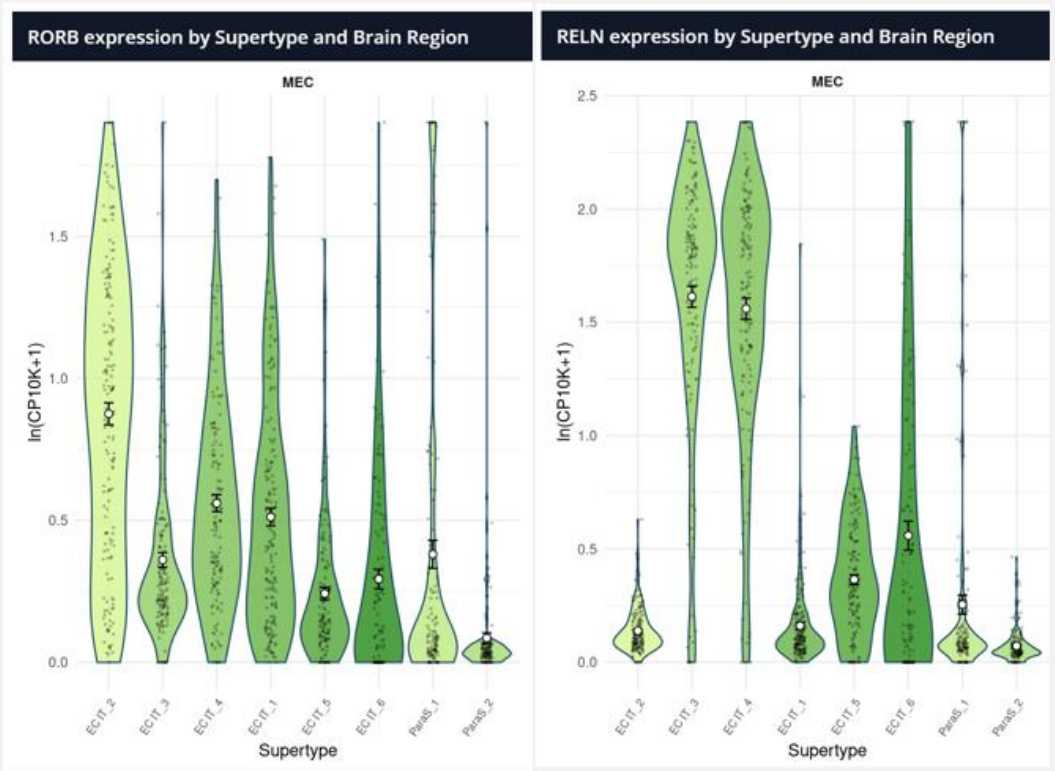
NEW: SEA-AD multiregion viewer/



https://alleninstitute.github.io/seaad_multiregion_trajectory_viewer/

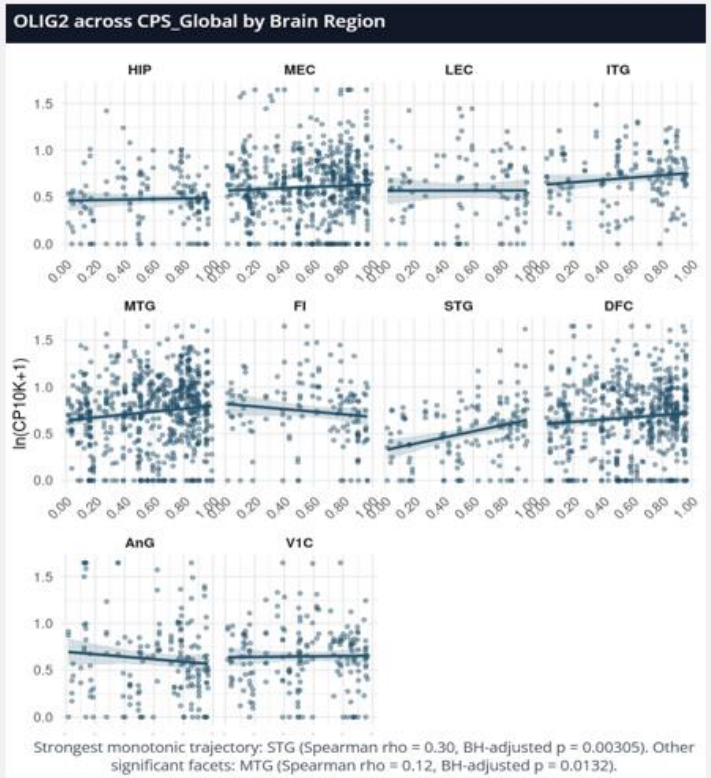
NEW: SEA-AD multiregion viewer/

Cell type selectivity in Entorhinal cortex



These genes were part of a panel used to distinguish EC IT_2 from EC IT_3 supertypes with RNAscope.

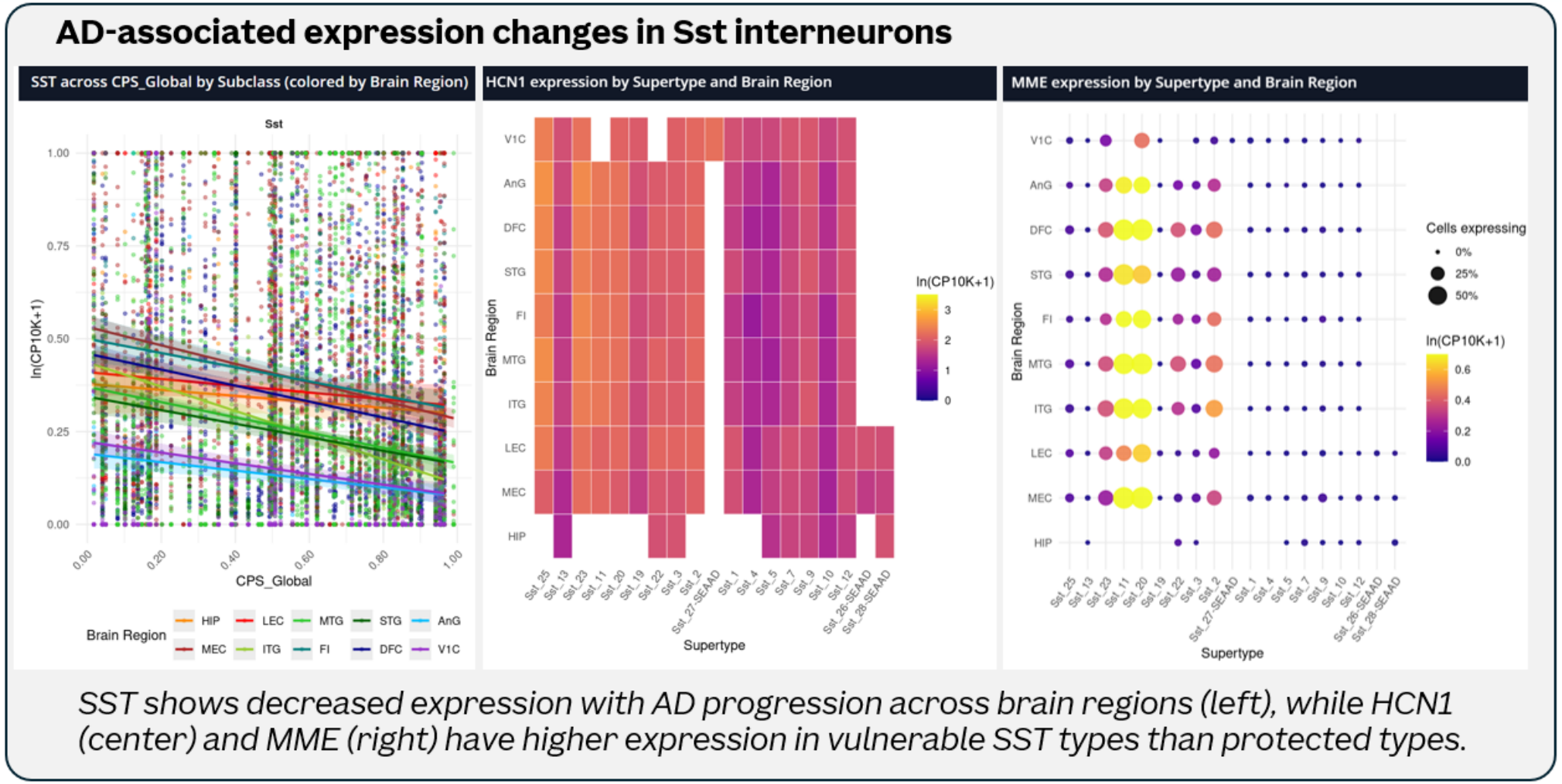
Region-specific changes in OPCs



While many AD-associated gene changes are conserved, this one is region-specific.

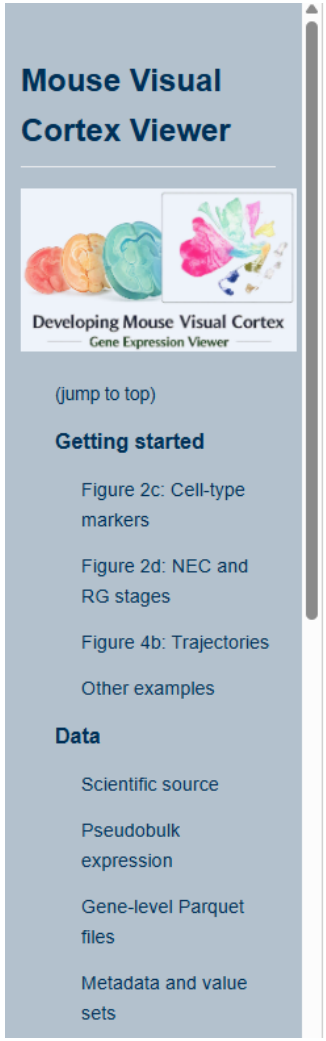
https://alleninstitute.github.io/seaad_multiregion_trajectory_viewer/

NEW: SEA-AD multiregion viewer/



https://alleninstitute.github.io/seaad_multiregion_trajectory_viewer/

parallel viewer for studying mouse development/





Developing Mouse Visual Cortex Gene Expression Viewer

The **Developing Mouse Visual Cortex Gene Expression Viewer** is an interactive R Shiny application for exploring pseudobulk gene expression and the fraction of cells expressing each gene across developmental age, developmental stage, cell type, donor, library, and anatomical metadata. Users can identify candidate genes from a sortable statistics table, retrieve genes from public AWS storage, apply multiple simultaneous filters, create trajectory, heatmap, dot, violin or boxplot, and two-gene correlation views, and share configured analyses through bookmarked URLs.

The underlying resource is part of the Allen Brain Cell Atlas, an open-science platform for visualizing and analyzing multimodal single-cell data across the mammalian brain.

Launch the interactive viewer | Go to GitHub

Getting started

Explore what the viewer can do through examples that revisit findings from Continuous cell-type diversification in mouse visual cortex development. Each linked example opens a gene with its filters and plot settings already configured.

Figure 2c: Cell-type markers

Explore prenatal and developmental expression patterns across cell populations.

Tnc: prenatal glioblast expression

Examine prenatal Tnc expression in glioblasts using an embryonic-stage heatmap.

Show this example

Neurog1: prenatal nonIT IP expression

Explore prenatal Neurog1 expression in nonIT intermediate progenitor (IP) cells using a dot plot.

Show this example

Hmga2: enriched expression in NEC cells

Compare Hmga2 across development, with its highest expression in neuroepithelial (NEC) cells.

Show this example

https://alleninstitute.github.io/mouse-developing-vis-app/

other omics modalities/

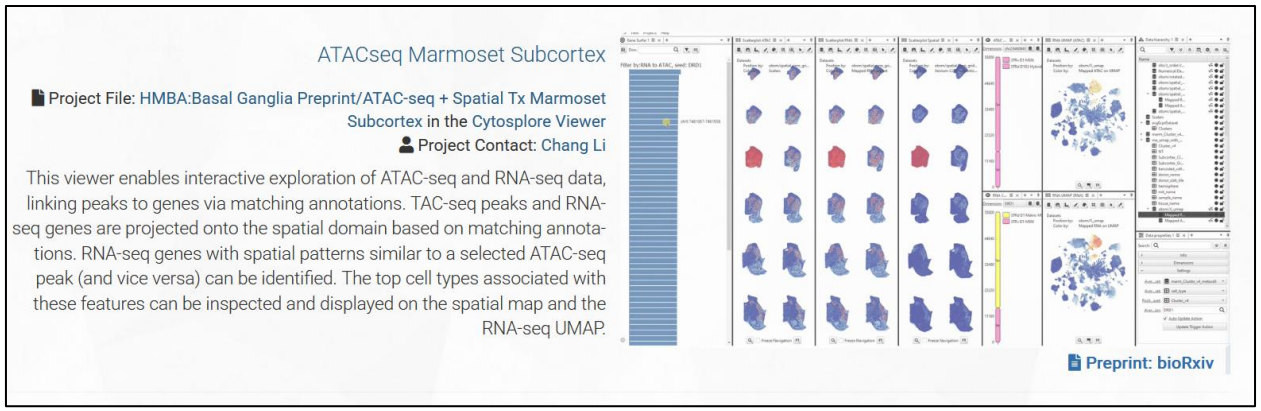
- Single cell ATAC-seq
- Spatial transcriptomics
- Patch-seq



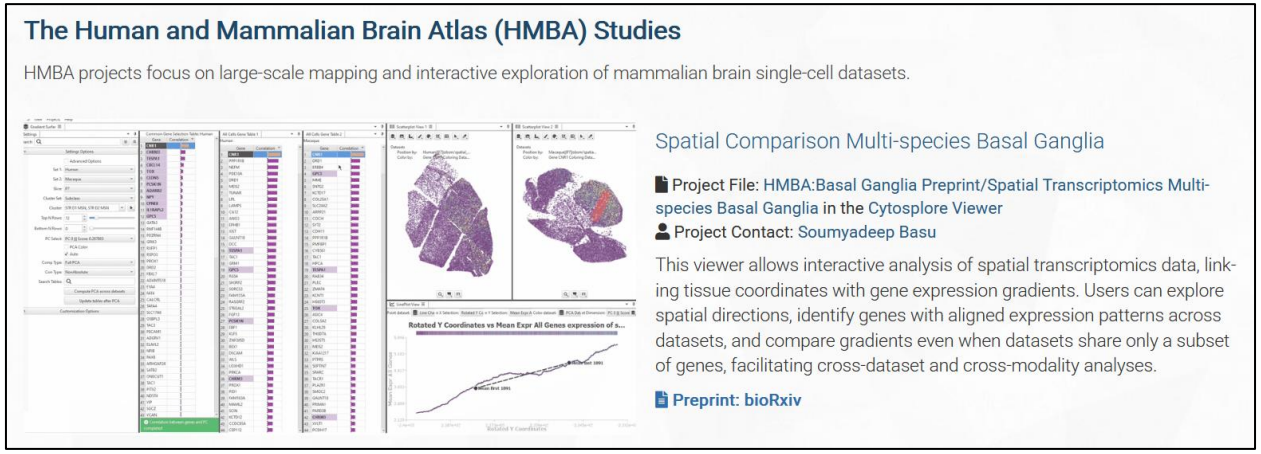
alleninstitute/

cytosplore viewers for multi-modal data/

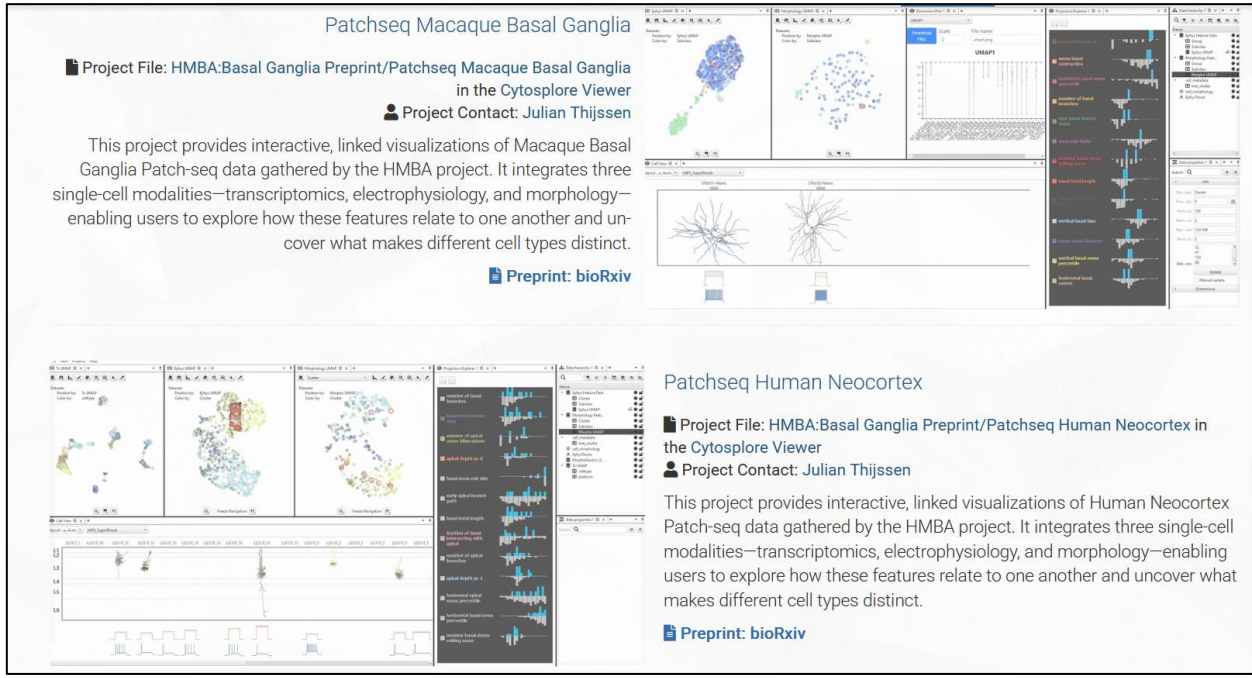
Spatial Tx + ATAC-seq (Marmoset Subcortex)



Cross-Species Spatial Tx (Primate BG)

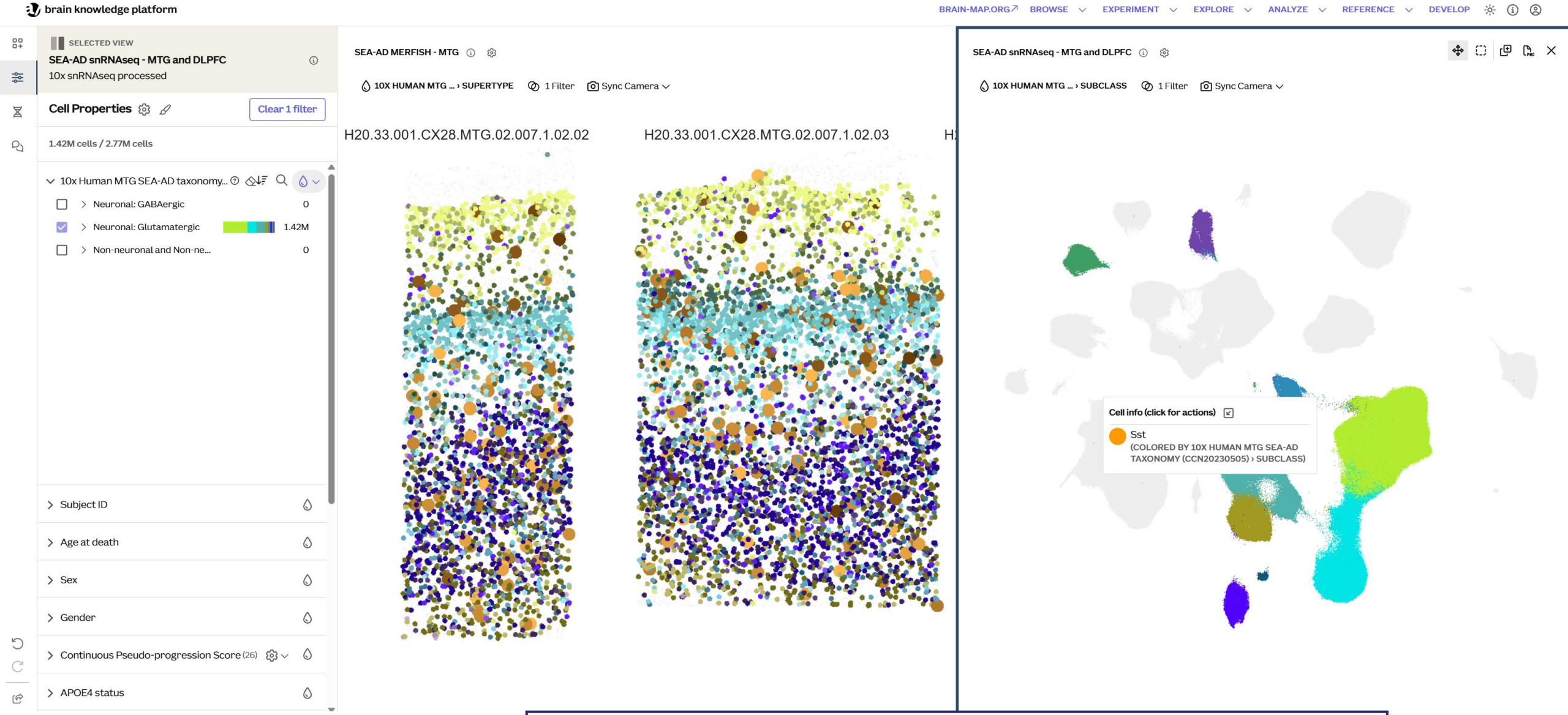


Patch-seq (Macaque BG, Human Neocortex)



https://viewer.cytosplore.org/projects/ (download desktop app + data bundles)

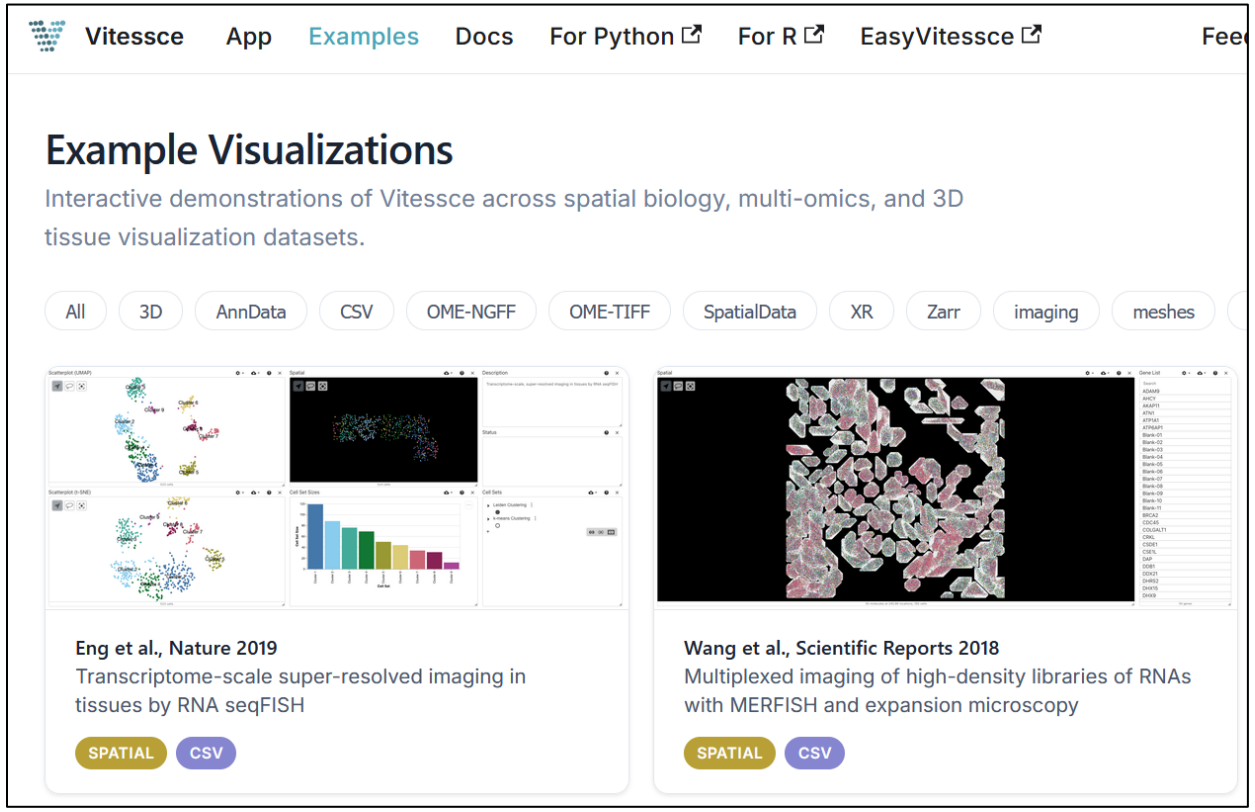
explore spatial & sn –omics (ABC Atlas)



Currently the ABC Atlas is the best tool for visualizing spatial transcriptomics data from the Allen Institute!

other visualization tools for spatial -omics/

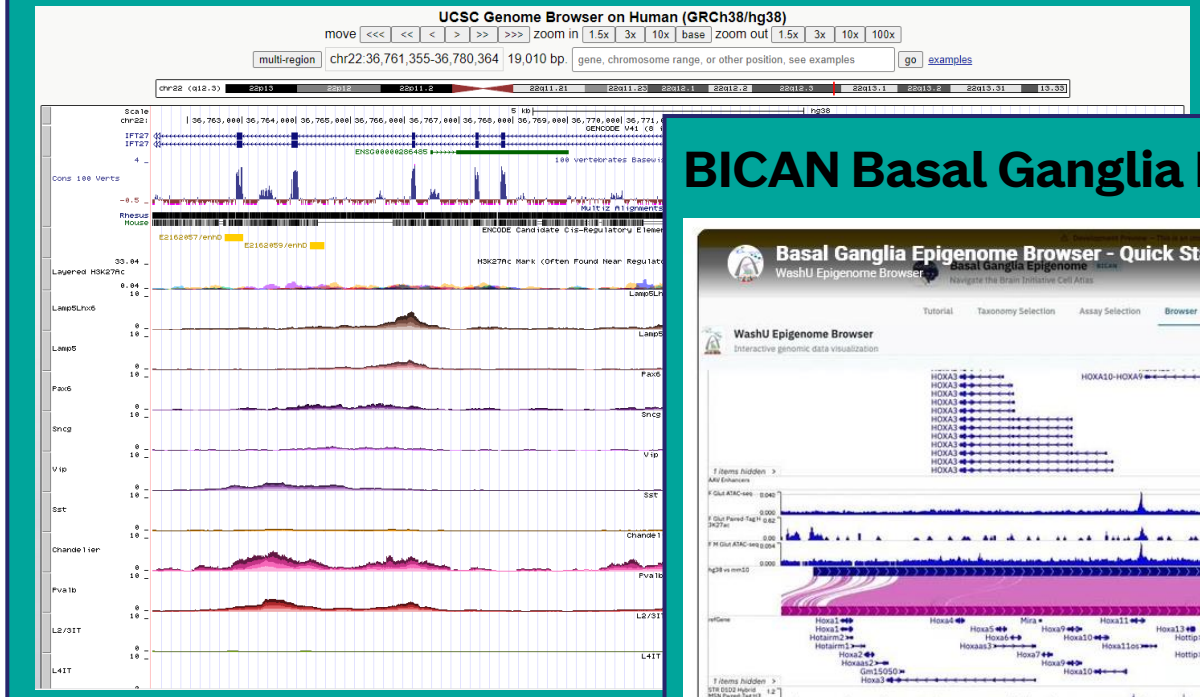
Vitessce has limited web-hosted example datasets



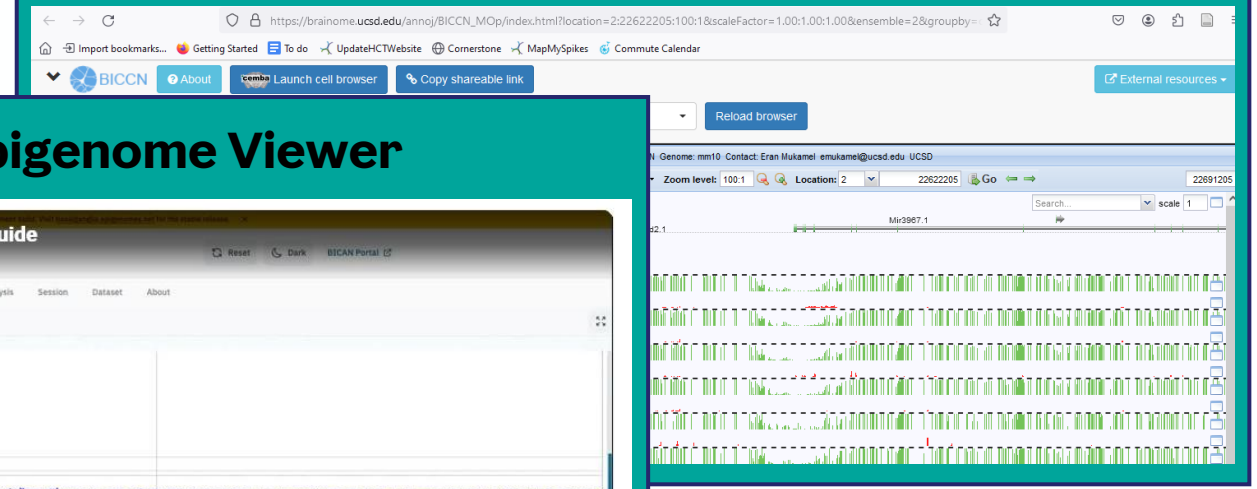
exploring chromatin accessibility data/

- Several related tools for finding, prioritizing, and viewing potential enhancers based on data that has gone through the standard analysis pipeline, and for creating custom plots and tracks in UCSC genome browser

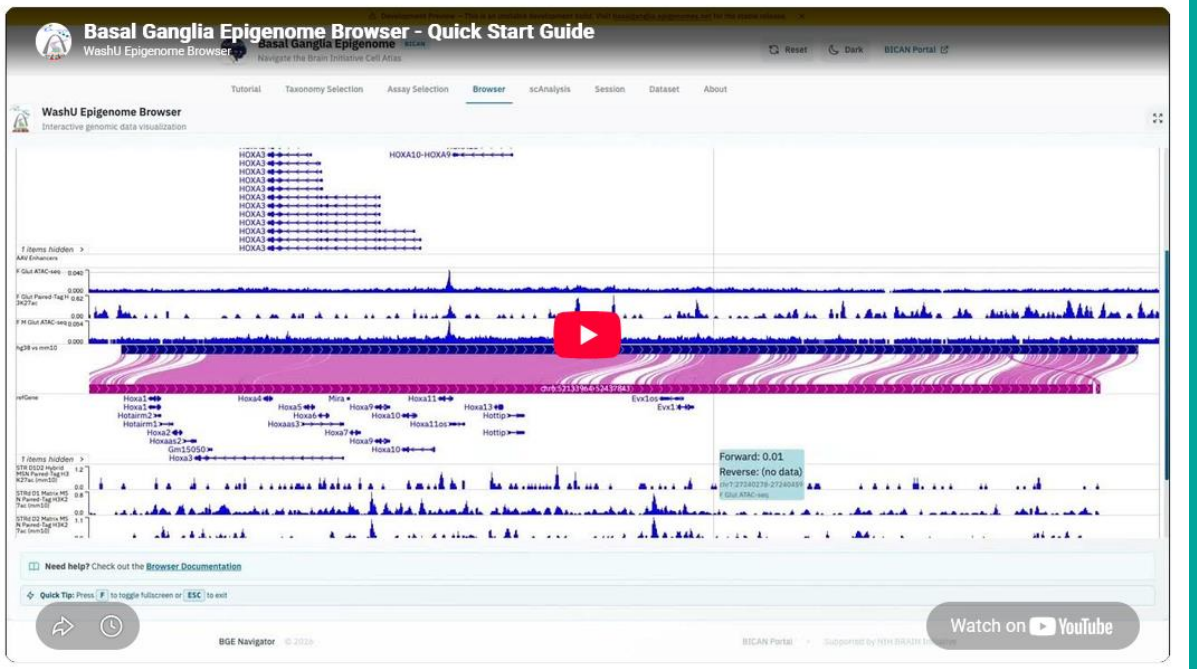
SEA-AD data in UCSC genome browser



BICCN AnnoJ Epigenome Browser tool



BICAN Basal Ganglia Epigenome Viewer

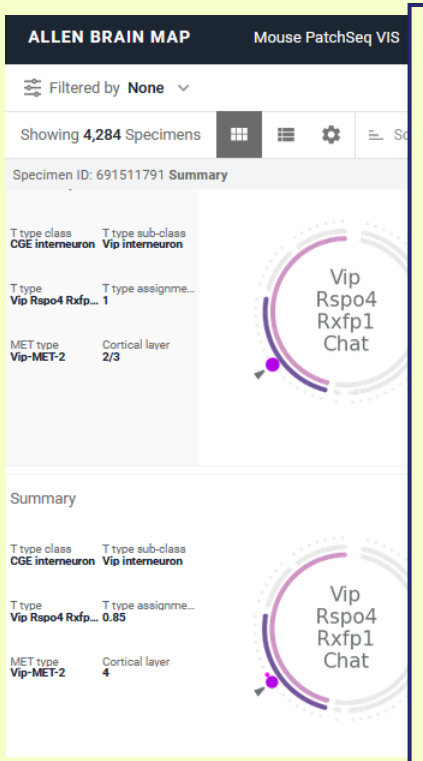


https://basalganglia.epigenomes.net

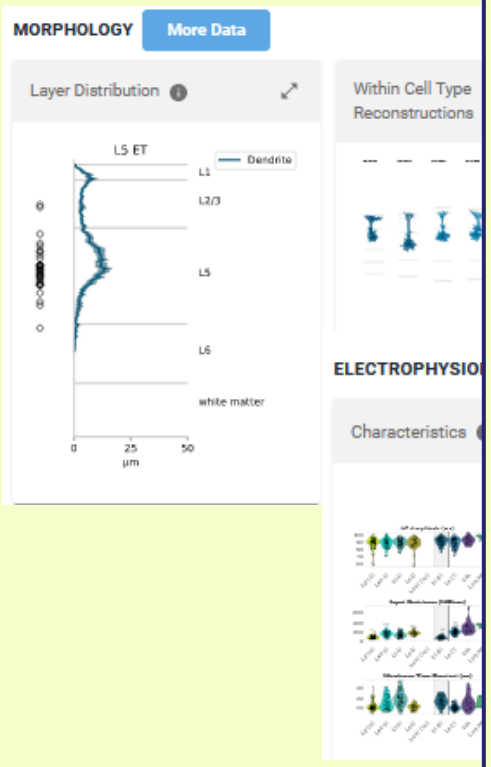
visualization tools for patch-seq data/

- Patch-seq is a method for collecting electrophysiology, morphology, and transcriptomics (and therefore cell type information) from the same cell
- Different visualization tools have been developed for different data sets

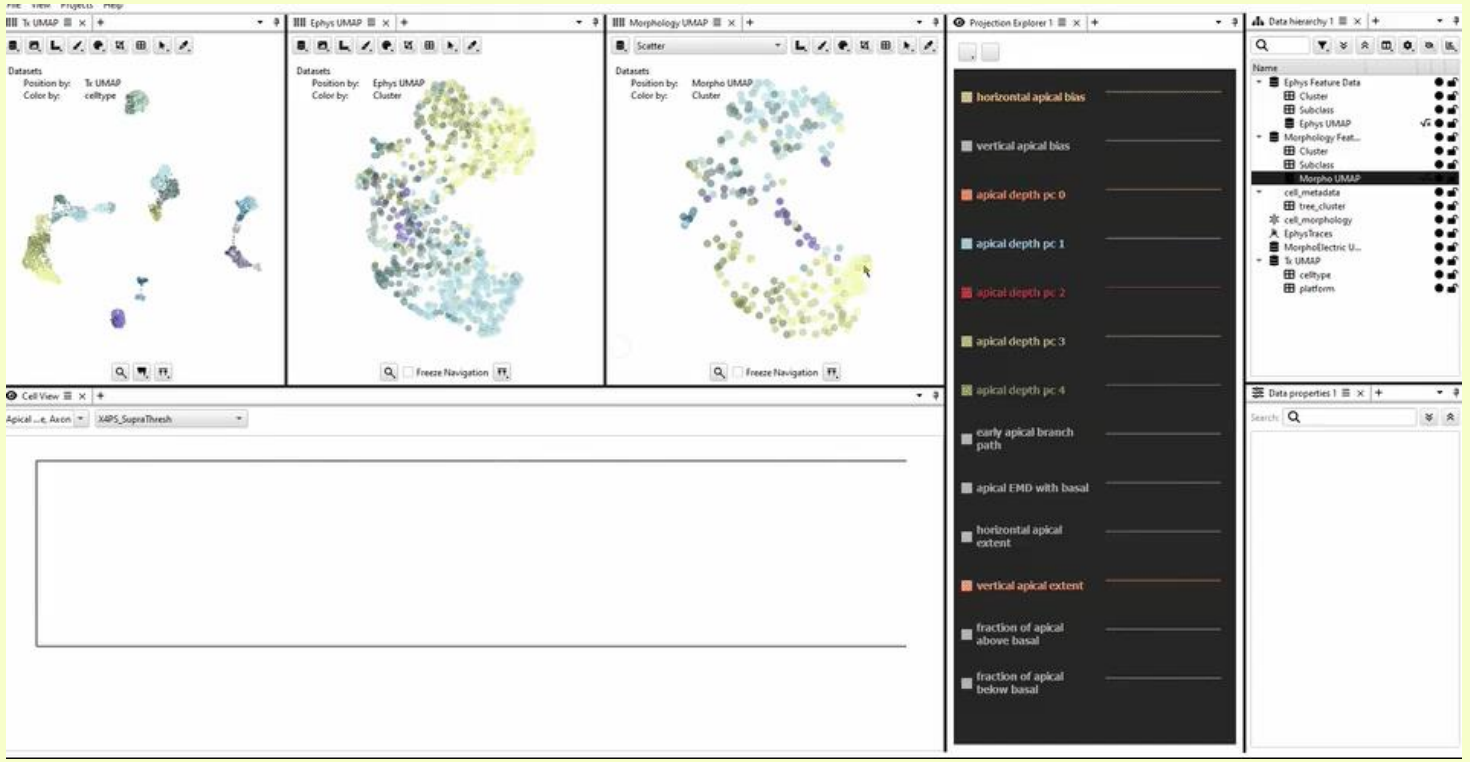
Patch-seq Viewer (mouse VIS GABA)



Cell Type Knowledge Explorer (mouse M1)



Cytosplore (human neocortex / macaque BG)



questions?

allen institute/

agenda/afternoon **– breakout rooms**

Session 2: analysis of your own sequencing data using allen institute resources.

Room 1165 – 1185

1:10-1:35 | Using a reference cell type taxonomy to understand Alzheimer's disease

1:35-2:00 | cross-species alignment and comparative analyses

2:00-2:20 | other interactive tools for -omics data exploration (CellxGene, Cytosplore, etc.)

2:20-2:30 | break

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ABC Atlas demo & external use case examples

allen institute/

community use of BKP tools and data/

Title: Evolution of Cajal-Retzius Cells in Vertebrates from an Ancient Class of Tp73+ Neurons

Authors: Elias Gumnit[1], Giacomo Gattoni[1], Jamie Woych[1], Astrid Deryckere[1#], Prabesh Bhattarai[2,3], J Andrew Gillis[4], Caghan Kizil[2,3], Maria Antonietta Tosches[1*]

Next, we investigated whether the molecular and spatial heterogeneity of salamander *Tp73*+ neurons has counterparts in mammals. We isolated mouse *Tp73*+ neurons from the Allen Brain Cell (ABC) atlas (*34*) and compared them to the salamander *Tp73*+ neuron dataset. The

A unified analysis of atlas single-cell data

Hao Chen,[1,2] Nam D. Nguyen,[1] Matthew Ruffalo,[1] and Ziv Bar-Joseph[1,3]

To further test the ability of GIANT on integrating data from different modalities, we collected single-cell mouse brain data from the Allen Brain Cell Atlas (Yao et al. 2023), which contains 2,341,350 cells from scRNA-seq and 3,938,808 cells from MERFISH spatial transcriptomics (Chen et al. 2015). Both modali-

Morphine regulates astrocyte transcriptional dynamics in the ventral tegmental area by stimulation of glucocorticoid signaling

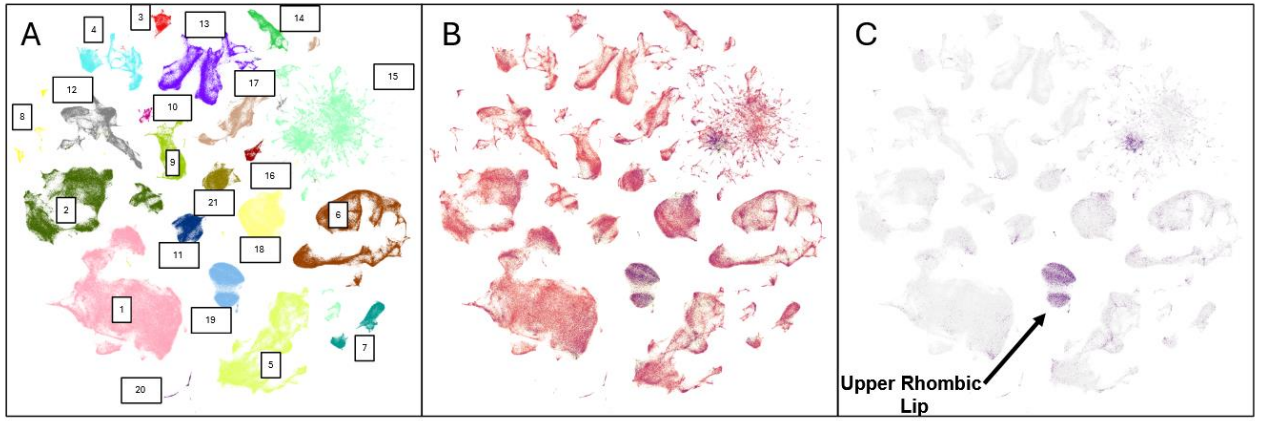
Jennifer J. Tuscher[1,3]*, Angela Cleere[1,3], Robert A. Phillips III[1,4], Catherine E. Newman[1], Guy Twa[1], Nathaniel J. Robinson[1], Lara Ianov[1], Robert E. Sorge[2], & Jeremy J. Day[1]*

clusters using lisi v1.0 in R[83]. To refine resolution of VTA neuronal subtypes and assign cells to classes, we followed the Allen Institute's Whole Mouse Brain taxonomy using the MapMyCells web interface[84] (RRID:SCR_024672). After removing additional low-

Review

Using the Allen Brain Cell Atlas of the Human Brain to Gain Insights into C-Terminal-Binding Protein 1 (CtBP1)'s Potential Function

Suhjin Lee [1] and Uthayashanker R. Ezekiel [2,*]



Use cases for ABC Atlas

“Are genes X and Y co-expressed in the same cell type?”
“Where is gene X expressed in the brain?”
“Is gene X expressed in a specific cell type?”
“Where is cell type X in the brain?”
“What is the spatial relationship between cell type X and cell type Y?”

However,

- MERFISH data in ABC Atlas is based on cell segmentation; cell-to-cell or synaptic contacts are not shown.
  - “Does cell type X synapse onto cell type Y?” cannot be answered
- MERFISH data in ABC Atlas shows centroids of cells, not cell shape or size
  - “Is the soma of cell type X larger than the soma of cell type Y?” cannot be answered

ABC atlas data/ experimental design

New Results

Follow this preprint

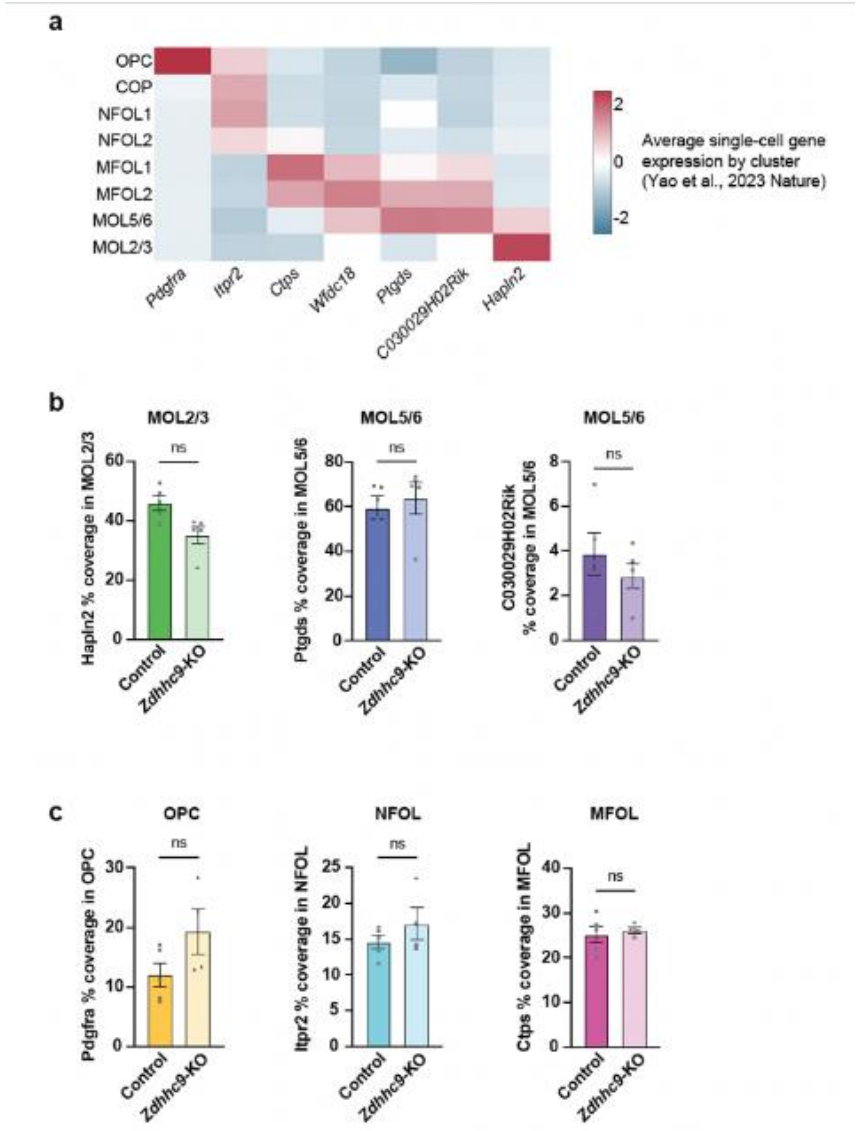
The X-Linked Intellectual Disability gene, *ZDHHC9*, is important for oligodendrocyte subtype determination and myelination

Experimental design: designing FISH probes to label oligo subtypes based on marker genes found in ABC Atlas cell types

Altered oligodendrocyte subtype proportions in *Zdhhc9*-KO mice

The changes in gene expression identified with RNAseq predict alterations in proportion of oligodendrocyte lineage cells in *Zdhhc9*-KO mice (Figure 1d). We therefore designed fluorescent *in situ* hybridization (FISH RNAscope) probes for the dominant MOL subtypes in the corpus callosum, guided by expression data from previously published scRNAseq and FISH studies(15, 28, 32)(Figure S4a). We chose marker genes with highly enriched expression in specific oligodendrocyte lineage cells, including *Hapln2* (MOL2/3 marker), *Ptgds* (MOL5/6 marker), and *C030029H02Rik* (MOL5/6 marker, Figure S4a). Importantly, within each cell type identified we

Hollman et al., 2024 (preprint)



ABC atlas data/ data analysis

Home > The EMBO Journal > Article

Rhythmic astrocytic GABA production synchronizes neuronal circadian timekeeping in the suprachiasmatic nucleus

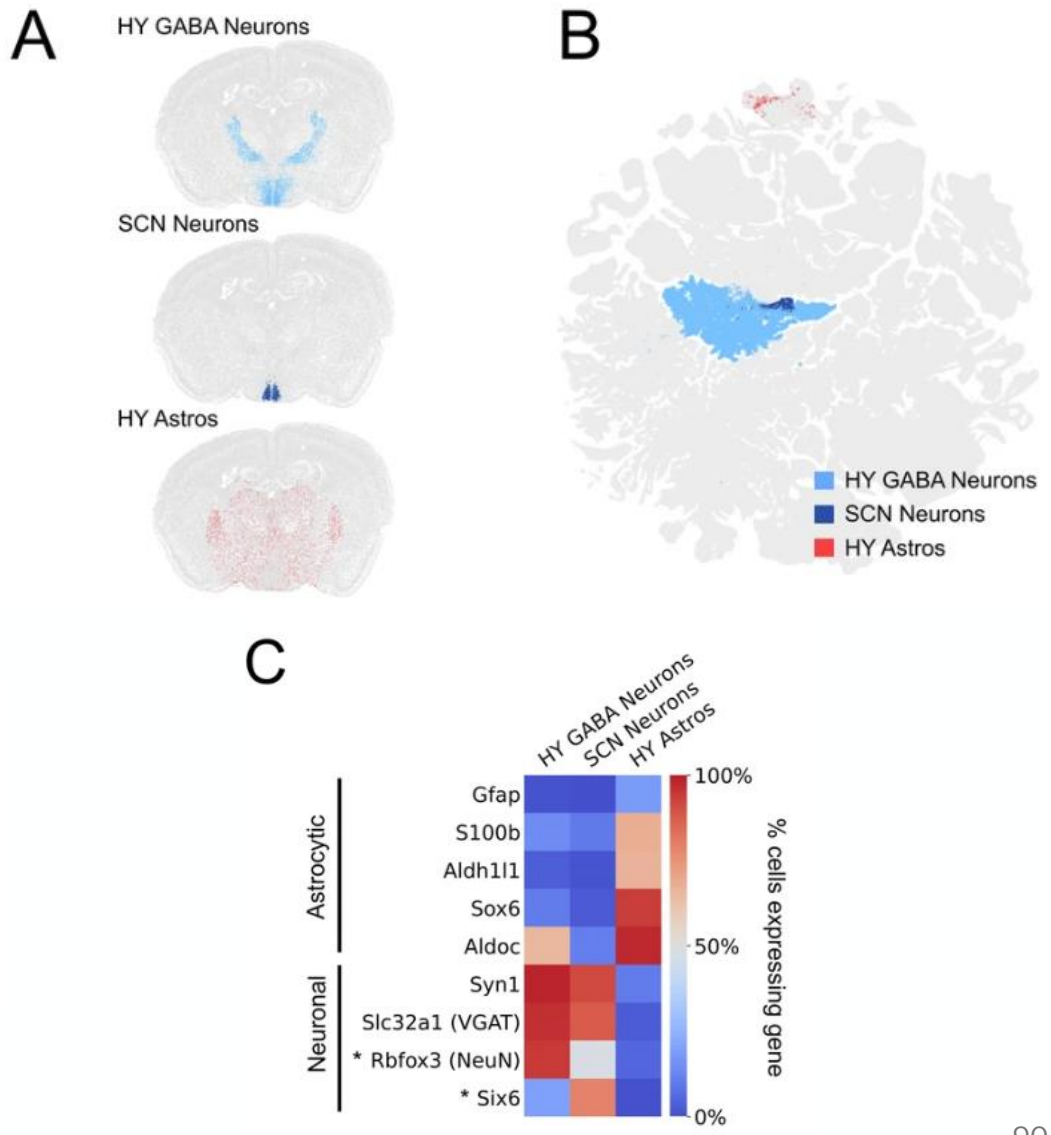
Article | Open access | Published: 02 December 2024

Volume 44, article number 2 (2025) Cite this article

Data analysis: examining synthesis pathways in astrocytes and neurons from ABC Atlas

To examine whether these astrocytic GABA synthesis pathways are expressed in SCN astrocytes, we analyzed publicly available data from a recent large single-cell RNA sequencing dataset of the mouse brain, consisting of a total of ~2.3 million cells, by the Allen Brain Institute (Yao et al, 2023). We compared gene expression of relevant biosynthetic pathways in hypothalamic astrocytes, SCN neurons and hypothalamic GABAergic neurons (Fig. 4A,B). Hypothalamic GABAergic neurons, SCN neurons and

Ness et al., 2024



ABC atlas data/ novel tool generation

New Results

Follow this preprint

3D multimodal histological atlas and coordinate framework for the mouse brain and head

Novel tool: mouse brain and head atlas combining MRI & CT imaging with histology data

Tward et al., 2025 (preprint)

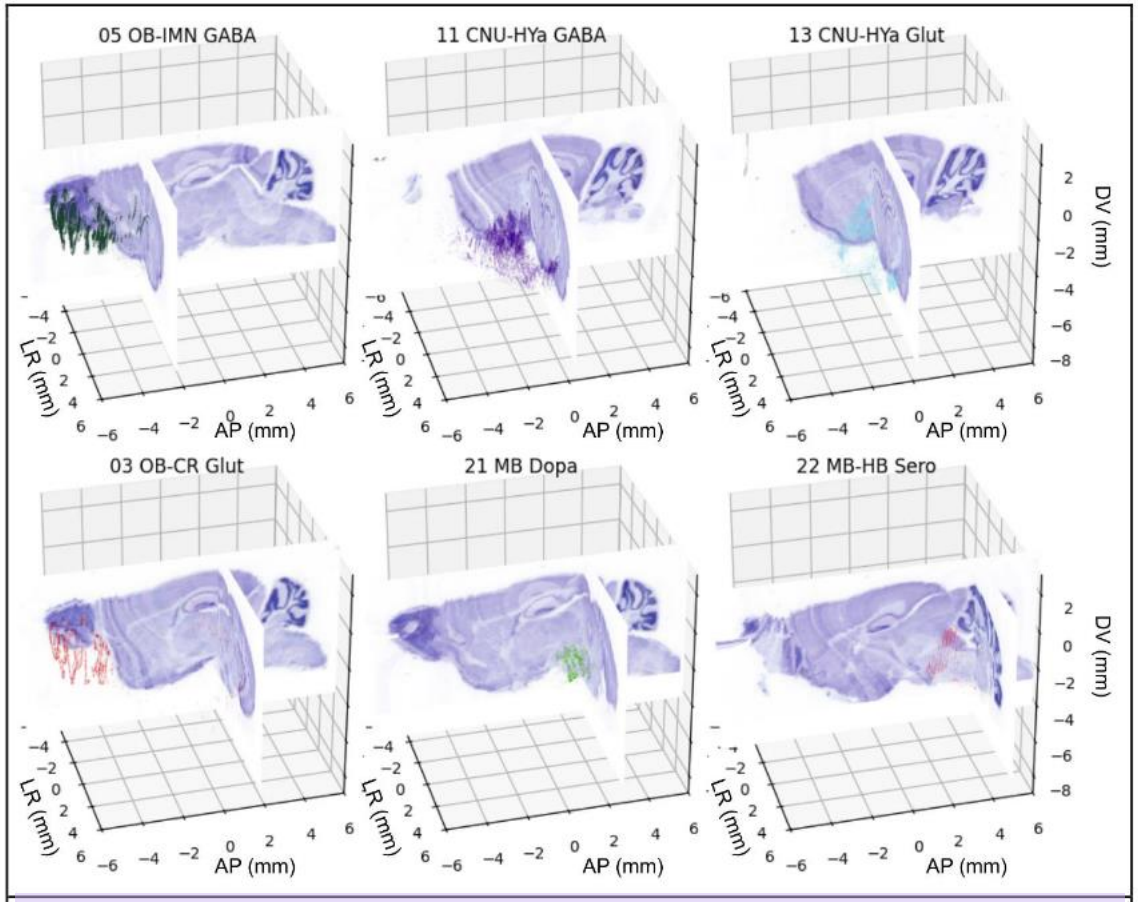


Figure 5. Spatial distributions of 6 selected cell types (see plot titles) from the ABC dataset, with coordinates mapped into our coordinate system. A 3D reconstructed Nissl reference brain dataset is shown in our coordinate system. Coronal and sagittal sections are chosen such that that 90% of cells appear in front of the slices. The visualization illustrates the intact brainstem present in the Nissl reference volume. Spatial distributions of a larger set of cell types are shown in **Extended Data Figure 6**.

ABC Atlas demo

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feedback survey/ please fill out!

www.surveymonkey.com/r/7VMVBKR


alleninstitute/
thank you/

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